

# Universe24: an integer cellular automaton read through clicks, the formulas it shares with nature and the ones it adds

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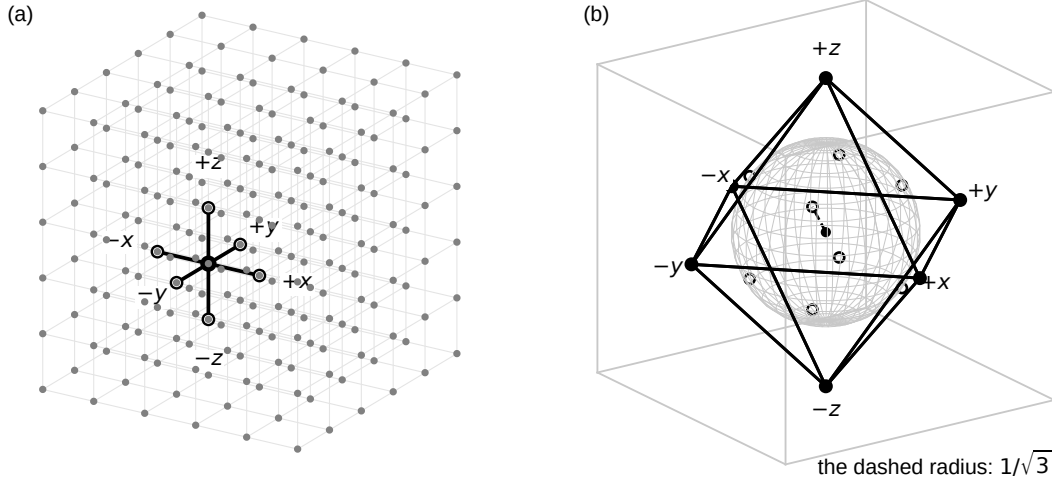
## Abstract

Universe24 is an integer cellular automaton: a cubic lattice of Nodes stepped by one recurrence taking a Node's present, past and six neighbours through one integer division, reversibly. The lattice is read only by NodeDetectors, declared regions of Nodes writing the clicks; Universe24 is those clicks. Measurement is a meeting at a NodeDetector of the record run forward with its transition read backward from the click, two quanta, never one. One line, Rule3, gives the lattice Klein-Gordon equation, Schrödinger's as its slow limit, Planck's relation in the band's form and, under one named assumption fixed by one click, gravity's static weak-field metric. A family's pair  $[\text{num}, \text{den}]$ ,  $0 < \cos \omega_0 = \text{num}/\text{den}$ , gives its inertial mass  $m^* = 3 \tan \omega_0$  and the post-Newtonian  $\gamma_K = \text{den}/\text{num}$  and  $\beta_K = (\text{den} + \text{num})/(2\text{num})$ , closing to 1 as the gap closes. The exponential metric gives  $\gamma = 1$  exactly; the lattice's frame gives  $\alpha_1 = -8$  and, for a static source,  $\alpha_2 = -1$ , both excluded by nature's bounds, a mending carrier a named hypothesis. Universe24 is one world built by the method, others closer to nature by it. Three predictions: its shadow's radius  $2em$  ( $m = GM/c^2$ ), within the Event Horizon Telescope's bounds; light's gravitational source at  $E/c^2$ ; its metric's second order. The claim: Universe24 helps compute things in nature; each computed quantity stands against measurement with its status and number, misses included; what it does not compute is named.

**Keywords:** integer cellular automaton; reversible lattice dynamics; measurement as a meeting of past and future; lattice dispersion and inertial mass; weak-field gravity on a lattice; limiting speed on a lattice

## 1 Introduction

**The question.** This paper is about one method of computing physics. A lattice of Nodes is stepped by one local, reversible rule in whole numbers, and it is read only through clicks, the one kind of measurement there is. In the words of the title: the integer cellular automaton is the lattice stepped by Rule3; Universe24 is its clicks, written by the NodeDetectors from the records of its declared files; the formulas it shares with nature are the comparisons of Sections 8 to 10; and the ones it adds are the lens's one formula for light and matter with its lattice chromatic term (Section 8, (p)) and the model's predictions no measurement has read, its shadow's radius and its metric's second order (Section 8) and light's integrated write at  $E/c^2$ , its field open (Section 10.2), stated as such there. Those comparisons are the claim's content, each quantity with its status mark and its number against measurement. On this lattice the classical wave equation, the quantum of action and the metric of gravity are three readings of one line, Rule3, the metric under the one assumption of Section 3.4. The line itself is the lattice wave equation. Its count, read in quanta, is the click. Its paces, read as a clock and a ruler, are the metric. The question is whether this method can give the formulas that nature's measurements show, and how little state, non-locality and randomness it needs. The formulas named in the abstract, the one computed file set beside its blind row, and the back-in-time gate below are examples of what the method yields. They are not its scope, and Section 10.3 lists what it does not give. The paper's one claim is that Universe24 helps compute things in nature: the main text states what the model computes of nature against measurement, each quantity with its status mark and its number, and the derivations stand in the supplement. The method, a lattice of integers stepped by one rule and read into the world through clicks, is how it computes and not the claim. The paper does not claim that nature is such a lattice. Universe24 is one world built by the paper's method; the method and the claims that support it are the work, and no more; other worlds closer to nature can be built by the same method, and building one needs the calculations named at the end of Section 10.3, which this work does not do. Every comparison and prediction below is one such quantity with its status, a hit or a miss named with its number; the derivations behind them stand in the supplement. The paper computes one click pattern, the two slits', against its blind, and sets the formulas of its clicks beside nature's measurements. The paper starts from one statement and never contradicts it (ASSUMPTION): Universe24 is the meeting of the past with the future; a click is that meeting, the one measurement there is in the model; and in it only clicks are seen. The clicks are the mechanism's own. In an experiment of nature an apparatus, of whatever kind, registers something, and the model's click stands for that registration; but the click is defined by the model, one quantum credited to a NodeDetector, and not by nature, so the two are not the same thing, and the paper compares what its clicks give with what nature's counters register. Every physical number in the paper is a reading of one integer rule with its declared integers: the pairs,  $\Gamma$ ,  $T$ , the two holders' weights, gravity's  $E_g$ , the sign holder's  $k$ , and the vacuum content (Section 6.6). For nature's other measurements it gives the forms of Section 8 at the computed pair, with the findings of Section 10.3 set against them. A Node holds a record's present and its past, two integer levels. The rule makes the record's future from these two and from the six neighbouring Nodes. The rule is covariant under the translations of the lattice and under the symmetry group of the cube, which has order 48; no boost is among these symmetries (Section 3.1). The lattice holds nothing but events that split: at every Node, a level splits by the rule to the six neighbours. Run forward from the root, the lay, the split events are the future, every path at once. Read backward from a click that happened, by the same rule, they are the past, and through two slits both of them pass through both gaps. A click is where the two meet at a declared NodeDetector. Which meeting happens is the one thing the model does not



**Fig. 1** (a) The lattice in space: a cubic array of Nodes, here  $6 \times 6 \times 6$ , each joined to its six neighbours by Links through the Ports  $\pm x$ ,  $\pm y$ ,  $\pm z$  (the faint lines). One Node and its six neighbours are marked; these are the Nodes one interval away from it. A Node keeps each family's two levels and its remainders, and nothing else. (b) The Nodes one interval away from a Node form the octahedron  $|x| + |y| + |z| \leq 1$ , whose six vertices are the six Ports. Its symmetry group is the cube's,  $O_h$ : the 24 rotations and, with the reflections, the 48 signed permutations of the axes. Rule3 is covariant under this group (Section 3.1). The inscribed sphere, of radius  $1/\sqrt{3}$ , touches the eight faces on the diagonals of the cube, and that radius is light's speed in Link per interval (Section 3.3). Both are drawn from the definitions; no run

determine, and it is the one non-local element of the description (Section 5.2). In the model, what looks like a wave is the future; what looks like a particle is the present, the click; the track is the past (Section 5.1).

**The words.** The physical location is a *Node*, and its complete local information is its *NodeState*. Nodes are joined by *Links* through six directional *Ports* per Node, and the cubic array of Nodes is the *lattice* (Fig. 1(a)). A local change is an *Event*, and there is one local rule, Rule3. A *family* is one kind of physical value, with its pair and with what it holds. Its *record* is its levels over the lattice, two per Node with a remainder. A family of quanta has a *dimension*: 1 for one real line, 2 for a plane of two lines. A *body* is its count and its record over its Nodes, standing bound in a holder's well; a record of count one may stand at one Node, a NodeDetector with Nodes alone never; it has no law of its own. A *cluster* is bodies standing in one another's wells. The *lay* is the file's initial writing of the records, and the *root* is the lay's Node; the *credit*, the *draw*, the *window* and the *books* are the NodeDetector's bookkeeping of Section 5.3. A family that holds a value for others to read is a *holder*, and its *field* is its level at every Node; a holder's *well* is the dip of its level around the body that sources it, and a record's *tail* is its levels beyond its main support, and a holder's tail its level's fall beyond its source. A region's *front* is its boundary Ports, through which its inflow is read; the erasing front of a click is the spreading erasure of the clicked record's other paths (Section 5.1). The *hole* is the absorbed record's levels and remainder set to 0 at the one Node of the click's write. A *packet* is a travelling record in no body, free: laid by the file or written by a click, in flight until something absorbs it, if anything does. *Universe24* is the world the method computes: the clicks the NodeDetectors write from the lattice, never the lattice itself and never nature; the paper's claim is what Universe24 computes of nature, its formulas set beside nature's with their status and their numbers; the method is how. A *file* declares one experiment of it and the implementation runs it: the families with their numbers, the holders, the bodies and the packets laid, the NodeDetectors with their declarations, and the faces; the engine's documents call a file's run a world, a word this paper keeps for the world we live in. A *NodeDetector* is the model's counter: a connected region of Nodes declared in the file, the draw the world's one (its seed, generator and window) shared by every NodeDetector with Nodes alone in turn and a body's NodeDetector's own, two or more where it reads with Nodes alone, one or more where the file gives it a record of its own; it reads its Nodes and writes the clicks that happened, naming itself and never a Node, and, given a record of its own with parts and a lifetime, it emits as well. It is the model's own: the one place of the draw and the one non-local element, the gate between the lattice and Universe24, standing where an experiment's detector stands in nature and defined by the model alone. Its *click* is one quantum credited to it, the one measurement, the model's name for what an experiment's counter registers. An *interval* is one step of the whole lattice. A *level* is the integer at a Node, a family's field value there, and the *remainder* is what its integer division keeps; a *pace* is the step factor of a Node (its clock, the metric's lapse) or of a Link (its ruler); the *content* is the level a family reads into its paces, the potential in the level's unit; a record's *share* is its energy in quanta and its *count* the integer the books hold (Section 4.4 names the share and  $W_{\text{rec}}$  a count where it says so); a Node's *tension* on an axis is its stress reading, a Link's the mean of its two ends'; a *booking* is any reading entered into the write or the books; a Link's booking is its tension's; the *blind* is the expectation written before a run. The file's *row* is a holder's declaration. A *quantum* is a record's count of one. The words in full are in S.49 [1, the objects][2, the words].

## 2 Related work

**What this paper adds.** Three things, each with its mark where it stands. The method: one line of integer arithmetic, stepped reversibly on a lattice and read only through clicks, with measurement as the meeting of the future, the record run forward from its root, with the past, the transition read backward from the click, at a NodeDetector, the one place of the draw (Sections 3 and 5). The forms it reaches: the continuum's known formulas, in lattice form, from that one line, each with its inputs and its status named (Section 8, Table 1), and the lattice's own, which depart from the continuum or have no counterpart there, each a prediction at the measurement its row names (the conclusion's table). The method of claim: every statement carries its mark and its fence, every input and every miss is named, and the derivations, the files and the

implementation are public, so that each step can be checked and each reading repeated.

**Why the world is read in clicks, and never on the lattice.** In the model, whoever sees is itself a cluster of bodies on the lattice, and a Node reads its six Ports and nothing else. Inside a cluster everything is seen: a lattice reading is the local language of the cluster's own record. Beyond the cluster's tail nothing reaches it except what a NodeDetector reports, one quantum credited at a meeting; a click is the only thing that passes between a cluster and its neighbour [1, the observer is a cluster; the click is the pointer]. What stands on the lattice between clicks is not a fact of the world but its possibilities: the split events of every path at once, the levels Rule3 makes at the Nodes, the future. A fact is a realised event, a quantum credited to a NodeDetector, and that happens only at a click. So in Universe24 there are clicks and nothing else, and nothing of the lattice is measured, only its clicks through a NodeDetector [1, no waves on the event board; what produces a click is not known]. This is why the paper sets beside nature only what the clicks give: a lattice reading, a level, a support, a count at a Node, is a diagnostic of the computation and is labelled so, and only a click pattern is compared with a measured pattern of nature [1, readings and measurements] (ASSUMPTION, the first postulate of Section 3.6). So the paper has three words and keeps them apart. The lattice is the lattice stepped by Rule3; it computes. The NodeDetector is the gate between the lattice and Universe24: it reads the lattice and writes the clicks that happened. Universe24 is those clicks, the world of the model. The paper's claim is what Universe24 computes of nature: its formulas are set beside nature's measurements, each with its status and its number and never by proof, and how close they come the reader judges from them; the method is how it computes.

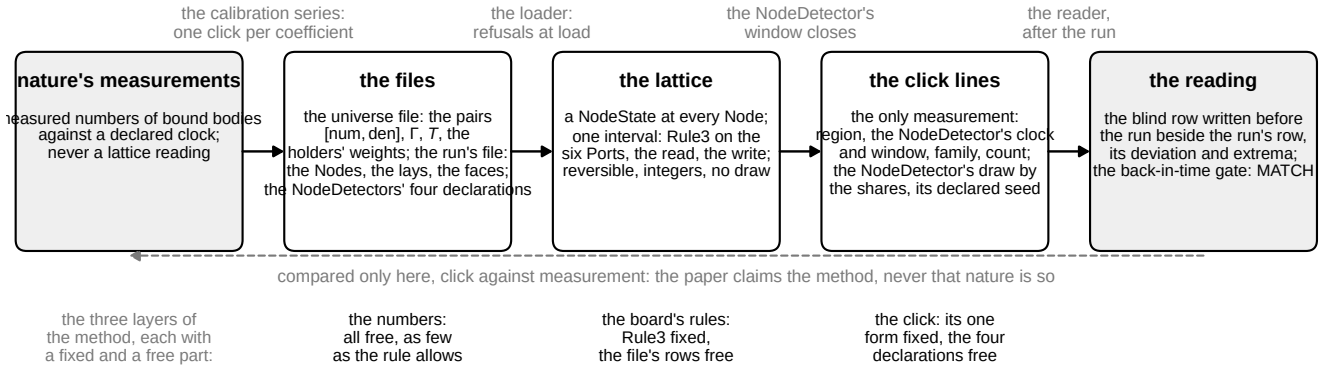
**The rule in one line.** Everything in the paper follows from one line of integer arithmetic, Rule3, together with one write, three readings, the one assumption of the forms (Section 3.4), the credit and the declarations listed in Section 10.3 and named where they stand. The line holds at every Node, in every interval and for every family of physical values [1, the line; the booking; the primitives]:

|                |                                                                                                                                                                                                                                                                                                                                                |     |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Rule3:         | $w a_{\text{next}} + r' = \sum_{j \sim i} R_{ij} \text{arr}_j + S a_{\text{now}} - w a_{\text{before}} + r, \quad 0 \leq r' < w,$ $w = 6 \text{den} \Gamma^2, \quad R_{ij} = 2 \text{num } p_a(i, j)^2 = 2 \text{num } p_i^2 q_{ij}^2 / \Gamma^2,$ $S = 12 \text{den} \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - \sum_{j \sim i} R_{ij};$ | (1) |
| the one write: | $\text{a held part at a Node gains } \left( \sum_f w_f q_f(\text{Node}) + r \right) \text{div } E, \text{ the remainder } r \text{ kept there;}$                                                                                                                                                                                               |     |
| the readings:  | $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i, \quad W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$                                                                                  |     |

In the first line,  $a_{\text{now}}$ ,  $a_{\text{before}}$  and  $a_{\text{next}}$  are the family's level at the Node now, one interval before and one interval later: the present, the past and the future.  $\text{arr}_j$  is the arrival through the Port to the neighbour  $j$ , which is that neighbour's level; the six Ports are summed.  $r$  and  $r'$  are the remainder kept at the Node before and after the step, and  $\Gamma$  is the Node clock.  $[\text{num}, \text{den}]$  is the family's pair, and  $\cos \omega_0 = \text{num}/\text{den}$  is its rotation per interval at wave number zero.  $p_0$  is the Node's clock pace,  $p_i$  its own pace, and  $p_a(i, j)$  is the pace of the Link through the Port: the Node's pace times the Link's factor  $q_{ij}/\Gamma$ . The factor reads the same from both ends; the pace is the reading Node's own. All of these are the Node clock, lowered by the holders' levels that the family reads there (Section 3.4). At the vacuum's paces, the two Links of an axis share one  $R_a = 2 \text{num} \Gamma^2$ , and  $S = 0$ .  $\text{div}$  is floor division. The second line is the only way a holder's level is ever written. It sums, over the sourcing families  $f$ , each family's own quantity  $q_f$  at the Node, weighted by  $w_f$ ; adds the part's one remainder; and divides by the part's wall  $E$ . The third line is what is read from a record and never written: its form  $D_i$ , which sources the holders; its Wronskian  $W_i$ , which is its charge (for a plane family  $\text{re}$  and  $\text{im}$  are its two level pairs, Section 3.2); and its current  $F_{ij}$  into the Node  $i$  from its neighbour  $j$ . A NodeDetector reads that current through its boundary in quanta of the family's unit  $W_c = 3 \text{den} T$ ; one quantum's share of the inflow is  $W_c \sin \omega_0 = \text{isqrt}(9T^2(\text{den}^2 - \text{num}^2))$ ,  $\text{isqrt}$  the integer square root, (its energy  $T \sin \omega_0$ , S.49 (22)), and a massless family's quantum is read as  $W_c$  itself, where  $T$  is the quantum's norm, the normalisation of the quadratic form.

**The three layers of the method.** The method has three layers, and each has a fixed part and a free part (Fig. 2). The first layer is the lattice's rules (Section 3): which families exist and of what dimension, which holders, who reads what and how. Here Rule3 with its reads is fixed, and the rows of the file are free. The second layer is the numbers (Section 6):  $\Gamma$ , the pairs, the holders' weights, gravity's  $E_g$  and the sign holder's  $k$ ,  $T$ , the vacuum content and the ranges. All of these are free, and there are as few of them as the rule allows. The third layer is the click (Sections 5, 5.3 and 7.4): how the lattice becomes a click, what the NodeDetector declares, what is credited, and how a mass on the lattice is read as a mass in clicks. Fixed here, and common to every file, is the one form of the click: the meeting, the bilinear credit, the draw by the shares, the NodeDetector's one write at the Node of the exchange, and the postulate that only a click is a measurement. Free, and belonging to the file, are the NodeDetector's declarations, its Nodes, the world's one draw (its seed, generator and window) shared by every NodeDetector with Nodes alone in turn, a body's NodeDetector's own seed (and its record's parts where it has one); its unit is the record's own, derived.

**The formulas derived.** The following formulas come from Eq. (1), in the order of the paper. Each has its mark in Table 1 and its algebra in the supplement. Light's band, its dispersion and its index in a well (Sections 3.3 and 3.4). The lattice Klein-Gordon equation and its Schrödinger limit (Section 3.3). The conserved form and the share identity (Sections 4.3 and 4.4). The Wronskian as the charge, and the rests of the holders (Sections 5.5 and 6.4). The functional of a bound body and its stability (Section 7.2). The adiabatic invariant and the drift of the count (Section 7.4). The rate of a clock, the bending of light, Shapiro's delay, the perihelion (Einstein's forms, recovered) and the capture radius of the shadow (the paper's own, the conclusion's table), the moving clock (Lorentz's), de Broglie's and Planck's relations and  $E = mc^2$  in the band's form, the index of the content for every family, and the fall as refraction (Section 8). Bell's  $E$  and  $S$ , Tsirelson's bound and the Greenberger-Horne-Zeilinger (GHZ) correlation's  $M$  under the declared credit (Section 9.4). The fine-structure constant as a coefficient of the file, the hierarchy of the two forces between families, the units and the highest photon energy, and Newton's  $G$  as a reading (Section 10.1). The formulas of Tables 1 and 2, several with their companions, cover five areas,



**Fig. 2** The method, from nature's measurements to the paper's reading (Sections 1, 3.7, 5.3 and 10.2; drawn from the definitions, no run). Nature's measured values on bound bodies would be read into the integers of a file, one click per coefficient, the calibration series of Section 10.2; the archived file's integers are the rule's own. The files declare Universe24. The lattice is stepped by Rule3, one interval of acts on the six Ports, reversible, in whole numbers and with no draw. The NodeDetectors write the click lines, the only measurement. The reader prints the row of the run beside the blind row, which was written before the run, with the MATCH of the back-in-time gate. Only there, click against measurement, does the computed Universe24 meet the formulas of nature it is set beside. The paper claims what Universe24 computes of nature, each quantity with its number against measurement, never that nature is so

and every one is an algebraic consequence of the line and of the inputs named beside it, the file's rows of Table 3 among them.

**What stands, and on what.** Five things, each with its mark. (i) The core. Rule3 and the boxed lines after it, the paces (their composition, the one assumption of Section 3.4, excepted), the conserved form and the Wronskian with the paces' weights  $1/p_i^2$  at static paces, the share, Kepler's potential and  $G$  as a reading are theorems of the line, or are derived from it with no number of nature in them, and the step is reversible to the bit (Section 4). (ii) From the composed paces alone, under the one named assumption of Section 3.4, come the exponential metric, the bending  $4U_b$ , Shapiro's delay, the redshift and the push  $E(1 + \beta^2)\nabla U$ , with  $\gamma = 1$  and  $\beta = 1$  exactly in the clock's potential, and  $\beta_K = 1$  as the gap closes, a weak-field sector under the static read and not the full first post-Newtonian metric (no  $g_{0i}$ , no frame dragging, Section 10.3). The assumption is fixed by one of those clicks, the bending's, and the others are its predictions (Section 8). (iii) The click is the one measurement, and the non-locality is accounted for in one place. Bell's 478/169 (the credit's rational value of  $2\sqrt{2}$ ) and the GHZ's  $-4$  follow from the declared credit; the law's line at the file's own lay gives the two slits' 284.7, and the two slits' 285 at both seeds stands against it within the integer step's walk, 12 over the Huygens blind 273 written before the run; the one non-local element is the credit's one draw through the root. (iv) Every claim carries its mark and its fence, and where the model misses nature's measurement the miss is named with its number: its  $\alpha_1 = -8$ , its hydrogen  $1s$  at  $-0.043$  of Rydberg's unit under the self-read (S.46), and its telegraph of the shelved ion (Section 10.3). Nothing is claimed beyond its mark. (v) Everything can be recomputed. The files, the engine, the scripts and the documents are public (Section 3.7).

**What is put in, and the three kinds of claim.** The paper's claims are of three kinds, and the text keeps them apart. *Found*: what is proved or derived from the rule with no input chosen for it. These are the formulas listed above and the click formulas of Table 2 marked derived. The findings against nature are comparisons, listed in Section 10.3. *Assumed, chosen or declared*: the assumption and the credit below, the rows of the universe, the NodeDetector's declarations (its Nodes; the world's one draw, its seed, generator and window, shared by every NodeDetector with Nodes alone in turn; a body's NodeDetector's own seed), the electron as the free quantum of matter, and the denominator of the matter pair as the family's own integer (Section 10.1). *Held to be computable and not computed*: the bodies at the closing gap, the moving fixed point of a body, and the absorbed fraction with the body's write. The explicitly named hypotheses are stated where they stand; among them are light's gravity, derived under the static kernel with the moving packet's field open (Section 10.2), the carrier that would mend the lattice's frame, and the nuclear holders (Section 10.3). One assumption is named with the forms: a Link is clocked twice by the clock of the Node that reads it,  $hN = 1$ , fixed by one click, the bending's, and tested by the perihelion (Section 3.4). The forms against Kepler's potential reach Einstein's only as the gap of the matter family closes (Section 8). Three choices were made with a known result in view, and we state them as inputs, not predictions: the one assumption  $hN = 1$  (one click, the bending's; Section 3.4), line (1) of S.60 (the atom's levels: a record reads every sign field but its own), and the credit: a click's share is the square of the meeting. This credit was chosen to reach nature's correlations, so Bell's  $2\sqrt{2}$  follows from it and does not test it. The numbers of the file are nature's and not the rule's: the vacuum content, a declaration; the rate of the vacuum write,  $\Lambda$ , the one number the hypothesis of Section 6.5 would declare (the hypothesis is not claimed); the binding range, a file's choice; gravity's weight  $E_g$  and the write weight  $k$  of the sign holder, read from  $G$  and from  $\alpha$ ; and the gaps, read from the masses (Section 10.2 and its table). Table 1 lists every result of the paper with its kind, its fence and its inputs. What is not claimed: that nature is a lattice; that any number of the computed pair  $[2, 3]$  is nature's; that a comparison with nature is more than a consistency statement at the stated order; or that a formula known in the continuum is new where the paper gives its lattice form. What is claimed is that those known formulas are reached from the one line in lattice form, each with its mark and its section (Section 8, Table 1), and that the paper's own additions (the conclusion's table) are unconfirmed. Where a status line or the supplement's derivation table (S.61) says two derivations, the result was obtained twice by independent routes; where it says one, once. Where the text says exact, it means exact in the rational arithmetic before the rounding, with the integer remainders bounded as stated.

**How a Universe24 file is built and tested.** A Universe24 file declares the following, and nothing else: a family's pair and dimension; a holder's sources, divisor and rest; the bodies and the packets; a NodeDetector's Nodes, the world's one draw (its seed, generator and window) shared by every NodeDetector with Nodes alone in turn, a body's NodeDetector's own seed (and its record's parts where it has one), the floor being the credit's own  $\max(s_R, 0)$  and the whole the record's own  $W_{\text{rec}}$ , derived; and the faces. The engine holds no number, no formula, no family name and no flag, and every primitive is one folder found by its name [2, how to run a world; how to add a family]. One change, in numbers the paper keeps, shows what this means: replacing the matter pair [2, 3] by [999, 1000] moves  $c_m/c$  from 0.867 to about 0.9997, and the computed pair's  $\gamma_K$  and  $\beta_K$  from 1.5 and 1.25 to about 1.001 and 1.0005, approaching 1 as the gap closes (Section 8); a binding holder's range  $R$  turns a declared body from unbound to bound (Section 6.4). The procedure, with the two slits as the example, is this. The blind row is written from the law's lines before the run. The run is made. The back-in-time gate reports MATCH or MISS on its reversibility. The reader prints the row of the run beside the blind row, with its deviation and its extrema (Sections 3.7 and 9). One file is the experiment, the two slits; Bell's and the GHZ's values are derived from the declared credit and not run for this paper. What ran on the implementation for this paper is that one file, stepped by Rule3 alone at every Node from its lay, with the wall's face and the screen's NodeDetectors the file declares and nothing else (the experiment; Section 9 shows it mid-run). Every other comparison the paper sets against nature is derived from the law's lines, Rule3's own, and not run, each with its mark and fence where it stands. These are Einstein's weak-field forms, the bending, Shapiro's delay, the redshift as the clock at two heights and the perihelion at the small gap (S.15, S.19, S.20); the shadow (S.30); the push on a moving record (S.58); the two slits' own blind row, written before the run (Section 9.2); the moving clock (S.21) and de Broglie's relation (S.22); the magnetism of moving charges (S.33); the Eötvös ratio (S.47); the one-quantum image of the lens (Eq. (17)); the deceleration of the clusters (S.29); the atom (S.46); Rabi's absorption line (S.43); the gravity of light (Section 10.2); the anticoincidence (S.57); the Zeno effect (S.59); the preferred frame's  $\alpha_1$  and  $\alpha_2$  against Gravity Probe B and the Sun's spin (S.35), derived and failing nature's bounds; and, derived under a declaration, the telegraph of the shelved ion and the emission of an excited body (the rate declared; Section 10.3) and the nuclear binding (S.45, the weight declared). Rule3 is fixed and no rule is added. A new line of physics enters as a hypothesis under its own name, after the three tests, generic, vector and local, which is how every family and holder of this paper entered (Section 4.2).

**How to read the paper.** Every claim carries one of eight marks. THEOREM: proved for every lattice. DERIVED: obtained from theorems by argument. COMPUTED: a number obtained from the rule's lines by derivation, or by a named run of the implementation. ASSUMPTION: a line held by decision and named where it is used. HYPOTHESIS: a line outside the model, under its own name. EXPERIMENT: a number the model does not fix, with the experiment named that would fix it; the paper performs none. FITTED: an input chosen with a known result in view. INSPIRATION: a direction of Section 10.4, named with no result claimed. The status columns of the tables use the same marks, and beside them calibration (a number taken from nature), declaration (a file's own number) and definition (a word fixed by the law's text). Every formula carries its fence beside its mark. A derived mark names in ... the module of `tools/derivations/` of the archive [43] that computes its numbers from Rule3's line alone, importing nothing of the engine and reading no run's file, or the paper's own check script; a computed mark names the run's reader; the marks that name no script are the qualitative statements. Some letters serve twice and are named at each first use:  $N$  the books' count and the clock's exponent  $N = e^{-U}$ ;  $E$  the energy, the conserved form of Eq. (6), the part's wall in Eq. (1) and the level weights  $E_s, E_g$ ;  $p_0, p_i, p_a$  the paces and  $q_{ij}$  the Link's factor, with  $(p, q)$  Bell's setting pair;  $m$  the rest term of the lattice Klein-Gordon equation,  $m^*$  the inertia,  $m = GM/c^2$  in Links in the shadow's radius and Rule3's margin  $m(c)$ ;  $k$  the wave number and the sign holder's write weight;  $R$  the read coefficient of Rule3, a body's radius in Links and the binding range;  $L$  and  $\ell$  both a holder's level in levels,  $L$  also a length in Links where so named;  $G$  Newton's constant,  $G_K$  as the orbits read it and  $G_0(r)$  the lattice's Green's function (S.39), and the Link's unit  $G$ ,  $G_{ab}$  the stress reading;  $\alpha$  the fine-structure constant, the redshift's  $\alpha_U$ , the bending angle where so named and the frame's  $\alpha_1, \alpha_2$ ;  $\beta$  the post-Newtonian parameter and a speed over light's;  $c$  light's speed, and the content in level units where so written ( $c, c_{\text{vac}}, c_{\text{law}}$ );  $h$  the ruler of Section 3.4 and Planck's constant where nature's is meant;  $P$  the momentum density, a probability, the guard's edge on a pace and the momentum  $P = cN^2k$  at the composed paces;  $Q$  the two-level form and  $q_{ij}$  the Link's factor;  $T$  the invariant's unit and  $T_{aa}$  the tension;  $W$  the Wronskian,  $W_c$  and  $W_{\text{rec}}$  the credit's units;  $S$  Rule3's self coefficient in Eq. (1) Bell's  $S$  in Section 9 and  $S_t$  the step map;  $U$  the potential, the clock's fraction  $c/\Gamma$ ;  $\Omega$  a transition's rotation and the moving centre's rotation  $\Omega(k)$ ;  $\kappa$  a holder's inverse reach;  $R$  a NodeDetector's region from Section 5.2 on;  $m$  also matter's numerator in  $[m, \text{den}]$ ;  $e$  Euler's number, the share  $e_i$ , a port's coefficient  $e_k$ , the unit charge (one quantum's Wronskian  $T/2$ , Section 10.1), an orbit's eccentricity, a record's ellipticity and the part label  $e$ ;  $g$  the rest term's coefficient, the metric's components, the ground part  $g$  and Landé's  $g_L$ ;  $n$  a count of intervals or quanta where so named;  $r$  Rule3's remainder and a radius;  $E(a, b)$  Bell's correlation;  $\Delta$  the six-neighbour second difference and, as  $1/m^*$ , the band's curvature;  $F$  the current, the functional, a force, each where so named;  $D$  the form of a record and a length where so named;  $h_a(i)$  the tension's reading of Eq. (8);  $c_m$  the band's kinetic scale (Section 8 (e));  $n$  also light's index  $n = e^{2U}$ ;  $Q_R$  a region's inflow;  $\epsilon$  the step's remainder term  $(r - r')/w$  and a resonance's modulation depth where so named;  $\sigma$  a holder's source per interval, a sign and a spread where so named;  $\rho$  Bell's  $2r/(1 + r^2)$  and the dust density;  $K$  Friedmann's constant, the start's kick and a pressure's weight where so named;  $s$  a share, the refinement scale, the family index and a scaling power where so named;  $d$  a radius, the slits' distance and a dimension where so named;  $\lambda$  the wavelength and a Lagrange multiplier;  $\mu$  a growth rate and  $GM/c^2$  where so written;  $x$  a content fraction, a level or a coordinate where so named;  $C$  a body's count in quanta;  $M$  the mass in  $GM$  and Mermin's  $M$ ;  $A$  an amplitude, a mass number and Bell's side;  $z$  the redshift and the complex level where so named. LATTICE: a reading of the lattice's own numbers that no NodeDetector reports, a diagnostic. CLICKS: a formula for what a NodeDetector reports, the share  $Q_R/W_c$  on the forward orbit (Section 5.2) or a ratio of two such shares; this is the only kind that can be set beside nature. The algebraic steps of every derivation that the text states without its proof are in the supplementary material (Online Resource 1), Derivations S.1 to S.65. They are numbered independently of the paper's order, each names its section, and they are cited as (S. $n$ ) where used. Every sentence of the paper that says what a thing is, light, mass, an atom, gravity, the direction of time, is a sentence about the model and its files, and never about nature. Nothing in the paper is a proof about nature: the theorems are theorems about the rule, and every comparison with nature is a supporting finding and not a proof. The paper cites three documents of its archive by the names of their sections: the law [1], the engine [2] and the decisions [3]. A large language model was used as a tool throughout the work, as the Statements and Declarations state.

**Table 1** The results of the paper, each with its kind, its fence and what it rests on. A calibration is a number taken from nature; EXPERIMENT a number the model does not fix.

| Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Kind                                                                                                              | Fence                                                                  | Rests on                                                                  | Section    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------|------------|
| Rule3's step is reversible to the bit; the interval under the condition of Section 4.1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | theorem                                                                                                           | lattice                                                                | Rule3                                                                     | 4.1        |
| The back-in-time gate says MATCH on the two slits' file over 40 intervals at both seeds; the two slits' NodeDetector $N = 285$ units of $W_c$ at both seeds (674 photons of energy $T \sin \omega$ at $\omega = 0.4366$ , the light's own frequency at the declared vacuum content's paces, that is $285/\sin \omega$ , a lattice reading) against the law's line's 284.7 at the same lay, within the integer step's walk, and 12 over the Huygens blind 273, the exact lay's own number, the maxima at the blind's regions 2, 6 and 9, both sums deterministic (Section 9.2) | computed, a run                                                                                                   | clicks for the NodeDetector's counts; lattice for the gate and the 674 | the implementation                                                        | 4.1        |
| The rule's acts form no product of levels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | theorem                                                                                                           | lattice                                                                | Rule3                                                                     | 4.2        |
| A conserved form at static paces with the weights $1/p_i^2$ , its current, and the momentum flux at uniform coefficients as the wave's stress                                                                                                                                                                                                                                                                                                                                                                                                                                 | theorem at static paces, weighted                                                                                 | lattice                                                                | Rule3                                                                     | 4.3        |
| The count is the record's share; the rational total share is never negative on any lattice whose Links read the same from both ends and whose every beyond reads 0, at $ \text{num}  \leq \text{den}$ and the vacuum's coefficients (negative shares at a lay's edge named), the floors' total within $N$ of it                                                                                                                                                                                                                                                               | theorem under the two conditions                                                                                  | lattice                                                                | Rule3                                                                     | 4.4        |
| Three exact massive rotations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | theorem                                                                                                           | lattice                                                                | Rule3                                                                     | 6.1        |
| Light's speed $1/\sqrt{3}$ Link per interval, its dispersion, and no dispersion along the cube's diagonal                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | derived                                                                                                           | lattice                                                                | Rule3 for the pair $[1, 1]$                                               | 3.3        |
| A click is the meeting of two quanta, decided in the credit, written at one Node; no signal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | assumption, derived                                                                                               | clicks                                                                 | the meeting, the NodeDetector's draw and write                            | 5          |
| A holder's reach from its gap, Yukawa's form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | derived                                                                                                           | lattice                                                                | Rule3 for a gapped pair                                                   | 6.4        |
| The stability of a body under the dilation, to first order in the well                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | derived (the continuous functional, the dilation continuous; the integer step's version open)                     | lattice                                                                | the body's functional                                                     | 7.2        |
| A body's clicks: no light from a steady rotation, light at a breathing body's beat, the resonance at $2\omega_b$ under the pace read (none under the sign holder's rotation, S.43), the share's drift                                                                                                                                                                                                                                                                                                                                                                         | derived                                                                                                           | clicks; the drift lattice                                              | Rule3 at the body's Nodes                                                 | 7.4        |
| A cluster: the three levels at a Node as the nesting, three kinds of binding, the click the only thing between clusters, the hierarchy recursive; dark matter in kind                                                                                                                                                                                                                                                                                                                                                                                                         | definition, derived                                                                                               | lattice, clicks                                                        | every body is bound, records may be free; Rule3                           | 7.5        |
| The gravitational redshift, $z = f(\omega_0) U$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | derived                                                                                                           | clicks                                                                 | the clock's read                                                          | 8          |
| The bending, which fixed the one assumption; the perihelion, the delay, the redshift's factor and the shadow's capture radius $2e m$ ( $m = GM/c^2$ in Links), its predictions, the last conditional on the interior's fate                                                                                                                                                                                                                                                                                                                                                   | derived under the one assumption, fixed by the first                                                              | clicks                                                                 | a Link clocked twice by its Node's clock, $hN = 1$                        | 8          |
| The same forms against Kepler's potential: $\gamma_K$ , $\beta_K$ and $\alpha_K$ from the pair alone for $0 < \text{num} \leq \text{den}$ , closing with the gap; the $[2, 3]$ values an illustration                                                                                                                                                                                                                                                                                                                                                                         | derived (the formulas)                                                                                            | clicks; the computed pair's values lattice                             | the fall's formula                                                        | 8          |
| The moving clock: its fourth order and its kinetic scale $c_m$ , Lorentz's as the gap closes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | derived                                                                                                           | clicks                                                                 | Rule3's band                                                              | 8          |
| The force between moving charges, Coulomb's over $\gamma$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | derived                                                                                                           | clicks                                                                 | the rule's continuity identity, the sign's covector                       | 8          |
| De Broglie, the fall, Newton's pull; the content's index for every family, the lens                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | derived                                                                                                           | lattice, clicks                                                        | the band and the holder's rest                                            | 8          |
| Newton's pull, the clock's $G$ and Kepler's $G$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | derived                                                                                                           | lattice, clicks                                                        | Rule3 for $[1, 1]$ and $E_g$                                              | 10.1       |
| $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ between two quanta, $k$ the sign holder's declared weight; charge universality; the two forces' ratio between families $(m_p/m_e)^2$ ; the units; the share's ceiling; the gap below $10^{-27}$ at MAGIC's bound                                                                                                                                                                                                                                                                                                      | derived; $k$ declared; charge universality and the forces' ratio in the small-gap limit, by construction, no test | clicks                                                                 | the rotation, Poisson's rest, de Broglie; the quantum of a family; LHAASO | 10.1       |
| The two-slit fringe row and the lattice's arrival interval                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | computed                                                                                                          | clicks                                                                 | Rule3 on the declared file                                                | 9.2        |
| Bell's $E = \cos 2(a - b)$ , $S = 2\sqrt{2}$ , $S(\rho)$ and Tsirelson's $2\sqrt{1 + \rho^2}$ ; every local credit at most 2 (derived under the uniform phase)                                                                                                                                                                                                                                                                                                                                                                                                                | fitted; derived                                                                                                   | clicks                                                                 | the credit rule, the meeting's square                                     | 9.4        |
| The vacuum content 60 at $\Gamma = 6,000$ , bounded below by the lattice's own floor and the measured swing; no click bounds it above; gravity's events at light's speed in every level                                                                                                                                                                                                                                                                                                                                                                                       | declaration (the content, above its floors); derived from the law's line (the events' speed, S.12)                | clicks                                                                 | declared above two floors; every family reads the content                 | 6.5        |
| The binding range, 20 Links                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | declaration calibration, experiment                                                                               | lattice                                                                | declared, a file's choice nature's measurements                           | 6.4        |
| Gravity's weight $E_g$ and the sign holder's $k$ , the matter pair's gap, the Link's length                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | derived; the comparisons supporting findings                                                                      | clicks                                                                 | the principle; Rule3                                                      | 6.6, 10.2  |
| The two clocks shift alike; the findings against nature of Section 10.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                   |                                                                        |                                                                           | 10.2, 10.3 |

**Reversible cellular automata and digital physics.** The idea that the physical world is a discrete computation goes back to Zuse [4]. Fredkin, Toffoli and Margolus developed it as reversible cellular automata [5][6][7]. Wolfram developed it as a programme of cellular automata [8], and 't Hooft as the cellular automaton interpretation, a deterministic foundation beneath quantum theory [9][10]. Quantum cellular automata put a unitary step on a lattice. They range from the lattice Weyl, Dirac and Maxwell equations [11] and quantum lattice gases [12] to the structure theorems of the reviews [13][14] and the derivation of Weyl's and Dirac's equations from an automaton's symmetries [15]. The step of the present rule is an integer recurrence on two levels, not a unitary on amplitudes. Rule3 is a second-order cellular automaton in the sense of

Toffoli and Margolus,  $a_{\text{next}} = f(a_{\text{now}}) - a_{\text{before}}$  [7]. Its alphabet is the integers, and it has one Euclidean division whose remainder stays at the Node. This is what the title calls an integer cellular automaton. Feynman noted that a local classical computer reproduces no correlations beyond Bell's bound [16]. The model agrees, and it places the excess in the declared credit of the click, not in the lattice (Section 9.4).

**Time-symmetric accounts of measurement.** Wheeler and Feynman described radiation as an interaction between the source and the absorber, with advanced and retarded fields together [17]. Cramer's transactional interpretation has an offer wave forward from the emitter and a confirmation wave backward from the absorber, and the transaction forms between them [18]. The two-state vector formalism evolves one state forward from the preparation and one backward from the outcome [19][20]. Price argued that a time-symmetric account is what Bell's theorem leaves open [21]. Wharton and Argaman showed that every locally mediated reformulation of quantum theory must let some mediating parameters depend on measurement settings in their future [22]. The model belongs to this class: the backward reading from a realised click depends on the NodeDetector's setting alone. It differs in three respects. Both directions are one integer line, exact to the bit; their product is a reading and not a wave; and the choice among the meetings is the one non-local element. The record is an ontic state in the terms of [23].

**Bell's theorem and local models.** Bell's inequality bounds every local account of a pair's correlations [24], in the form that experiments test [25]. A local model, a joint distribution and the inequalities are one condition [26]. The model's local credits give  $S = \sqrt{2}$  and  $S = 2$ , as they must (Section 9.4). Its credit by the contrast reaches the quantum correlation only by discarding pairs. This is Pearle's road [27], and it runs below the efficiency bounds of Eberhard and of Garg and Mermin [28][29]. The loophole-free tests of 2015 close that road for nature [30][31][33].

**Discrete gravity and its tests.** Regge calculus [35] and causal sets [36] discretise the geometry of space-time itself. Here the lattice is never curved, and the weak-field forms are read from the coefficients (Section 8). In the parametrised post-Newtonian framework [38], the one assumption gives  $\gamma = 1$  and  $\beta = 1$  exactly in the model's own potential by its metric; the orbit reads them as the post-Newtonian ones at the quadratic band. Against the potential that the tests read, the matter band's factors close as its gap closes. A lattice breaks Lorentz invariance at order one unless its terms are tuned [39]; the bounds are reviewed in [40]. The paper's answer is no loops and the Link's bound of S.24, with the closing gap for matter (Section 8).

### 3 The model

This section states the model: the lattice, the families, the rule, the paces by which one family acts on another, the one write, the readings and the order of acts in an interval [1, the lattice and the families; Rule3; the primitives].

#### 3.1 The lattice

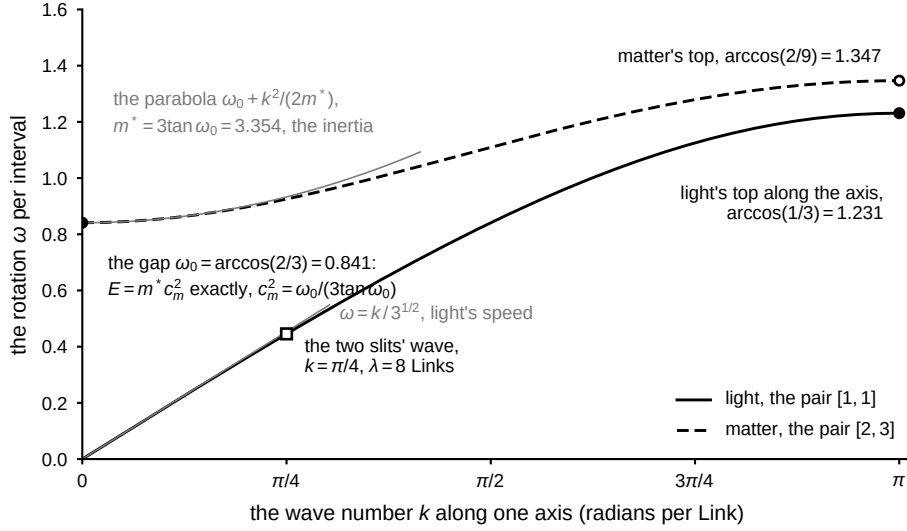
The lattice is a box of  $N_x \times N_y \times N_z$  Nodes. Two Nodes one step apart along one axis are joined by a Link. Each Node has six Links, one through each of its six Ports,  $\pm x$ ,  $\pm y$  and  $\pm z$ . Each axis is declared periodic, open or folded. Beyond an open face there is no Node, and a level there is read as 0. A folded axis has one layer, and its two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps; this is how a wall with slits is made. A face may also be declared receding (ASSUMPTION). The box then grows by a layer of zeros whenever a nonzero level reaches the Node next to the face. So no event ever meets the face, and a run on such a box is a run on the unbounded lattice. The box grows and nothing stretches, so a receding face gives no cosmological redshift [1, the objects; the unbounded board; the declarations by name; the receding face].

**The group and its action.** The symmetry group of the cube has 48 elements: the 24 rotations and, with the reflections, the signed permutations of the three axes. It is  $O_h$ , the group of the octahedron, which is also the group of the cube (Fig. 1(b)). With the translations it is the space group of the simple cubic lattice. It acts by permuting the Nodes and their Ports, and it carries every level pair with its Node. The time reversal exchanges a record's present and its past and carries the step to its inverse (Theorem 1). Rule3 commutes with the action of the space group, since its line reads the six Ports symmetrically and names no axis. Under the group, the form  $D$  is a scalar, even under the time reversal. The current  $F$  is a vector, odd under it. The Wronskian  $W$  is a scalar under the cube's 48, odd under the time reversal and under the plane's reflection. The click is a scalar: it names a region and an interval, never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group. So Lorentz's group and the continuum's rotations are no symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (Section 8) (THEOREM for the covariance and the kinds of the readings; LATTICE; S.4) [1, the direction; the four acts][3, one algebraic line per rule].

The NodeState holds the model's own numbers and nothing else: for every family and every part, two levels and one remainder; and for every part of a holder, the remainder of its one write. A count, a current, a charge or a tension is a reading, computed from these numbers when it is needed. No register, table, formula or random draw is kept at a Node [1, the objects][3, nothing is kept at a Node beyond the law's own numbers].

#### 3.2 Families and records

A family is declared by two keys: its pair [num, den] and its shape. A family of quanta, such as matter or light, declares its dimension, 1 for one real line or 2 for a plane of two lines, and it holds nothing. A family of dimension 3, three real lines with a rotation among them on each Link, is declared by no file. It enters only as an explicitly named hypothesis of Section 10.3 (HYPOTHESIS). A holder declares its sources, the form, the form with the three tensions, or the Wronskian, together with its divisor  $E_s$  (the wall  $E$  of Eq. (1)) and, for the massless holder, its rest (its declared level far from every source, Section 6.4). The sources fix its parts, one real line per source. A holder sourced by the form or the tensions holds the content, which slows the clocks of the families that read it. A holder sourced by the Wronskian holds the sign, the charge's holder. A family's record is its levels over the lattice: at every Node a level now and a level one interval before, with a remainder. A family of dimension 2 lays two level pairs (re, im), two solutions of the same line, and only a plane carries a charge (Section 5.5). A body is its count and its record over its Nodes, standing bound in a holder's well, and a packet is a travelling record in no body, laid by the file or written by a click [1, the objects].



**Fig. 3** The bands of light and matter along one axis, from the law's line (DERIVED, LATTICE; drawn from the formula, no run; `bands.py`):  $\cos \omega = \frac{\text{num}}{3 \text{ den}} (2 + \cos k)$  at the wave number  $k$  along one axis, for light's pair [1, 1] (solid) and matter's [2, 3] (dashed). At  $k = 0$  the gap is  $\cos \omega_0 = \text{num}/\text{den}$ ,  $\omega_0 = 0.841$  for matter, with the parabola  $\omega_0 + k^2/(2m^*)$  of the inertia  $m^* = 3 \tan \omega_0 = 3.354$  (grey). At  $k = \pi$  the top of the band is  $\arccos(\text{num}/(3 \text{ den}))$ , 1.231 for light and 1.347 for matter. Light's slope at the origin is its speed,  $1/\sqrt{3}$  Link per interval (grey). The wave of the two slits,  $k = \pi/4$  (the square), sits on light's band at  $\omega = 0.4456$  at the content 0, with the period 14.1 intervals, and at  $\omega = 0.4366$  at the declared vacuum content's paces, where the packet is laid (Section 9.1).  $E = m^* c_m^2 = \omega_0$ , the rotation per interval, the one band read three ways, its curvature ( $m^*$ ), its Lorentz form and the moving clock's kinetic scale ( $c_m^2 = \omega_0/(3 \tan \omega_0)$ , Section 8 (e)); the energy a click credits is  $T \sin \omega_0$ , 0.8862 of  $T \omega_0$  at [2, 3], and  $\sin \omega_0 \rightarrow \omega_0$  is the small-gap limit where the two agree. Along the cube's diagonal light has  $\cos \omega = \cos k$ , no dispersion; it is not drawn

### 3.3 Rule3

At every Node, in every interval, every family steps by the first line of Eq. (1). One whole number is formed: the six arrivals through the Ports, each weighted by the coefficient  $R_{ij}$  of its Link, plus the Node's own level times  $S$ , minus  $w$  times the level before, plus the remainder the Node kept. That sum is divided once by the wall  $w$ . The quotient is the next level, and the remainder,  $0 \leq r' < w$ , stays at the Node and never travels. Nothing is rounded away, and the step back is the same division. This is why every step is exact and reversible to the bit (Theorem 1); the NodeDetectors' writes stand outside the reversal (Section 4.1). The name comes from the three axes: at the vacuum's paces one coefficient per direction serves its two Ports, and the wall's factor 6 is the three directions twice; in a content the six Ports carry six coefficients, Eq. (1). For light the next level is the sum of the six arrivals over 3, minus the level before [1, the line].

$$a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a}), \quad (2)$$

the line at the vacuum's paces, whose rotation at the wave vector  $\mathbf{k}$  is  $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$ . At wave number zero  $\cos \omega_0 = \text{num}/\text{den}$ . For the massless pair [1, 1] along one axis  $\cos \omega = (2 + \cos k)/3$  (Fig. 3), so  $\omega = |k|/\sqrt{3}$  as  $k \rightarrow 0$ . Along the cube's diagonal  $\cos \omega = \cos k$  exactly, with no dispersion (DERIVED, LATTICE; S.1; `bands.py`). Light's speed at long wavelength is therefore  $1/\sqrt{3}$  Link per interval, below the causal bound of one Link per interval. Its group speed falls at shorter wavelengths as  $1 - 0.083 k^2$  [1, the lattice constants]. In nature's names the vacuum's line is the discrete Klein-Gordon equation, exact but for the remainder term. With  $S_6 = 6 + \Delta_{\text{lat}}$  it reads  $a_{\text{next}} - 2a_{\text{now}} + a_{\text{before}} = c_K^2 \Delta_{\text{lat}} a_{\text{now}} - m^2 a_{\text{now}} + (r - r')/w$ , with  $c_K^2 = \text{num}/(3 \text{ den})$  and  $m^2 = 2(1 - \text{num}/\text{den})$ ; the rest rotation is the mass. ( $c_K$  is the coefficient of the equation. The band's kinetic scale  $c_m^2$  of Section 8, (e), is  $\omega_0/(3 \tan \omega_0)$ , and its largest group speed is  $c \cos \omega_0$ . The three are one as the gap closes.) Its slow limit at small wave number is Schrödinger's equation with the inertia  $m^* = 3 \tan \omega_0$  and, in a content  $U$ , the potential  $-\omega_0 f(\omega_0) U$  of Section 7.4, the well lowering the rotation: the slow limit with its three conditions named, the remainder within the budget clause, the band's  $k^4$  term dropped at the order  $k^2/12$ , the envelope's second time difference dropped at the order  $(\omega - \omega_0)/\omega_0$  (DERIVED, LATTICE; S.63, `schroedinger_check.py`). And  $m^* c_m^2 = \omega_0$  exactly for every pair. This is the model's  $E = mc^2$  at the band's kinetic scale, read in three click numbers; nature's is its small-gap limit. The departure is  $\omega_0/\tan \omega_0$ , which is  $1 - \omega_0^2/3$  to the order  $\omega_0^2$  (DERIVED, CLICKS; S.18, S.21; `bands.py`). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, whose dispersion is the known one. What the law adds is one rational [num, den] that fixes mass and speed together, an equation that is exact, departures that are predictions with bounds (Section 10.2), the integers with the remainder, and the paces (DERIVED, LATTICE; `bands.py`).

### 3.4 The paces: how a holder enters a record's paces

A holder enters a family's step by changing the family's paces at a Node, the coefficients of Rule3 there:

$$\boxed{\begin{aligned} p_0 &= \Gamma (1 - 1/\Gamma)^c, & p_i &= \frac{p_0(c_i)^2}{\Gamma}, & p_a(i, j) &= \frac{p_i q_{ij}}{\Gamma}, & q_{ij} &= \Gamma - t_a(i, j), \\ c &= \sum_{\text{reads}} \text{weight} \times \text{the read family's time level}, \end{aligned}} \quad (3)$$

Here  $c_i$  is the content the family reads at the Node: the sum, over its reads, of the weight times the read holder's time level (the level of its time part, Section 3.2); the content and the clock's potential  $U = c/\Gamma$  are one quantity in two units, the level's and the potential's. (The symbol  $c$  names the content in this section and light's speed elsewhere.) The clock  $p_0$  composes: each unit of the level slows it by the same factor  $1 - 1/\Gamma$ , to the unit. The law's integer roundings, which the implementation holds:  $p_0 = \Gamma(\Gamma - 1)^c$  over  $\Gamma^c$  rounded half up in one division;  $p_i = p_0^2/\Gamma$  rounded once from the clock's integer; the Link's factor  $q_{ij}^2/\Gamma^2$  rounded once with the half wall;  $R_{ij}$ ,  $S$  and  $w$  formed from these with no further rounding; the step's one division Rule3's, its remainder kept at the Node. The Node's pace  $p_i$  is the clock twice,  $p_0^2/\Gamma$ , so that the ruler goes with the clock. The Link's tension  $t_a(i, j)$  is one number per Link, from the parts at its two ends (Section 4.3); it is read into the Link's factor  $q_{ij}$ , the same from both ends. And the Link's pace is the product,  $\Gamma - 2c_i - t_a$  to first order [1, every row reads the content; the clock is the Node's, the tension is the Link's]. A positive weight makes a hollow, an attraction; a negative weight makes a hill. The pace is the slowing of the clock: a record at a Node of pace  $p$  rotates and moves as it would at  $\Gamma$ , with its intervals  $p/\Gamma$  as long. Every family reads the holders of the content that its declaration names, at the weights it names. Gravity's holder and the binding holder's are read at the read weight 1 by every family (the weight of a read, not the divisor  $E_g$ , which the paper also calls gravity's weight). The holders are among the readers, and a holder of the content reads its own level among the rest (the law's line; the nuclear holders of Section 10.3 are read by the nucleon families alone). So the content in units of  $\Gamma$ , written  $x$  here and  $U$  in the metric below, is  $x = (L_b + L_g)/\Gamma$ , with  $L_b$  the binding holder's level and  $L_g$  gravity's. A family of dimension 2 also reads the holder of the sign plainly. Light, the record the sign holder's write bears (Section 5.5), reads the holders of the content as every family does, and not the holder of the sign. And a family sources what it reads, at the weight it reads with, so every pair of families that read each other is reciprocal [1, the paces; the reads are the held row's pair's].

**The one assumption of the forms: a Link is clocked twice by its reading Node's own clock.** Nothing in Rule3 fixes how a level enters the paces. Beside the line, the model holds that a level is a count of units and that every act is measured in the Node's own units; this is the one assumption itself,  $hN = 1$  in its units form (ASSUMPTION). Two parts of the paces follow from that. The clock composes,  $p_0 = \Gamma(1 - 1/\Gamma)^c$ : each unit slows the clock as it stands by the same fraction, since the write of one unit is an act measured in the Node's own units, the read included (DERIVED from it; LATTICE; paces.py). And every act of the interval, Rule3's step, the write of the holders and the turn of the sign holder, is Rule3's act measured in the cell's units. So a uniform level  $U = c/\Gamma$  changes only the units: time by the lapse  $N = p_0/\Gamma$  (the letter that counts a run's clicks in Section 9 is the same letter in another role) and length by  $h = p_0/p_a$  per Link. The write carries  $p_x p_y p_z / (p_0 \Gamma^2)$  on the form and  $p_x p_y p_z / (p_0^2 \Gamma)$  on the Wronskian, and the turn carries  $p_0/\Gamma$ , per proper volume and per proper interval (DERIVED, S.34; LATTICE; paces.py). One sentence is not in Rule3. It is the model's one assumption of the forms, and it is named as the credit is (Section 5): a Link is clocked twice by its reading Node's own clock,  $p_i = p_0^2/\Gamma$  (the Link's pace where no tension stands), that is  $hN = 1$ , the ruler going with the clock (ASSUMPTION). The lattice says why twice, since a Link has two ends. It does not say why the reading Node's clock stands at both ends rather than the product of the two ends, which gives  $\beta_K = -1$ . The sentence is one discrete choice (the reading Node's clock at both ends of the Link, against the product of the two ends, which gives  $\beta_K = -1$ ), fixed by one click, the bending's  $4U_b$  ( $\gamma = 1$ ), and this is where Einstein's  $\gamma = 1$  enters the model by hand (ASSUMPTION, the one assumption of the forms, the first of the introduction's three choices made with a known result in view, fixed by the bending's click; every form of Section 8 is conditional on it, and none is a consequence of Rule3 alone; the bending fixed it and is no test of it, while the perihelion's  $\beta = 1$ , Shapiro's delay, the redshift's factor  $f(\omega_0)$  and the shadow's  $2em$ ,  $e$  Euler's number, are its predictions). Under it the model's static metric for light is  $ds^2 = -e^{-2U} dt^2 + e^{2U} d\mathbf{x}^2$ , to  $O(U/\Gamma)$  and in the ray limit. This is the exponential metric [42], Einstein's to the first post-Newtonian order. The clock's rate is  $N$ : exactly in the lattice's variable  $2\sin(\omega/2)$ , and in  $\omega$  up to the finite-rotation correction  $\omega_0^2/12$ , which is 5.9 percent at the computed pair by that estimate and 6.3 percent exactly,  $3 \times 10^{-29}$  at the bound of the MAGIC telescopes on light's dispersion [72] and  $10^{-20}$  at the LHAASO observatory's highest photon [71] (Section 10.1). The metric is stated in the variable in which it is exact (DERIVED; S.31; LATTICE; paces.py). The shadow's capture radius  $2em$  is its second-order click, against Schwarzschild's  $3\sqrt{3}m$  (Section 8), in the lattice's frame, a conditional geometric result: the ray's capture follows from  $n(r) = e^{2m/r}$  alone, while the fate of the rays inside is open, the law having no horizon, a frozen region returning a wave as a fixed boundary and not absorbing it, so that a shadow needs absorption by a body's clicks, not computed; the exponential metric is a wormhole metric in the literature [91]; that frame is a second residue, not derived from Rule3 and read by the binary pulsars'  $\alpha_1$  (Section 10.3). The quadratic form  $(\Gamma - c)^2 + c^2$  is the second-order truncation of the composition; the law uses the composed form [1, the same clock and the same ruler for every act; the paces compose; what is open].

**The band at a pace.** From the coefficients, the rotation of a record at a Node with clock pace  $p_0$  and Link paces  $p_a$  is (DERIVED, LATTICE; S.1; rule3.py)

$$\boxed{\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a)}, \quad (4)$$

The mass term scales with the clock's square, and the band term with the square of the Link's pace. Two readings follow. First, at  $k = 0$  the rest rotation of every massive band obeys  $1 - \cos \omega_b = (p_0/\Gamma)^2 (1 - \cos \omega_0)$ , the clocks' shifts of Section 7.4.

Second, along a Link the speed of every wave falls with the Link's pace,  $\Gamma - 2c - t_a$  to first order, so light is delayed and bent by twice the clock's share. At a Node of content  $x = L/\Gamma$ , light's band is  $[2 - 4N^4, 2]$  in  $2\cos\omega$ , with  $N = (1 - 1/\Gamma)^L$ , and at long wavelength  $\omega = N^2|k|/\sqrt{3}$ . So light's speed is  $N^2 = e^{-2x}$  times its vacuum speed, to the unit: an index  $n = e^{2x}$ , which is  $1/(1 - 2x)$  to the first order, with  $n - 1 = 2x$  (DERIVED, LATTICE; S.2; paces.py). A record whose rotation lies outside the band at a Node is evanescent there and reflected whole by a thick barrier, a thin one passing the evanescent tail. Light meets a total mirror where  $p_a < \Gamma \sin(k/2)$  (DERIVED, LATTICE; paces.py) [1, the band at a pace; the rows against nature, (b)].

**The guard.** A pace is positive by the composition, and it must lie below an edge  $P$ , with  $P^2 = 2 \text{den} \Gamma^2 \text{div} (\text{den} + |\text{num}|)$ , which is  $\Gamma^2$  for light. Beyond  $P$  the extreme mode grows without bound, the mode at wave number  $\pi$  on every axis for a numerator at or above 0 and the uniform mode  $k = 0$  for a negative numerator; at or below  $P$  no mode grows exponentially at coefficients fixed in time (THEOREM: the extreme mode has  $2w\cos\omega = S - 6R$  at wave number  $\pi$  on every axis and  $S + 6R$  at  $k = 0$ , the lower of the two by the sign of num at any paces, and  $\cos\omega \geq -1$  exactly up to the edge, at coefficients uniform over the lattice and Node by Node under the composed paces; LATTICE). The guard is the Courant, Friedrichs and Lewy condition of the explicit scheme that Rule3 is at the vacuum's pace [41], and it is checked once at load. Under the composed paces no Link ever closes. The clock  $\Gamma(1 - 1/\Gamma)^c$  is positive at every content, and the Node's pace is its square over  $\Gamma$ . So a deep well freezes a clock and closes nothing, and the horizon of the quadratic form at  $\Gamma/2$  is a feature of the truncation and not of the law (DERIVED, LATTICE; S.3; paces.py). A Node whose Link pace rounds to 0 is a frozen Node. It reads nothing, and no event enters it. Its level is kept: matter's rotation is nearly stopped, and light's drifts linearly. Its active neighbours still read it at their own pace, so the cut is one-way. It has no share, by the law's declaration by name (the frozen Node has no share [1, the count is the record's share]): Eq. (9)'s weight  $1/p_i^2$  is undefined at the pace 0 (S.3), the weighted share's theorem is not extended through it, and the engine reads no share over a lattice holding a frozen Node (no number invented); so it is dark (ASSUMPTION; LATTICE; S.3, S.41). Two thresholds carry the word: the Link's pace at  $(\Gamma/2) \ln 2\Gamma$  levels, 28,178 at  $\Gamma = 6,000$  (28,206 with  $p_0$  rounded to the unit before squaring, S.41); and the clock's at  $\Gamma \ln 2\Gamma$ . The guard's bound,  $|2\cos\omega| \leq 2$  at the band's edges, admits the edge itself, where a linearly growing solution exists, for both signs of the numerator. At that edge the vacuum's checkerboard mode, driven by the remainders, grows as  $t^{3/2}$  under the model of independent remainders (the double root's response, the variance  $\sigma^2 t^3/3$ ; the uniform mode is not driven) [1, the guard]; no click reads it, and it reaches the load's amplitude bound (LATTICE). For a negative numerator the lowest  $2\cos\omega$  lies at  $k = 0$  (S.3). At a negative content light's zone-corner mode grows, so light refuses a content below 0, and the vacuum's content lies at or above 0, this world's declared 60 (Section 6.5) [1, the guard; the frozen Node has no share].

### 3.5 The write, the readings and the interval

**The one write.** Every write into a holder's level is one act, the second line of Eq. (1). At a Node, in each interval, a held part gains  $(\sum_f w_f q_f + r) \text{div } E$ : one division per part, with one remainder  $r$  carried at that Node. Here  $q_f$  is the sourcing family's own quantity at the Node: its form  $D_f$  for a holder of the content, its Wronskian  $W_f$  for the holder of the sign, or its tension  $T_{aa,f}$  for an axis part.  $w_f$  is the weight with which that family reads the written holder.  $E$  is the part's wall:  $E_s T$  for the time part and  $E_s W_c$  for an axis part. The sources are summed in the numerator and never divided apart, so there is no remainder per sourcing family [1, the primitives][3, one algebraic line per rule].

**The readings.** A reading is a bilinear form of the levels with declared coefficients. It writes nothing on the lattice. The law reads three of them, the third line of Eq. (1). The form  $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$  sets the present against the product of the future and the past. For a standing record of peak  $B$  it is  $B^2 \sin^2 \omega_b$  at every interval. Its quotient over  $T$ , the record's write, sources the holders. The Wronskian  $W_i$  is the charge (Section 5.5). The current  $F_{ij} = \text{num}(\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$  into the Node  $i$  from its neighbour  $j$  is antisymmetric. With the Link's factor  $q_{ij}^2/\Gamma^2$  it is the Link term of the conserved form (Theorem 3); where no tension stands it is the plain current. A wave travelling from  $i$  to  $j$  has  $F_{ij} < 0$ . A static record has  $\text{now} = \text{next} = \text{before}$ , so its form and its current vanish: a static record carries no quanta and no NodeDetector sees it [1, the booking; the conserved form].

**The interval.** Every read takes the values at the interval's start. Rule3's line reads one Link per interval, and nothing in zero time. Every reading of the interval, the step, the tensions and the write, is read from the record as it stands at the interval's start. So a Node's NodeState after one interval depends on the Node and its six neighbours alone: the interval's dependency radius is one Link (the law's line, LATTICE) [1, the interval's dependency radius is one Link]. The interval has four acts, in order: (i) the read, which takes each family's paces from the holders' levels, the reading holder's own among them, and each Link's factor from its tension; (ii) Rule3 on every record; (iii) the readings, which take each record's currents, tension, form and Wronskian at the levels the step started from, and each NodeDetector's report; (iv) the write of each holder from its summed sources. [1, the interval][2, the interval].

**What is conserved, and where.** On the lattice the conservations are per family. They are: the weighted form of a family while its reads are static (Theorem 3); the Wronskian of a plane family, its total with the weights  $1/p_i^2$  and its plain total where the paces are uniform, exactly in the line for any level in time under the law's turn of the time Link, the sign holder's turn (Section 5.5); the quanta through the Ports; the wave number of a free record; and reversibility bit for bit. Energy is conserved within a family by theorem while its reads are static, with the paces' weights, and between a charged body and its light by the energy line, a declared line of the law (Section 10.1). Between a body and the holders of the content, its conservation is replaced by the mass defect (Section 10.3). The lattice conserves no total: the sum of the families' forms is not conserved and not assumed, since the share moves with the paces and no current compensates (Eq. (13)). The click layer has books of its own. A click credits one whole quantum at the NodeDetector's own whole, drawn by the inflow of the record's share through its front over its window. A frequency of nature is read as the inflow's beat within the window or as the resonance of the NodeDetector's own record. The lattice's quantities, the form, the content, the level and the rotation, are read by no click except as the share's inflow. And the credit's bookkeeping is that the shares sum to what the screen saw, one whole per click, where the whole is the NodeDetector's own transition, up to the absorptions from undepleted records, read in the books and printed as the deficit, the lattice's share of such a record standing within its declared tolerance while

its count falls (Sections 5.3 and 7.4). In the clicks there are conservations other than the lattice's, and neither layer keeps a total energy (THEOREM for the lattice's, LATTICE; ASSUMPTION for the credit's, CLICKS) [1, the conserved form].

### 3.6 The postulates

The model rests on the law's seven postulates [1, the postulates], with the NodeDetector's write added by decision [3, A click writes on the lattice]. (1) The model's world is Nodes and Events. An Event is a level stepped by Rule3 with its remainder. A click is a NodeDetector's report and, by that decision, its one write at the Node the quantum reached it from. Nothing else happens. (2) A Node holds bounded local information, read from itself and its six neighbours, and nothing of the past is kept beyond the one level before. (3) Consistency is local and causal, one Link per interval for every reading of the interval (Section 3.5), with one exception: the credit of a click by the shares over every declared NodeDetector, the one non-local element, not derived. (4) The causal speed is one Link per interval. (5) Every physical calculation is on bounded integers, bounded for the run's length by the load's bound (the width's room, the amplitude bound and the register's room checked at load, the uniform light state  $a_t = A + Bt$  the case the bound names), with no floating-point number, no root and no random draw in the lattice's step; the NodeDetector draws with the world's one draw, a body's NodeDetector with its own seed. (6) A measurement is a NodeDetector's click: the net current into a declared region through its front boundary over one interval, accumulated over the window, and never a Node. (7) The lattice, where no one measures, is deterministic and reversible to the bit; a NodeDetector's write is the NodeDetector's act, forward only and outside the reversal; and the NodeDetectors' clicks are the only thing claimed to represent nature. The line and its clock are fixed: every mend of the model is a family, a holder, a declaration or a read, never a change of Rule3 (ASSUMPTION) [3, Rule3 is not changed].

### 3.7 The implementation

The implementation, Universe24 [43][2], is the law's implementation and nothing else; the file's choices stand in Table 4. Every statement of this paper about the implementation is of the archive's main branch at the commit 1fe3790a of 2026-10-05, the commit the two slits ran at. The worlds' lay files are the lay tool's output at that commit, the two body worlds laid at the declared rest; one finding of the runs is named open (the growth of a body's breathing, Section 10.3). The three documents this paper cites are at the tagged release 1.1.0, whose commit names that hash and changes nothing of the implementation [2, how to run a world]. It is a lattice of Nodes, with every family's NodeState at every Node, and one interval of acts in the law's order. Each act is a call of Rule3 on whole-lattice arrays of integers at a declared width, every neighbour read through a Port. The call is the same at every Node and for every family; a family's pair, dimension, reads and weights enter it as arguments from the universe file and from nothing else. It holds no number of physics, no formula, no family's name and no flag; every physical value comes from the run's files. A run is headless and writes the click lines, the measurement, and it runs for results only after the check of Section 4.1 says MATCH [2, the engine; the NodeState; the output; the gates]. It holds the law's lines of Section 3.4 [2, the NodeState; the interval]. The one run this paper reports is the two slits: the Huygens prediction 273, written before the run; the run  $N = 285$  at both of its seeds in the NodeDetector's unit  $W_c = 3 \text{ den } T$  (674 photons of energy  $T \sin \omega$  at  $\omega = 0.4366$ , the light's own frequency at the declared vacuum content's paces, that is  $285/\sin \omega$ , S.61, a lattice reading), 12 units over the blind, the exact lay's own number, the maxima at the blind's regions 2, 6 and 9, both sums deterministic; the law's real line at the same lay 284.7 in the paper's labelling, the pair after  $t$  steps labelled  $t$  (284.5 in the engine's labels, the click at the interval  $t$  reporting the pair after  $t - 1$  steps), printed beside the paper in `two_slits_real_line.txt` by the paper's script `two_slits_real_line.py`, which runs from the archive on the two slits' files at that commit and on no run of the implementation; and the back-in-time check, MATCH on the file over 40 intervals at both seeds (COMPUTED, a run of the implementation read by `back_in_time.py` and `click_counts.py`; CLICKS for the NodeDetector's counts, LATTICE for the gate and the 674) [1, the rows against nature, (g)]. The paper reports that one run and derives the rest: a run of a declared file computes what the algebra of that file states, and the implementation is held to the law's lines by its tests and the gate. The implementation is public, with the documents, the files and the scripts, in the Git repository <https://github.com/Closer24/Universe24>, archived at Zenodo [43]. It installs in a clone with `python -m pip install -e..`. The two slits' file runs with `PYTHONPATH=src python tools/run_inputs.py --out runs/first examples/events/two_slits/two_slits.json` and is read with `tools/click_counts.py`; the back-in-time gate runs with `tools/back_in_time.py`; each as the engine's document states it [2, how to run a world]. The paper's own scripts stand in `paper/general_formula/`. Every act of the engine is derived in the supplement from the law's lines, or is named there as declared, a named assumption or open, in one table of the two derivations (S.61), with no claim of nature.

**The rules we took.** What this paper computes is one chosen set of rules: Rule3 with the one write and the three readings (Eq. (1)); the postulates of Section 3.6; the primitives the files declare, each under its own name (the families with their pairs and dimensions, the holders with their sources and divisors, the NodeDetectors with their Nodes, the world's one draw with its seed, generator and window and a body's NodeDetector's own seed, the faces); and the integers of `rule.json`,  $\Gamma$ ,  $T$ , the width and the Link unit (Section 6.6). The engine fixes Rule3's form, the recurrence, and nothing else of the set. Another universe file, other families and primitives and other integers can be taken, and other things built with it [2, the interval; how to add a family]; there are many worlds to build under the rule, and this paper builds one. Everything the paper derives is derived from this set, and it claims nothing beyond it. Where a derived form coincides with a known one, the text says derived and names the form it meets; it never presents the known form as the law's own.

**The computation is elementary.** The computation reported here is elementary, and it is offered so that others may carry it further. The two slits' lattice is a plane of  $45 \times 48$  Nodes. A body spans a few Links at about  $10^3$  levels per Node. The integers are kept to 63 bits. The step is a Python loop over whole-lattice arrays, and no symmetry of the lattice and no derived resolution is used. The lattice's pixel is therefore coarse in every computed body, and its budget,  $(\text{Link}/R)^2$  and  $\sqrt{n}/A$  with  $R$  the body's radius in Links,  $n$  the run's intervals and  $A$  its peak level, is of the order of percent, stated against nature's pair in Section 7.3. What this paper derives, it derives by the law's lines. What it computes, it computes on this elementary engine, and marks as such. The gates of the bodies and the atom are named open and not claimed (Sections 7.3 and 10.3). The method's room for improvement is the paper's invitation.

## 4 Exact properties

This section states what holds exactly on every lattice, each item proved in S.48: the reversibility of the rule, the kind of function its acts can form, the conserved form with its current and its momentum flux, and the count as a reading of the record (S.5, S.6).

### 4.1 Reversibility

Let  $\sigma$  be the direction, +1 forward and -1 backward; let  $\text{div}$  and  $\text{mod}$  be the floor division and its remainder in  $[0, w)$ ; and let  $\rho$  be the remainder carried in. With

$$u = \sigma \left( \sum_j R_{ij} \text{arr}_j + Sx - wy \right) + \rho, \quad z = \sigma(u \text{ div } w), \quad \rho' = u \text{ mod } w, \quad (5)$$

the forward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$  gives  $(z, \rho') = (a_{\text{next}}, r')$ , and the backward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$  gives  $(z, \rho') = (a_{\text{before}}, r)$ .

**Theorem 1** (the direction). *The backward call returns  $(a_{\text{before}}, r)$  bit for bit; the reversibility of every step needs no second function.*

(Proved in S.48.) Theorem 1 holds for the lattice's step at every interval. A NodeDetector's write and a body's window, click or not (Section 5.4), are the NodeDetectors' acts outside the reversal, as the lay is. The back-in-time gate is exact between clicks, and runs across a click and the erasure intervals that follow it, treating the values a NodeDetector presented at its Ports as fixed boundary values; those values are read from the lattice-labelled lines of the run's output beside the click line, with the lay's own acts removed from their lines, and the click line itself names no Node (THEOREM for the step; the write ASSUMPTION; LATTICE). The whole run therefore goes back:  $N$  intervals forward and  $N$  back return every array of every family bit for bit, provided every number an act read at an interval's start is one the state after the interval still holds (THEOREM where the writes change no level, by construction from Theorem 1; LATTICE). The implementation checks this as a test: it runs a file forward and back and compares every array at every interval; on the two slits' file it reports MATCH over 40 intervals at the commit 1fe3790a, a file whose write stays under one level (Section 6.6) and whose NodeDetectors write nothing (screens with no record of their own; the write belongs to a NodeDetector given a record, Section 5.3) (COMPUTED, a run of the implementation read by `back_in_time.py`; LATTICE) [1, the inverse; the objects][2, how to run a world][1, the Node knows no family]. The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the step back. So a record's rotation sense is never read at a Node, and the holder of the sign is read by the record's shape, a plane or a line (Section 5.5) [1, the interval][3, Every world passes the back-in-time gate][1, there is no sense].

### 4.2 What the acts can form

Rule3's line is one act, and declared coefficients make four special cases of it, which the law also calls acts (not the interval's four acts of Section 3.5): (a) the read, the line with its six arrivals; (b) the one-Node step, the line with the arrivals declared 0, which rotates a level by Chebyshev's recurrence; (c) the division, the line with  $R = S = 0$ , Euclid's division with the remainder kept; and (d) the load, a declared integer in the numerator, which is how a count enters a level [1, the four acts].

**Theorem 2** (what the acts can form). *Every fixed finite composition of the four acts with fixed coefficients is an affine form of one family's levels plus a term bounded by a constant of the composition, since a floor is its argument less a fraction in  $[0, 1)$ ; so no product of two of its levels is written by the step on the whole lattice, a product growing as the square along a diagonal where the composition grows at most linearly. The products across families enter through the coefficients the read turns from the held levels: the read multiplies. On a bounded domain every integer function is a finite sum of floors of affine forms, and the theorem says nothing there.*

(Proved in S.48.) So every product of two levels is a reading and never a write by the step; the one write writes the declared bilinear readings, and the three readings of Section 3.5 are the only products the law takes. A line enters the law only if it passes three tests. It must be generic (one primitive with declared integers, no family's name), vector (an act on the whole state, with no root and no floating-point number) and local (its own record and the six neighbours, with nothing kept at a Node beyond the law's numbers). A line that needs more is a hypothesis under its own name. No act of the interval takes a square root. The integer square root appears only when a file is laid, as the stop of the division act  $x \leftarrow (x + n \text{ div } x) \text{ div } 2$ , the law's fixed point, where the iterate stops falling (at  $n = 3$  the iterates 2, 1, 2 alternate and the stop is 1) [1, the three tests of every rule; the six verbs; the root leaves the run, and everywhere].

### 4.3 The conserved form and the momentum flux

The read coefficient  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  carries the Node's factor and the Link's. Where the paces differ, the Node's factor is not symmetric and the Link's is. So the conserved form carries the weight  $1/p_i^2$  on its Node terms and the Link's factor  $q_{ij}^2 / \Gamma^2$  on its Link terms:

$$E = \sum_i \frac{w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i}{p_i^2} - 2 \text{ num } \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{ now}_i \text{ before}_j. \quad (6)$$

**Theorem 3** (the conserved form). *At constant paces for which a positive Node weight  $d_i$  with  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$  on every Link exists,  $E$  with the weights  $d_i$  is exact under the line without the rounding. Under the law's read coefficient, the Node's pace squared times the Link's factor squared,  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  with  $q_{ij} = q_{ji}$  and  $p_i = p_0(c_i)^2 / \Gamma$  the*

composed Node factor, the weight  $d_i = 1/p_i^2$  cancels the Node's factor on both sides of every Link. So the weighted form,  $d_i = 1/p_i^2$ , with  $\mathbf{M}$  the matrix of Rule3's step,  $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ ,  $\mathbf{D} = \text{diag}(1/p_i^2)$  and  $\mathbf{DM}$  symmetric at every interval, is exact for any static content and static tensions. Its Link term carries the Link's factor, so the current of Rule3's own conservation law through the Link  $ij$  is  $\text{num}(q_{ij}^2/\Gamma^2)(\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ , which is the plain current  $F_{ij}$  where no tension stands,  $q_{ij} = \Gamma$ . The form with the plain current, the Link factor dropped ( $q_{ij} \rightarrow \Gamma$  in the Link term), holds where no tension stands alone; with a tension it is not conserved (two witnesses, S.32).

(Proved in S.48.) With the rounding, each Node's remainder adds the term  $(r - r')(\text{next} - \text{before})/(w p_i^2)$  to the walk of the step's own two-level form  $Q$ , the form of now and before (S.48, without the wall), which equals the three-level form  $\sum_i(\text{now}_i^2 - \text{next}_i \text{ before}_i)/p_i^2$  exactly without rounding, Eq. (6) being  $w$  times it; the term is a coefficient below one in magnitude times the level's change, and the three-level form itself walks by  $\sum_i[\epsilon_i(t) \text{ next}_i - \epsilon_i(t+1) \text{ now}_i]/p_i^2$  with  $\epsilon_i = (r_i - r'_i)/w$  (DERIVED, exact in integers, two derivations; S.5; LATTICE; rule3.py).  $E$  is a diagnostic. Where the paces move, the total form  $D = |\mathbf{a}_{\text{now}}|^2 - \langle \mathbf{a}_{\text{next}}, \mathbf{a}_{\text{before}} \rangle$  changes exactly, in the rationals, by  $D(t+1) - D(t) = \langle \mathbf{a}_{\text{next}}, (\mathbf{M}(t) - \mathbf{M}(t+1)^\top) \mathbf{a}_{\text{now}} \rangle$ ;  $\mathbf{M}$  is symmetric only at paces uniform over the lattice, since  $R_{ij}$  carries the reading Node's own pace, and there the transpose drops. For the weighted form, written here without Eq. (6)'s factor  $w$ ,  $E = \sum_i(\text{now}_i^2 - \text{next}_i \text{ before}_i)/p_i^2$  with the weights fixed in time,  $E(t+1) - E(t) = \mathbf{a}_{\text{next}}^\top \mathbf{D} (\mathbf{M}(t) - \mathbf{M}(t+1)) \mathbf{a}_{\text{now}}$  with  $\mathbf{D} = \text{diag}(1/p_i^2)$  the weights of the paces at  $t+1$ , those that symmetrize  $\mathbf{DM}(t+1)$ , the weights' own change added where they move in time (the substitution above; S.48 proves the fixed-weights case); in the integers Rule3's floors add the remainders' term (S.5). The change is 0 at paces fixed in time, uniform or not, for the weighted form (for the plain form at fixed uniform paces alone), and negative where a well deepens under a standing record (THEOREM for the identity, by substitution; the sign under a deepening well COMPUTED, surplus\_check.py; LATTICE) [1, the conserved form].

**The momentum flux is the wave's own stress.** The momentum density along the axis  $a$  at the Node  $i$  is the current per axis,  $P_a(i) = (F_{i,i-a} - F_{i,i+a})/\text{num}$  for  $\text{num} \neq 0$  (at  $[0, 1]$  both currents are 0, and the density with them). At coefficients uniform over the lattice, the line without its division gives the identity below exactly; at non-uniform coefficients the exact identity is one line,  $w[P_a(t+1) - P_a(t)] = -Y_i \Delta_a(\text{now})_i + \text{now}_i \Delta_a(Y)_i$  with  $Y_i = \sum_b R_b(i) (\text{now}_{i+b} + \text{now}_{i-b}) + S_i \text{ now}_i$  the line's read, exact in the integers before the division for any coefficients, static or moving; Eq. (7) is its uniform case, and the difference between the two is the  $a$ -axis differences of the reading Nodes'  $R_b$  and  $S$  times the levels, while the share's identity (S.5) gains nothing at static paces. The integer step subtracts the remainder term  $\delta_i \Delta_a(\text{now})_i - \text{now}_i \Delta_a(\delta)_i$ , with  $\delta = r - r'$  and  $\Delta_a$  the centred difference, in this orientation of  $P_a$ , positive for a record moving toward  $+a$ , the one orientation of the law and the paper (THEOREM for the line, LATTICE),

$$w[P_a(t+1) - P_a(t)] = \sum_b R_b[G_{ab}(i) - G_{ab}(i-b)], \quad (7)$$

$$G_{ab}(i) = \text{now}_i (\text{now}_{i+a+b} - \text{now}_{i-a+b}) - \text{now}_{i+b} (\text{now}_{i+a} - \text{now}_{i-a}),$$

with  $G_{ab}(i)$  the stress reading on the  $b$ -Link ( $i, i+b$ ), from the levels now alone. The proof is in S.48. The tension on the axis  $a$  at a Node is the part a holder reads into the Link's pace, and the source of a holder's axis part. It is the Node's own reading, and the Link's booking is the sum of its two ends',

$$\begin{aligned} T_{aa}(i) &= -\text{num } h_a(i), & h_a(i) &= \text{now}_{i-a} \text{ now}_{i+a} - \text{now}_i^2, \\ G_{aa}(i) &= \text{now}_i (\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a} (\text{now}_{i+a} - \text{now}_{i-a}), \end{aligned} \quad (8)$$

and the axis part receives  $-\text{num } h_a(i)$ , one Link's reach, read from the Node's two neighbours along the axis at the interval's start;  $G_{aa}(i) = h_a(i) + h_a(i+a)$ , so the Link's booking  $-\text{num } G_{aa}(i) = T_{aa}(i) + T_{aa}(i+a)$  is Eq. (7)'s flux in the booking's unit, the sum of its two ends' tensions, and the pace reads the Link's tension as the mean of its two ends' (on light's wave the written stress over the written energy is then  $(\text{num}/3 \text{ den}) \sin^2 k / \sin^2 \omega$ , 1 at long wavelength, nature's  $T_{xx} = T_{00}$ , 0.90 at  $k = \pi/4$  3/5 at  $k = \pi/2$ , where  $\sin k$ , the Link's current per amplitude, is largest, and 0 at the band's top along an axis; the sum would write 2 at long wavelength). The Link's tension is one number per Link, read the same from both ends by the pace at the next interval. A plane wave of amplitude  $A$  at the wave number  $k$  has  $h_a = -A^2 \sin^2 k_a$  and  $G_{ab} = -2 A^2 \sin k_a \sin k_b$ , so the tension along the wave,  $T_{aa} = \text{num } A^2 \sin^2 k \geq 0$ , and the Link's booking  $2 \text{ num } A^2 \sin^2 k$  are positive, as a wave's pressure is (S.7). A narrow-band packet of count  $n$  and velocity  $\mathbf{v}$ , its components' cross terms averaging out, has  $2n v_a v_b$ , the two Links' currents summed, which is the stress of dust (DERIVED, LATTICE; S.7 for the plane wave; tension.py). With the sign as written, a wave's stress deepens the content around it, so light gravitates by its pressure too. The off-diagonal parts and the vector part, which Rule3 never reads, are not written [1, the tension; the final unification].

#### 4.4 The count is the record's share

Three things are called a count, and the paper names them apart: the books' count  $N$ , an integer in the credit's books, moved by clicks alone, down one at an absorption and up one at an emission, and constant between clicks; the share,  $\sum_i e_i / W_c$ , a lattice reading of the record that drifts between clicks by the currents and the remainders within the floors, and where the paces move in time (Eq. (13)); and the quantum  $W_{\text{rec}}$ , the record's share per quantum read once at its lay (Section 5.3), constant along its passage by the adiabatic invariant (S.14). The conservation claims are the books' count's, exact by construction, and the share's, within the floors. A family's share of quanta is a reading of its record and no line of its own. At every Node the record's share is

$$e_i = \left[ w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i - \text{now}_i \sum_{j \sim i} R_{ij} \text{ before}_j \right] \text{div}(2p_i^2), \quad (9)$$

the floor of the rational share  $\tilde{e}_i$ . Here  $\tilde{e}_i$  is the conserved form's term at the Node with its weights (Theorem 3), the Node's  $1/p_i^2$  and each Link's  $q_{ij}^2/\Gamma^2$  inside  $R_{ij}$ ;  $e_i$  is its floor, the Node's integer term; and the credit reads  $\sum e_i$  in quanta over  $W_c$ , rounded half up to the nearest whole. At the vacuum's paces it is exactly  $3 \text{ den}(\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before})_i$ , with  $S_6$  the six-neighbour sum. Read with the vacuum's coefficients instead of the Node's own paces, the same expression is the vacuum's share, the unit of the readings of Section 7. Over one step of Rule3 the share's rational value, before its floor, changes by exactly the six currents into the Node, each weighted by its Link's factor, plus Rule3's own rounding (THEOREM for the rational identity, LATTICE; S.5). The integer share of Eq. (9) adds the NodeDetector's own floor, a residual strictly between  $-1$  and  $1$  per Node and level pair, distinct from Rule3's remainder:

$$\tilde{e}_i(\text{next}, \text{now}) - \tilde{e}_i(\text{now}, \text{before}) = \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} F_{ij} - \frac{(\text{next}_i - \text{before}_i)(r'_i - r_i)}{2p_i^2}, \quad (10)$$

with  $F_{ij}$  at the levels the step started from,  $r, r'$  Rule3's remainder before and after the step, and the integer share  $e_i$  differing from  $\tilde{e}_i$  by the NodeDetector's floor residual at each of the two level pairs. So over any region the total share changes only by the weighted currents through the region's boundary, the remainder terms and the NodeDetectors' floors (THEOREM, LATTICE) [1, the count is the record's share; the share's change is the currents]. A formulation that moved whole quanta by a division of the current would put numbers at every Node that no act of Rule3 reads back. The law does not carry it (ASSUMPTION) [1, the count's line and the sense's line leave the law]. Where the paces move, the share drifts with no current crossing any Link, Eq. (13); this is a diagnostic [1, the paces' own change moves the share with no current].

**The signed integers.** The levels are signed integers. The currents are signed by their direction and the Wronskian by its sense, and a click is one whole quantum, never negative. The total share of any lattice whose Links read the same from both ends and whose every beyond reads 0 (a closed, a periodic or an open-faced lattice) is non-negative for  $|\text{num}| \leq \text{den}$  at the vacuum's coefficients, with equality for a static uniform level of light and, on an even periodic lattice, for its checkerboard (THEOREM, LATTICE; S.6). The archived files' massless holder, at its rest 60 read beyond every open face, lies outside the theorem, and the sign of its total share there is a diagnostic. One Node's share can be negative at a lay's edge, and a region's by at most half a level-square per boundary Port. Over the whole passage of one forward packet, the inflow accumulated at a region's front equals the region's final share plus what left through its back, and each of these is bounded below by S.6 (Eq. (10) summed over the window), at the vacuum's paces, where the inflow's coefficients are the identity's and the credit's unit is read (in a well the balance carries the paces' weights, S.42, inputs). So the inflow is non-negative up to the rounding, and the credit by the shares is well defined. Where a window is cut inside a passage, the NodeDetector credits  $\max(s_R, 0)$ , the floor at 0 the credit's declared rule (ASSUMPTION, CLICKS) [1, the signed integers contradict nothing].

## 5 Measurement: the click as a meeting

The click is where the method meets Universe24, the clicks it computes. The click is the meeting of the future with the past at a declared NodeDetector; it is no act of the lattice. A click is a fact: one whole quantum passed from the arriving record to the record that absorbs it, written once and never undone. It is always the meeting of two quanta, so every click has the record absorbed and the record that absorbs, the arriving record and the transition of the record that waits for it, and never happens by itself (S.49 (16)). In the law's one line, a NodeDetector drives Rule3 from outside the lattice, as a face and a start do. It presents one value at one Port of the click's Node, in the place of what the Node would see from the lattice, and lays whole quanta there; the Node steps Rule3 as always and knows neither. A click is this act: one whole quantum of one record is absorbed, Rule3 writes the hole, and whole quanta of others are emitted, the rotations summing at the Node. The conversion, the weak force under the hypothesis of Section 10.3, is the same act with the family changed on the out side; every other click is of its kind with the family kept. The draw alone chooses whether and where, which future continues. In the words of the division, a click is Rule3's division act on the families. The shares that entered a NodeDetector over its window, with the remainder the books carry, are divided by one quantum  $W_c$ ; the quotient is the clicks and the remainder is kept. What Rule3 does to the levels at a Node with the wall  $w$ , the credit does to the shares at a NodeDetector with the wall  $W_c$ , with the draw in the place of the Port (the three storeys of the one act, S.50). A NodeState is written by Rule3, by the lay and by the face, and by nothing else; the hole is written by the face value at the click's Port [3, A click writes on the lattice]. The credit rounds the window half up. The carried remainder is the law's form, stated under its own name with its own blind on  $N$ ; it is the top storey's "how many" and never its "where". The lattice's remainders cannot stand where the draw stands: a local carry per NodeDetector gives no shot noise, and Rule3's remainders used as pick numbers put every click of the two slits on two of twelve regions (a counter-check recorded in the law's document [1, the click is the meeting]). The credit and the write in this paragraph are the NodeDetector's declaration (ASSUMPTION; CLICKS). This section states what is on the lattice, the meeting, the NodeDetector, what a click leaves behind, the charge and light, and what locality costs [1, the click; the reading].

### 5.1 Events, and no waves

On the lattice there are events and amplitudes. An event is a level split by Rule3 to the six neighbours. An amplitude is the level Rule3 produces at a Node, the sum of the split events there. The record is the amplitudes over the lattice. In Universe24 there are clicks, the NodeDetectors' reports, and nothing else. In the two slits the lay's events pass both gaps, every path of the future at once; the fringes are where split events from the two gaps meet at the screen with the same sign. The past is made of the same split events. Read backward from the realised click by the same line, Rule3 at  $-1$ , they form the past's tree, which also passes both gaps back to the root. What stays on the lattice, the amplitudes no click realised, is not in Universe24. After a click they are a cancelled future at three levels, the credit's, the NodeDetectors' and the lattice's. In the credit, the record's count is at 0 and no NodeDetector credits it again. Where they reach a NodeDetector backed by a declared face, their share leaves the lattice there. On the lattice, the absorbed record is erased from the click's Node outward at the causal bound, one Link distance per interval, inside the click's light cone; beyond the front its levels remain, an empty wave that still sources the holders it writes to, until the front arrives (Sections 5.7, 5.4) (ASSUMPTION) [1, no waves on the event board].

## 5.2 The click is the meeting of the future with the past

**Assumption 1** (the click is a meeting, the law and not a hypothesis). A record runs on the lattice in both directions by the one line: forward from the root, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at  $-1$ . A click is the meeting of two quanta at a declared NodeDetector: the quantum of the incoming record run forward, the future, and the transition of the record that waits for it, the click's unit, the past read back from it. Its share is their product at the region, at resonance (Section 7.4): the form  $D = \text{now}^2 - \text{next} \times \text{before}$ , the now against the future times the past. Which meeting is realised is the NodeDetector's draw, the world's one (a body's NodeDetector's own seed), and the lattice draws nothing. The one non-local place is the credit's one draw over the NodeDetectors that read one record, through the root. The NodeDetector only asks whether the future and the past met at it.

There is one kind of reader, the NodeDetector, and the file decides what it holds. A NodeDetector with no record of its own reads the record that arrives at it, a bolometer's front; the two slits' screen is of this kind (Section 5.3). A NodeDetector whose file gives it a record of its own, with parts, transitions and a lifetime, stands for an atom or an ion of nature: it reads the arriving record against its own transition, and it emits as well (Section 5.4).

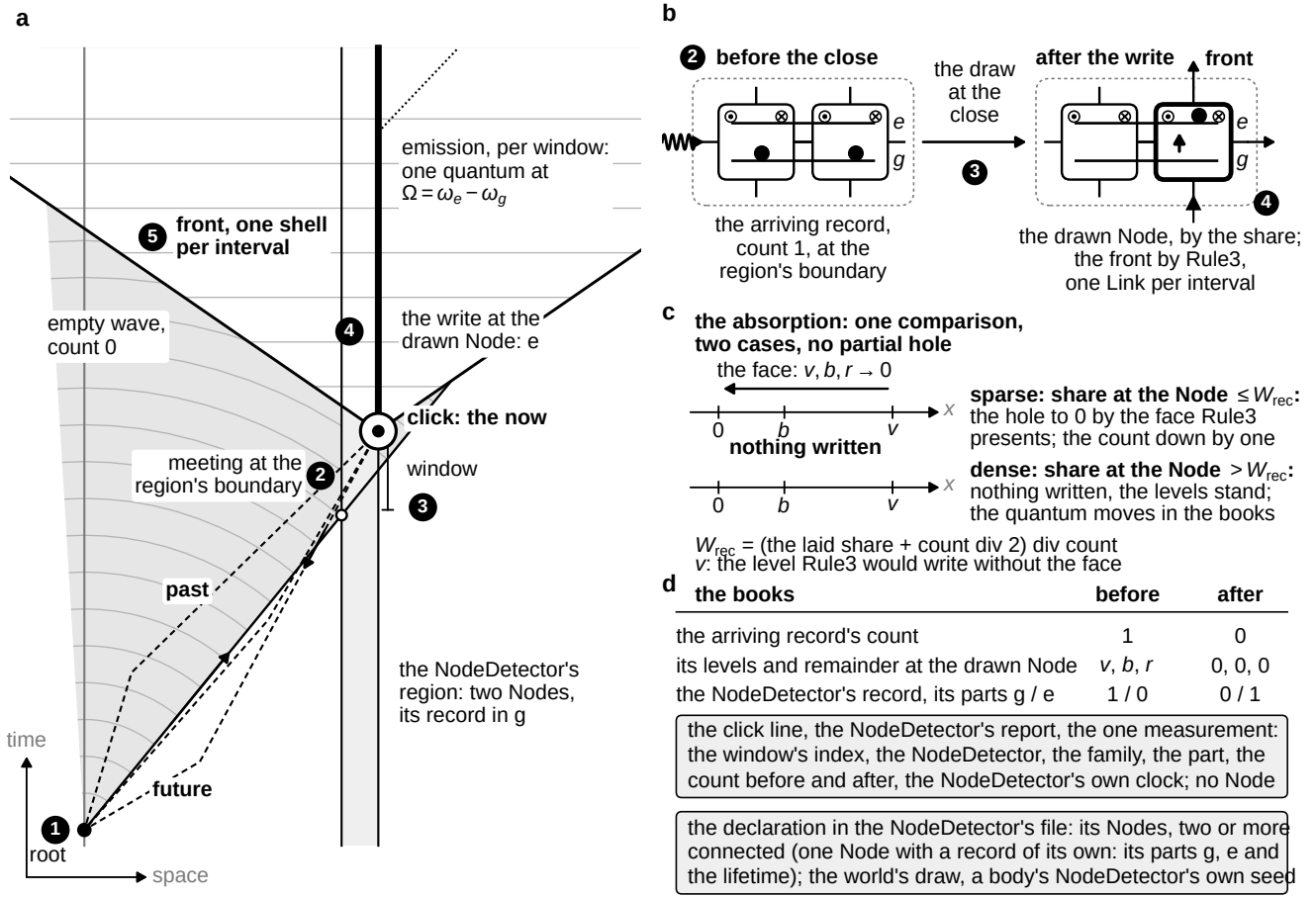
**The click, algebraically.** Let  $X$  be the state space: at every Node, for every part of every family, one integer level pair and the write remainders, and nothing else. The step  $S_t$  is Rule3 over the interval, with the paces read from the holders at its start; it is a bijection of  $X$  (Theorem 1). Its forward orbit is the future and its inverse orbit is the past. A NodeDetector  $R$  is a declared region with its front Ports and its coefficients, which define  $Q_R$ , the inflow through its front over the window (S.50). A click is the pair  $(R, t)$ : the credit of one whole quantum to  $R$  at the interval  $t$ , drawn with the probability  $s_R / \sum_{R'} s_{R'}$ , where  $s_R = Q_R(x_t) / W_{\text{rec}}$ , the record's own unit and the credit's divisor throughout ( $W_c$  for light and for a record laid by share with its count,  $W_c \sin \Omega$  for a born quantum,  $\Omega$  the transition's rotation defined below). The decision is the one map in the model that is not a bijection. It is a scalar under the group. Its write is one act on  $X$ , at the one Node of the region that the draw picks by the share (Section 5.4): the levels that arrived there are absorbed, the record's count falls by one and the absorbing record's count rises by one, the quantum is conserved between them, and the change spreads by Rule3 as an event (Section 5.4; ASSUMPTION, the NodeDetector's declaration). A frozen Node stands in no NodeDetector's front and is never clicked (ASSUMPTION for the definition, DERIVED for the precisions; CLICKS; `clicks.py`) [1, the click's algebraic definition].

**One form for every click.** At the credit (Fig. 4) the parts of a record are read with the NodeDetector's coefficients, each part paired with the same part through the common origin, then summed and squared. The share is that density, and the draw is by the shares. The two slits are the case with nothing to pair: light is one real line, and the share is the record's own inflow. Bell's check is the case through the common origin: a pair born as one event is two trees with one root, and the backward reading from the first click passes the root and continues forward on the partner's tree (Section 9.3). The joint share is computed from the forward records alone, by the linearity of the rule, and the backward reading exists by Theorem 1, the inverse orbit. The meeting is the law's reading of the click, laid over a credit that computes the forward record's share; the NodeDetector's transition enters as the unit it credits and the family and part it reads; the backward reading is Theorem 1's bijection, by which the root is one for the two trees (the credit ASSUMPTION; the equality of the forward and backward descriptions DERIVED for the line at fixed paces; CLICKS; `rule3.py`).

## 5.3 The NodeDetector, the one declaration kind

**Assumption 2** (the NodeDetector's declaration). A NodeDetector is one connected region declared in the file, drawing with the world's one draw or, with a record of its own, its own seed: one Node or more where the file gives it a record of its own, two or more otherwise; a single Node without a record is refused at load. Where it reads a travelling wave it is at least half the wavelength of the family it reports across the beam (4 Links at  $\lambda = 8$  Links); a narrower region, a single Node without a record of its own or a region in pieces is refused at load. A NodeDetector backed by a receding face (behind which the lattice grows, Section 3.1, so what passes the region continues outward and never returns) or by the wall's own face takes what enters it, since what enters leaves the declared lattice and never returns; a bare region in the lattice reports the net inflow through its front, and since what enters it leaves, its net over the window is nothing and it takes nothing, a reading beside the screen and no click (the two slits' gap, Section 9.1). Each interval it reads the inflow through its Nodes' front Ports, the Ports leading in from the declared lattice, Rule3's currents summed in integers into its window. Its click is its report, one line per quantum: the window's index, the NodeDetector's name, the family and the part, the count before and after, and the NodeDetector's own clock, never a Node. One unit of the family,  $W_c \sin \omega_0$  of the inflow's share ( $W_c = 3 \text{ den } T$  itself for a massless family, Section 4.4), of the inflow over a passage, booked as the conserved form's own current through each front Link with the Link's factor squared, the plain current where no tension stands on the Link, in the credit's unit  $W_{\text{rec}} G^2$  ( $G$  the Link's unit, a declaration of the file like  $\Gamma$ ), is one unit of count, one quantum's share: the invariant's unit  $T$  for a bolometer's whole and  $T \sin \omega$  for a counter's share (S.49 (22)), so a screen saw  $N$  units where the inflows of its regions sum to  $N W_c$ . The credit is one quantum to one NodeDetector, drawn by the shares of the inflows of the regions, read outside the lattice over every declared NodeDetector of the file; one credit per quantum and never two. The NodeDetector draws with the world's declared draw (a body's NodeDetector with its own seed), the lattice draws nothing, and the run is deterministic per seed. After a realised click the NodeDetector writes once, at the one Node of its region that the same generator's next draw picks, by the inflow booked through each Node (Section 5.4). A NodeDetector whose file gives it a record of its own lays that record over its Nodes in the weights' proportion, by the declared weight per Node and the count declared once, with its parts, its transitions and its lifetime, its outer Links cut (the Link's factor 0 at both ends, the Node's own coefficient recomputed from its open Links) and its inner Links open, so that its uniform mode rotates at  $\cos \omega_0 = \text{num}/\text{den}$  exactly whatever the region's size, at the vacuum's paces; it is held by its declaration and never bound by a well, and it emits as well as takes. The credit is the one non-local place of the description and no line of the law.

What produces a click is deterministic: the meeting. Which of the possibilities becomes the click, no line of the law gives; the NodeDetector draws. The draw decides a place and a combination, never a value and never a path. How many clicks is no draw: a NodeDetector reading the record credits at the window's close every whole quantum its inflow holds,  $N$  the sum of the shares over  $W_c$ , capped by the record's count, each quantum with its own draw of the region and of the Node and its own click line, and no cap of one click at a time is added; a NodeDetector with a record of several parts turns and



**Fig. 4** The click at a NodeDetector, in plain words (ASSUMPTION for the draw and the write, the rest DERIVED; CLICKS; drawn from the definitions, no run; solid lines run forward, dashed lines read backward; clicks.py; drawn by click.body.py). A NodeDetector is where the future meets the past: the arriving record's quantum, run forward from its root, is the future; the quantum of the transition that waits for it, read back from the realised click, is the past; the click is their meeting, the present, and there is no click of one quantum alone. (a) Space and time. 1: the arriving record runs forward from its root by Rule3, every path at once (the grey fan). 2: it meets the NodeDetector's region, two adjacent Nodes (the grey strip), at the region's boundary (the open circle); the window (the bracket) opens at the NodeDetector's last write and closes at its draw. 3: at the close the NodeDetector draws with the world's declared draw, a body's NodeDetector with its own seed, by the shares of its Nodes, once over every NodeDetector that reads the record, through the root: the click, the now (the ring). From the click the same line is read backward, Rule3 at  $-1$  (the dashed lines): a reading, never a write. 4: forward, the click's write, at the Node the draw picked by the share (the thick world line); the NodeDetector's record goes to its part  $e$ . 5: the change spreads by Rule3 at one Link per interval, the front (the white cone), and erases the absorbed record inside the click's light cone, the ball of radius  $d$ ,  $(2d+1)(2d^2+2d+3)/3$  Nodes of zeros, never faster (Section 5.7); beyond the front its levels remain, the empty wave with count 0, stepped on and never credited. The dotted line is the emission when it comes: one whole quantum of light laid at that Node at  $\Omega = \omega_e - \omega_g$ , drawn per window at the rate, the beat's current over  $W_c$  plus the rate's declared constant. (b) The region: two Nodes with their Ports, the dotted boundary the declaration's; the NodeDetector's own record is laid over both with equal levels, its parts  $g$  and  $e$  the ladder. Before the close (2) the record is in  $g$ ; the draw (3) picks the drawn Node by the share; after the write (4) the drawn Node (the thick border) carries the record in  $e$ , and the front leaves through its Ports. (c) The hole at the drawn Node: where the record's share there is at most  $W_{\text{rec}}$  the hole goes to 0 and the count falls by one; a denser record is undepleted, its count moving in the books alone and the deficit printed (Section 5.4); no factor scales the hole. (d) The books in whole numbers, the click line (the window's index, the NodeDetector, the family and the part, the count before and after, the NodeDetector's own clock, no Node: the report and the only measurement) and the file's declaration (the seed, the Nodes, connected, two or more with Nodes alone, the record with its parts and lifetime where declared, which may stand on one Node)

draws its outcome once per window. The two declaration kinds carry two rules, both the law's: a region with Nodes alone counts by the half-up rounding of the total over every region reading the record and draws each quantum's region by the shares, with no outcome in which none takes, so one region's count moves with the others' inflows through the rounding of the total alone, by at most one quantum per window, the one named non-local place; a body weighs each transition by its own transfer share and the rest of the unit by the outcome that none absorbs, so its marginal is its own share whatever the other bodies' (S.57 (b)); "no signal passes" below is the amplitudes' sentence. Which region is drawn by the shares; which boundary Node, by the per-Node inflow; which combination of the sides' ports, through the root. The generator is a declared linear congruence with its seed, of the full period  $2^{\text{width}}$  by Hull and Dobell's theorem [88] for every archived generator. The draw is the generator's next state  $x$  scaled to the weights' total, the pick  $(x \times \text{total}) \text{ div } 2^{\text{width}}$ , and the outcome the first index whose cumulative weight exceeds it. So Born's rule on the lattice is the proportionality of the weights to the shares, a declaration; the generator's uniformity over its period gives each outcome a frequency within  $2^{-\text{width}}$  of its weight's share

whatever the total, the weights the raw inflows in the share's integer unit,  $1.1 \times 10^{-19}$  at the width 63, and not exactness; the successive draws' independence is assumed and not proven, and the scatter  $\sqrt{NP(1-P)}$  is conditioned on it. Nothing of the Node and nothing of Rule3 enters the draw. Exclusivity, one click per quantum, is derived under the credit's one draw, the NodeDetector's declaration, from the count conserved (Section 5.4). Born's rule, the credit's chance equal to the intensity's share [44], is a declaration of the NodeDetector and not a finding of the model [1, what produces a click is not known]. The uncertainty relation is the NodeDetector's declarations and no rule of its own: the region (its Nodes),  $\Delta y k \geq \pi$  at half a wavelength; the window (the file's draw window),  $\Delta \omega \geq 2\pi/n$  over a window of  $n$  intervals; and the draw (its seed),  $\sqrt{NP(1-P)}$  over  $N$  clicks, with the credit's floor and the record's whole. The write is exact inside, at a Node and an interval that are the lattice's alone and never reported, and the report is uncertain outside; the two are one act. The NodeDetector is defined by the uncertainty relation, and that is why a NodeDetector with Nodes alone stands on two Nodes or more, never on one, since one Node has no wave number to resolve; a NodeDetector with a record of its own may stand on one Node, resolving the place and not the wave; the click line names no Node either way (ASSUMPTION). A NodeDetector is a face of the lattice that reports: the Nodes see a boundary, the world sees a click. It simulates the click's effect and not its cause. In nature what registers is bodies, and what it does happens in those bodies; our NodeDetector is not in the Nodes. Nature's reacting bodies, the Sun's nuclei among them, are in the model NodeDetectors with a record of their own, declared with their tables and rates; the model derives what their clicks write and at which of the body's Nodes by the share, and declares the bodies, their tables and how often they click (a supporting statement of what the model holds and no claim about the Sun; Section 10.3). Its acts are the loop's from outside the Node, as the lay and the receding face are. The share that reaches a region backed by a receding face, as the two slits' screen is, leaves the lattice there, credited or not; that is a declaration of the file and not a rule of the NodeDetector [1, the uncertainty relation is not a rule of its own]. What each declaration costs is derived in S.8 and S.50: a pattern in the histogram and never in one click (DERIVED, CLICKS; clicks.py) [2, the output].

**The NodeDetector is one declaration kind.** The NodeDetector is one declaration kind for every experiment: its Nodes and the world's one draw shared by every NodeDetector with Nodes alone in turn (a body's NodeDetector its own seed with its record), two or more for a NodeDetector with Nodes alone, and, where the file gives it, a record of its own, which may stand on one Node, and nothing else; the window is the file's draw declaration, one for every NodeDetector. Its window is declared and is never read off the wave: the file's window for a NodeDetector with Nodes alone (the archived gates declare it their run), and the interval of the probe's lay (the probe a second laid record whose arrival closes the window), or the declared window where no probe stands, for a NodeDetector with its own record; the declaration is set, explicitly, to cover the support of the arriving share at its Nodes, both levels, for a finite packet and for a record of its own alike, and for a continuous flow the cut is declared and stated as such. Its unit for an arriving record is the record's own quantum  $W_{\text{rec}} = (\text{the laid share} + \text{count div } 2) \text{ div count}$ , read once from the books' origin, or at the first lay for a record born later, and  $W_c \sin \Omega$  for a born quantum, so that no transition is declared on a region; for its own record the unit is that record's transition: one click is one quantum's change in the absorbing record,  $T \sin \Omega$  at the frequency of the NodeDetector's own record, never the event's division by its own  $\sin \omega$  and never a change of the share. For its own record the arriving level is read into each part's phase at the declared weight and resonance pair [num, den],  $\cos \Omega = \text{num/den}$ : the part's two time levels turn by Rule3's rotation act, and the parts' levels, squared, at the region's Nodes after the turn are the shares of the outcomes. It draws at the window's close by those shares; in the dark it draws at its clock's grain (the declared span of lattice intervals per draw), over that span or the lifetime  $\tau$  where that is shorter, with the weights [span  $\times p_0 \times \text{unit div } \Gamma$ , half up;  $\tau \times \text{unit} - \text{that}$ ], the hazard span/ $\tau$  in its proper time, so that the geometric survival, the exponential's discrete form, is read in proper intervals. A declared window's draw spans its lit intervals alone and the dark intervals are drawn at their own grain as they pass; every interval of the window enters exactly one draw's span, so the hazard summed over a window is  $n_w/\tau$  ( $n_w$  the window's intervals) whatever its dark intervals. It gives the quantum at  $\Omega = \omega_e - \omega_g$ , read from its modes. Its own clock, the carried sum of  $p_0$  over  $\Gamma$  at its Nodes, and its event clock, the windows closed, stand in the click line beside the lattice's interval; the click line is the experiment's whole reading, and every lattice number, the Node written, the levels taken, the erasure shells, the regions' densities, is a diagnostic labelled so (DERIVED, two derivations; CLICKS; S.61, lines 12, 13 and 17; clicks.py).

## 5.4 The click decides in the credit and writes at one Node

**Assumption 3** (the click's decision and its write). The lattice is deterministic and local: Rule3 with no draw, and the bijection of Theorem 1 at every interval. The click is the meeting of two quanta at the NodeDetector: the incoming record's quantum and the transition of the record that waits for it. The NodeDetector, declared in the Universe24 file, draws with the world's declared draw (a body's NodeDetector with its own seed) whether the meeting is realised; the randomness is the NodeDetector's, and the lattice has none. Forward, to the future, one write: the exchange at the one Node of the region that the draw picks. The NodeDetector knows that Node and we do not (the uncertainty principle: for us the click is a region and a window). That one Node is drawn after the outcome, one draw per click: for an absorption by the inflow booked per Node over the window, whose sum is the region's share; for a NodeDetector emitting from its own record, by that record's share over its Nodes; the Node is a lattice diagnostic beside the click line. The exchange: one whole quantum passes between the two records at that Node, the one generic act, a face presented at the Node's Ports. One quantum of the arriving record is absorbed there and its count in the books falls by one; the absorbing record's part rises by one, laid in its mode with the count's own scaling,  $A^2 = (N+1)T/(2\sin \Omega)$  on each of the part's two lines, with the phase of the product and the remainder at the origin of the lay; the part it leaves is laid with its count down by one. The hole in the arriving record is one rule: where the record's share at the drawn Node (the credit's own, one-sided) is at most  $W_{\text{rec}}$ , the spread quantum whose paths the click cancels, the hole goes to 0 at that Node, its levels and its remainder, and the record's count falls by one; a record denser than that is undepleted by an absorption: its count moves in the books, nothing is written on the lattice, and the deficit, the absorptions less the emissions back into it, in quanta, per family and per run, is printed explicitly. The booking identity  $\Delta(wQ) = w(x-v)(x-b)$ ,  $x$  the written level,  $v$  the level Rule3 would write at the Node and  $b$  the level the step leaves, the pair's before (S.50), is the share's change under any write and no recipe for the hole. Nothing is scaled by a factor and no number enters. The quantum is conserved between the two records up to the absorptions from undepleted records, read in the books and printed as the deficit, the lattice's share of such a record standing within its declared tolerance while its count falls, and the absorbing record's change spreads by Rule3 as an

event, one Link per interval. No act is faster than one Link per interval: the front erases the absorbed record, from that Node outward at the causal bound, one shell of Link distance per interval, the hole's act on each shell, the ball of radius  $d$ ,  $(2d+1)(2d^2+2d+3)/3$  Nodes of zeros; the clicked record alone is written and nothing else. The other paths are a cancelled future at three levels, and the front erases them inside the click's light cone (Section 5.1). Backward, to the past, a reading and never a write: the credit's reading through the root, which chooses which tree is the past and gives Bell at a distance (Section 9.3). It is the one non-local place, and it cannot be a write, since the lattice is deterministic forward.

After its click, a single quantum's count stands at 0. The count is conserved and is read by the credit through the root, the NodeDetectors' shared other side on the lattice; the Nodes' levels carry the shares and never the count. This gives antibunching, and one quantum, one click per record between far NodeDetectors [45], from the credit's one draw over all the NodeDetectors that read one record (ASSUMPTION for the one draw, DERIVED from it for the rest; CLICKS; the numbers of Sections 9.2 and 9.4 hold with one click per record; `clicks.py`). No signal passes: nothing at one NodeDetector changes the amplitudes at another except by Rule3 from the Node of the exchange. A NodeDetector is a declared region of the lattice with two sides, and it is entangled on both. One is the front, where the incoming quantum meets the transition it reads. The other is the side where its record connects to the apparatus's bodies [1, the observer is a cluster], and, through the record it absorbed, to that record's root. So two NodeDetectors reading one record are connected through the root, their common past. A bound body clicks when quanta from outside meet its own transition, never by clicks from outside; the meeting is the resonance of Section 7.4 (DERIVED, CLICKS; `bands.py`, `body_clicks.py`), and it clicks only where it is declared a NodeDetector. A bound body has no window of its own, and the now reaches it only through a NodeDetector's click; its own period as its window is excluded by the law, no click happening in a bound body by itself since Rule3 never writes a whole quantum, and nature's survival exponent  $n = 1$  stands beside it as the comparison (DERIVED, CLICKS; S.59; `zeno.py`) [1, what is open]. The window of a NodeDetector's own record is the packet's duration, as the law can read it; on the integer lattice a laid packet carries a precursor and a ringing tail that turn, so the support of the arrival closes no window under a laid wave, and no window closes on the wave itself. The window is therefore declared and bounded by the schedule of the lays: it opens at the record's last write and closes at the interval of the probe's lay, the probe's absorption in a measurement, and the probe's arrival is derived from its lay, from click times and from no level read. The declared window of an archived file is the experiment's protocol, the file's draw declaration, one for every NodeDetector, and no key of the NodeDetector's row, which declares its Nodes (and, for a body's NodeDetector, its own seed and its record's parts), the draw otherwise the world's one shared by every NodeDetector with Nodes alone in turn. Where no probe is laid, the declared window stands, explicitly named, as the probe. The arrival-window rule enters no archived file; the window is declared (ASSUMPTION, CLICKS). A body with no declaration does the Nodes' laws alone: the two-mode transfer and the radiation at its beat by the write, continuous, with no count moving and no jump. A NodeDetector with a record of its own draws over its parts at each window and writes its outcome at the one Node of its region that the draw picks, re-laid in the drawn part at its count there and nowhere else, click or not; into the light record it reads it writes nothing on a null outcome (S.59). A spread body's click is the exchange at its entry Node, and its settling is a reading, named open (Section 10.3). Of the past the click erases nothing on the lattice: Rule3 between clicks stays the bijection, and the back-in-time gate across a click restores the absorbed share from the lattice-labelled lines beside the click line, the click being the one act outside the reversal. Of the future: the record's remaining share spreads on as an empty wave, a wave with no quantum, interfering and never crediting, and the front carries the absorbed record. What is sent on: the changed record at the drawn Node steps by Rule3 from the next interval, so a later NodeDetector reads the changed lattice and the futures that cannot follow do not click. The emission is the lowering, at a Node whose record is in its upper part, drawn per window at the rate, the beat's current over  $W_c$  plus the rate's declared constant: one whole quantum of light is laid at that Node at the resonance  $\Omega = \omega_e - \omega_g$ , read from the modes, its total one quantum's form: on the open lattice the packet at  $T/\sin\Omega$  on the one real line laid, and in a guide the source in time at  $\sum_t A_t^2 = (2/3)T \sin k$ , the source's own time sum and not the standing record's  $T/(2\sin\Omega)$  (S.48), the direction an assumption stated as such; the conversion lays the table's rows at the one Node. The rotations sum exactly at the Node, and the shares sum to the booked remainder,  $T(\sin\omega_e - \sin\omega_g)$  against  $T \sin\omega_L$ , a finite-rotation correction booked explicitly (S.14, S.43). The conservation laws of a click are the write's own construction and no rule added: the count whole, the rotations summing, the charge's unit  $T/2$  of the Wronskian per quantum (Section 10.1), every line of a record together. The NodeDetector's one act of its own is the draw; the write that follows is derived. What clicks is this. Light clicks with a body's transition (the absorption and the emission). A quantum of matter clicks with a body's transition (the capture and its reverse). Light meets light by summing in one row and never by a click. The holders of the content, gravity and the binding holder never click (S.49; the conversion of a quantum between families is the hypothesis below). A body in the dark is only the future and still gravitates. Emission is the same meeting with the roles reversed: two local writes and one draw of the credit binding them. The Node stays generic: it knows no family, no NodeDetector and no click; the NodeDetector's acts are performed at the Node's Ports from outside Rule3, as the lay's and the face's are, and Rule3 is never changed. The write is the NodeDetector's act (ASSUMPTION, the NodeDetector's declaration [1, the click writes on the lattice]; the form of the decision is DERIVED, the one form of Section 5.2; LATTICE; `clicks.py`). The lattice holds only the future, every possibility, and has no arrow: Rule3 is a bijection at every interval (Theorem 1). The clicks have the arrow. Each is forward only, the one act outside the reversal. At each click what was future becomes past, and the present is the meeting. The chain of clicks is the past advancing into the future, written once and never undone. The direction of time belongs to the clicks and not to the lattice (THEOREM for the step; the clicks' arrow is the NodeDetector's declaration; LATTICE). The two slits' file declares NodeDetectors that report and write nothing (screens with no record of their own, Section 5.3), and the write at the window's end leaves its numbers unchanged (Section 4.1) (the model's rule: the NodeDetector alone writes, at the one Node its draw picked, and the lattice draws nothing) [3, A click writes on the lattice][1, there is no family of clicks at the Node; the click is the meeting of the future with the past; clicks go only forward; the backward run is exact].

**Only a click moves a quantum between families: a derived statement, and a hypothesis under its own name.** Two statements stand under this heading, and each has its own mark. The first is derived. On the lattice a record's levels change by Rule3 in its family's own plane and by the holders' reads alone, and the read moves no form; the turn is a gauge. A whole quantum's form passes between a family and a holder's record only through the write at a click: the emission's beat written to the holder's line, and the hole's levels absorbed at the one Node. A count passes between families only at a click: at an absorption the arriving record's count goes down by one and the body's part up by one at the Node; at an emission the

part goes down by one and one whole quantum of light, a record with count 1, is laid at the same Node; at a conversion the table's rows are laid at the one Node. Between clicks every count is constant. In one sentence: a quantum changes family or part only at a click's write, and between clicks every count is constant. This is a statement about quanta and the records' lines, not about form or a holder's content: a holder's own content is written every interval from the sources' bookings, and a change of the paces moves a record's form with no current, the mass defect (Section 10.3). Nothing writes a record's lines but Rule3, the lay and the face, and the credit's counts move in the click's acts alone (DERIVED, two derivations; CLICKS; `clicks.py`). The second is a reading of nature, a hypothesis under its own name, on which no number of this paper rests. The lattice computes the possibilities of the future locally and carries shares. The clicks are the present, where under the hypothesis the world's real computation is done, each between the past, the body's own transition read back to the record's root, and the future, the incoming record's paths. What passes between families on the lattice is form, a possibility; what passes at a click is a quantum, a fact. What tests it: a body laid in its excited part alone, with no beat, gives no light by the write alone, and under the hypothesis it gives one quantum at an exponential time, a single atom's lifetime. Its gate is named in Section 10.3 (HYPOTHESIS [1, only a click moves a quantum between families]; Section 10.3).

## 5.5 The sign is the rotation sense

A family of quanta laid as a plane carries its record as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder. A holder's parts stay real. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (11)$$

the Wronskian of the two pairs. The line without its division conserves it exactly, with the weights of the conserved form, under the condition of Theorem 3 on the paces (THEOREM, since **DM** is then symmetric; LATTICE; S.9). The integer step adds to each part the remainder term  $\epsilon = (r - r')/w$ ,  $|\epsilon| < 1$ . So the integer  $W$  changes at a Node by  $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$  per interval, not zero in general; one witness is computed in S.9 (COMPUTED, LATTICE). An exactly conserved integer charge is open (Section 10.3); the rational invariants above hold in the integers to the remainders' term of S.9. Its local change is the Link current:  $(2 \text{num}/w) (q_{ij}^2/\Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  per Port for the weighted  $W_i/p_i^2$ , and  $\text{num}/(3 \text{den})$  times the bracket for the integer  $W_i$  at the vacuum's paces (S.9). Its total over the lattice is invariant under any change of the paces that keeps them uniform over the lattice, and the weighted total  $\sum_i W_i/p_i^2$  is exact at any paces fixed in time, whatever their non-uniformity, and where the paces move it changes by  $\sum_i (d_i(t+1) - d_i(t)) W_i(t+1)$  with  $d_i = 1/p_i^2$ , since the bracket  $\text{im}_{\text{now}}^\top \mathbf{DM} \text{re}_{\text{now}} - \text{re}_{\text{now}}^\top \mathbf{DM} \text{im}_{\text{now}}$  vanishes for symmetric **DM** (THEOREM, the bracket's vanishing its proof; LATTICE). A real record ( $\text{im} = 0$ ) has  $W = 0$ , so light, the real record of the holder of the sign, is neutral. A rotating record has  $W$  of either sign by its sense [1, the sign is the rotation sense; the sign's count is the record's own].

Who reads the holder of the sign is decided by the family's dimension and by no name (ASSUMPTION, LATTICE): a family of dimension 2 has a Wronskian and reads it; a family of dimension 1 has  $W = 0$  and does not. The holder of the sign does not read its own level, so light does not read it and a static charge does not bend light. A reader's own sense never enters its coefficients, since a coefficient read from a Node's own state would break the exact inverse (Section 4.1). So the holder of the sign acts on its readers by the act its row declares, **rotation**. The holder's level over  $\Gamma$ , times the Node's clock  $p_0/\Gamma$ , is the angle  $\theta_t$  of the interval  $t$  that turns the two-part record in its own plane,  $e^{i\theta_t} z_{\text{next}} + e^{-i\theta_{t-1}} z_{\text{before}} = (S z_{\text{now}} + \sum_a R_a (z_{+a} + z_{-a}))/w$ , the level before under the previous interval's angle and the level next under this one's, each angle read from the holder's level at that interval's start, so the inverse reads the same numbers: the law's time-Link turn, the time Link's gauge, under which the record's form and Wronskian, booked from the turned levels  $u_i, v_i$  as S.33 reads them (the raw  $\text{Im}(\bar{z}_{\text{now}} z_{\text{before}})$  is not the conserved one under a varying turn), are conserved exactly for any level in time. These are three exact integer shears, one per factor of the rotation, inverted bit for bit; under them like and unlike senses are told apart by the record's own sense. Which sense moves apart is one bit the model does not fix; it is declared by nature's measurement that like charges repel. The exact angle of the shears is  $2 \arctan(L p_0 / (2 \Gamma^2))$ , which is  $(L/\Gamma)(p_0/\Gamma)$  at small angles, and the magnetic force follows as the Link's phase, Peierls' coupling (Section 8) (THEOREM for the conservation, S.33; ASSUMPTION for the declared bit, LATTICE) [1, there is no sense; every family has a dimension; the sign holder rotates the two-part record; the rows against nature, (f)].

## 5.6 Light is born by the write

In the model, light is the sign holder's own record: one record whose time part is a real level pair. It is stepped by Rule3 at the paces of its reads, so that light is bent by the content, and it is written by the one write from the Wronskian of the records that read it. A static rotating body writes a static level, Coulomb's field, and radiates nothing. A breathing or accelerating body writes a varying source at its Nodes; the sign holder carries it away at one Link per interval, and those spreading events are the light. A uniformly moving body writes no travelling light in the long-wavelength band (S.42 (b)) (DERIVED, CLICKS; `body_clicks.py`). A NodeDetector reads light's current through its boundary in quanta of  $W_c$ . In this reading a star shines by the write of its charges, continuous and clickless, emits whole quanta at its declared bodies' transitions and converts by clicks at its declared nuclei, the clicks' places and rates declared and not derived; no counter of ours is needed for any of it: the nuclei that convert and the bodies that emit are themselves NodeDetectors with a record of their own, declared with their tables and rates, and a counter is where a quantum of it is absorbed (a supporting statement of what the model holds and how clicks would read it, with no identification with the Sun claimed; the conversion a HYPOTHESIS under its own name, Section 10.3). The travelling events of the other holders enter a NodeDetector's region the same way and are never credited (Section 5.4); in gravity's case they stand beside nature's gravitational wave. A body's quanta that leave it stay its family, and no quantum changes family at a crossing. No record reads its own write of the sign; the emitted light is the light family's record, read by every reader, the emitting body included [1, light is born by the write].

## 5.7 The price of locality

Three consequences of the model's locality, stated against nature's measurements. Born's rule is the NodeDetector's declaration, and exclusivity is derived from the count conserved under the credit's one draw, the NodeDetector's declaration (Section 5.3). A clicked record's empty wave still writes its form into the holders it sources until the front reaches it: a Node

**Table 2** The formulas of clicks, each with its status. Every row is a formula of what a NodeDetector reports, CLICKS, except where a cell says lattice.  $U$  is the clock's share and  $U_K$  the Newtonian potential the orbits measure (Section 8).

| The formula                        | Status                                                                     | What the model gives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Section  |
|------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Born's rule, the count at a region | declaration; $N_R$ and the scatter derived under it                        | $N_R$ the whole part of $s_R = Q_R/W_{rec}$ , the inflow through the region's front over the window in the record's own unit ( $W_c$ for a packet laid at count-unit, $W_c \sin \Omega$ for a born quantum), the whole quanta by the division act; the pattern is in the histogram, $NP_i \pm \sqrt{NP_i(1 - P_i)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 5.2, 5.3 |
| The dead time                      | declaration                                                                | not derivable: the shares credit a region any whole number of quanta in one interval, and $W_c$ over the inflow's rate is the mean spacing of the clicks and not a least one                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5.3      |
| The two-slit row                   | derived; computed by <code>two.slits.real.line.py</code>                   | the near-field minima at the lines $y = 15$ and $33$ ; the screen's inflow peaks at the interval $69.0$ against the fourfold refined lattice's $62.0$ ; the wings, the regions $3$ and $8$ counted from $0$ (the four-row blocks holding the real line's minima rows), $18.2$ and $13.5$ on the lattice with both faces receding (the file's blind $16.4$ and $12.4$ at the same regions under the absorbing source side); per laid quantum the transmission $0.157$ against its continuum limit $0.136$ , the lattice $16$ percent above, converging as $1/s$ , $s$ the refinement scale (the script's section 6)                                                                                                                                                                                                  | 9.2      |
| The arrival and its spread         | derived; computed by <code>two.slits.real.line.py</code>                   | $t = L/v_g$ with $v_g = (\text{num}/(3 \text{ den})) \sin k / \sin \omega$ Link per interval; one path's packet spread<br>$\sigma_t = \sqrt{\sigma_x^2 + (\omega'' L \sigma_k / v_g)^2} / v_g$ from the band's curvature, for a Gaussian packet on the quadratic band with $\sigma_k \ll k$ , $4.1$ intervals at $k = \pi/4$ over $L = 34$ Links from the packet's centre to the screen (lattice: no NodeDetector reads one row); the screen's own arrival duration by the paths, root-mean-square $7.3$ intervals at equal weights over the six gap rows and the $48$ screen rows, each path's time $(10 + \sqrt{24^2 + (y_s - y_g)^2}) / 0.5469$ , and $10.9$ by the real line with the packet's spread and the tails (clicks); the centre's arrival $62.2$ by the real line, beside the refined lattice's $62.0$ | 9.2      |
| Bell's pairs                       | the credit declared (Born's rule for a pair); $E$ and $S$ derived under it | $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$ and $S = \sqrt{2}(1 + \rho)$ at the ideal angles, $\rho = 2r/(1 + r^2)$ with $r$ the parts' amplitude ratio, $478/169$ at the integer settings for equal parts ( $r = 1$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 9.4      |
| GHZ, three quanta                  | derived under the declared credit (S.40)                                   | $E_3 = \cos 2(a + b + c)$ , every pairwise $E$ $0$ , Mermin's $M = -4$ where every local credit gives $ M  \leq 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 9.4      |
| The clock's click rate             | derived                                                                    | $1 - f(\omega_0)U$ , $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , $1 - U$ at a small gap; the beat of two modes scales by $N$ exactly at long wavelength, to a correction $(1 - N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24$ at finite rotations (S.31); a cavity of bodies the same, declared faces $N^2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 7.4, 8   |
| The bending's clicks               | derived under the one assumption; in $U_K$                                 | $4U_b; (1 + \text{den}/\text{num}) 2G_K M/(c^2 b)$ against Kepler's potential                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 8        |
| The absorption line                | derived (S.16, S.43); the one-mode line under the pace read alone          | parametric at $2\omega_b$ , the growth $\epsilon/(4 \sin \omega_b)$ ; the transfer line at $\omega_j - \omega_i$ , Rabi's rate, under both reads                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 7.4      |
| Emission                           | derived                                                                    | a steady rotation writes no light, nor a uniform motion in the long-wavelength band (S.42 (b)); a breathing body writes at its beat, at the first order in the breathing's depth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 7.4      |
| The which-way sum                  | derived                                                                    | the continuity of the one inflow through the two fronts: a region behind one gap reads that gap's share and nothing of the fringe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 5.7      |
| The uncertainty relation           | declaration, with its inequalities                                         | $\Delta y k \geq \pi$ from the region, $\Delta \omega \geq 2\pi/n$ from the window of $n$ intervals, $\sqrt{NP(1 - P)}$ from the draw                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5.3      |
| No signalling                      | derived                                                                    | $P(A+) = (\cos^2 a + r^2 \sin^2 a)/(1 + r^2)$ , independent of $b$ for parts of any amplitude ratio $r$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 9.4      |
| A body's count in clicks           | derived; the unit the record's own, built; the drift a lattice reading     | the share summed over the body's Nodes, what a NodeDetector over all its Nodes credits; under a moving well it drifts by Eq. (13); a body's click one transition, $\hbar\omega$ per click                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 7.4      |
| Magnetism                          | derived under the holder's rotation (S.33); not under the plain read       | $F_C/\gamma_L$ (Lorentz's factor): the sign's halved current over the same wall as the time level, $E_s T$ ; $0.995$ at $0.1c$ and $0.866$ at $0.5c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 8        |

with six Ports erases nothing beyond itself, and the NodeDetector's front erases inside the click's light cone, never faster (Sections 5.4 and 10.3). And a pair is one laid record, its phase a shared classical variable inside Bell's hidden variable [24], so every local credit of it gives  $|S| \leq 2$  (THEOREM; CLICKS) [25]; the meeting through the root gives  $S = 2\sqrt{2}$  with nothing signalled, resting on the credit rule, a declaration (DECLARATION, FITTED; Section 9.4). A which-way measurement without absorption is no third price and is not computed here. The model's opaque NodeDetector in one gap loses the fringes and reads one gap's envelope, as nature's absorbing which-way markers do (DERIVED, the continuity of the one inflow through the two fronts; CLICKS; `two.slits.py`) [1, what is open; the rows against nature, (g) and (h); the click is the meeting of the future with the past][3, claims nowhere beyond what was shown].

## 5.8 The formulas of clicks

Table 2 collects the formulas of clicks that the model gives, each with its status and the section that derives it [1, the click's algebraic definition; what a body does, in the words of the clicks].

## 6 The families and the universe

The Universe24 the model computes is declared family by family. A family is a band of the one rule, not a separate law. Its row in the universe file holds only what the rule and the geometry do not fix: its pair  $[\text{num}, \text{den}]$ , with  $\cos \omega_0 = \text{num}/\text{den}$ , and its shape. The shape of a family of quanta is its dimension; the shape of a holder is its sources with its divisor. When a pair is written over the lattice's one denominator  $\Gamma$ , a mass is one integer  $m$ , with  $\cos \omega_0 = m/\Gamma$  and the wall  $6\Gamma^3$ . Reducing a pair to lowest terms is a reading and never an act of the load, so such families share one resolution of their remainders

**Table 3** The rule’s own universe, its four families as the file `examples/events/rule.json` holds them (DERIVED from the two keys of a row, its pair and its shape; LATTICE; `families.py`): each family with its pair [num, den] (in brackets over  $\Gamma = 6,000$ ), what it holds at its divisor, its shape, what it reads, and its free number. A further family, the pair family of light, two real lines never summed, is declared by Bell’s file and not by this file (Section 9).

| Family                                                                                                                                                                                                        | Pair                                               | Holds, at the divisor                                                                                                                                                             | Shape                                                                                                                                                                     | Reads                                                                                                                                                                      | Its free number                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| gravity, the time part and the three tensions                                                                                                                                                                 | [1, 1]<br>([6000, 6000])                           | the content, $E_g$<br>(1,000 in <code>rule.json</code> ;<br>$10^6$ in the two slits’<br>universe file<br><code>light.json</code> , gravity<br>off in all but name)<br>the sign, 1 | 1 + 3 real lines,<br>sourced by the<br>form and the three<br>tensions                                                                                                     | the content, its own<br>level among it (the<br>law’s line)                                                                                                                 | $E_g$ , the hierarchy; its<br>rest $c_{vac}$ , 60 by<br>declaration<br>(Section 6.5)                                                                                                                                                                                      |
| the charge, the holder of the sign, whose quanta are light, born by the write                                                                                                                                 | [1, 1]                                             |                                                                                                                                                                                   | 1 real line, sourced<br>by the Wronskian<br>of its readers (the<br>declaration); under<br>the rotation 1 + 3,<br>three odd axis<br>lines sourced by<br>the sign’s current | the holders of the<br>content, at the<br>weight 1; never its<br>own level                                                                                                  | $k$ , the write weight,<br>one key of the<br>universe file<br>( <code>write_weight</code> : 1 in<br>the rule’s own<br>universe file; the<br>calibrated values of<br>Section 10.2 in the<br>others), declared<br>the pair, its range<br>$R = 20$ Links, a file’s<br>choice |
| the binding holder, not the strong force (Section 10.3); its own events have the gap<br>$\cos \omega = \text{num}_b / \text{den}_b$<br>matter, the scalar; a quantum that leaves a body is a packet of matter | [2400, 2401]<br><br>[ $m$ , den]<br>([4000, 6000]) | the content, 1<br><br>nothing                                                                                                                                                     | 1 real line, sourced<br>by the form<br><br>dimension 1, one<br>real line; a charged<br>body of the same<br>pair is a second<br>row of dimension<br>2, a plane             | the content, its own<br>level among it (the<br>law’s line)<br><br>the holders of the<br>content,<br>$x = (L_b + L_g)/\Gamma$ ;<br>the plane also the<br>holder of the sign | <br><br>$m$ , the mass                                                                                                                                                                                                                                                    |

[1, the families from the rule; the mass is one integer; the wall is one]. A particle of nature is read as one quantum of a family, the electron of the matter family and the proton of its own; a family’s gap, its mass, is declared and not derived (Section 10.3; S.53).

**How a family enters.** A family enters the file as one row of the universe file, with two keys and nothing else: its pair and its shape. For a family of quanta the shape is its dimension, 1 or 2. For a holder the shape is its sources, which are the form, the form with the tensions, or the Wronskian, with the holder’s divisor  $E_s$  and, for the massless holder, its rest. Its rotation at rest, its band, its coefficients and its wall,  $w = 6 \text{ den } \Gamma^2$ , follow from Eqs. (1) and (4). Its parts number its sources, one real line per source. Its reach is set by its gap (Section 6.4). Its reads are those its declaration names, and gravity’s and the binding holder’s at the weight 1 (Section 3.4); it sources what it reads, at the weight it reads with. Every record of quanta can click, the exact bands at rest in the vacuum excepted (Section 6.1), and no row declares whether it clicks; the holders of the content, gravity’s and the binding holder’s, never click (Section 5.4). Table 3 gives the rows of the universe, and Section 6.6 the numbers of the file [1, a family’s declaration; the families from the rule][2, how to add a family].

## 6.1 The exact bands, and the two the universe does not use

The line of one Node at rest,  $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$ , is exact in integers only where  $2 \cos \omega_0$  is an integer. So the rule carries exactly three exact massive rotations:  $2 \cos \omega_0 = 1, 0$  and  $-1$ , with the periods 6, 4 and 3 and the pairs [1, 2], [0, 1] and  $[-1, 2]$  (THEOREM: an integer  $2 \cos \omega_0$  strictly inside  $[-2, 2]$  is 1, 0 or  $-1$ ; LATTICE). A record on an exact band, at rest in the vacuum, leaves no remainder in its rest rotation, so its quanta stand at every Node with no current through any Link; a NodeDetector reads no inflow from it, so it never clicks and only turns. The exact bands are exact only in the vacuum. At a Node of pace  $p_0$  the band’s rest rotation is  $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2$  with the rest term’s coefficient  $g = (\text{den} - \text{num})/\text{den} = 1 - \cos \omega_0$ , which is an integer at  $p_0 = \Gamma$  alone. So inside a well a record on an exact band leaves a remainder and, where its current through a front is not 0, it clicks like every family’s record [1, the exact bands; the exact bands are exact only in the vacuum].

The rule allows the bands [0, 1] and  $[-1, 2]$ , and the universe does not use them [1, the third and the colours, closed as absent].

## 6.2 The rule’s own universe

The universe of the model is the one in which every number the rule fixes is the rule’s own. The pairs come from the integers 1, 2 and 3 of the three exact massive rotations and the three axes; every divisor is 1 except gravity’s; and the remaining free numbers are named. Its rows stand in Table 3 [1, the families from the rule; the rule’s own universe][2, the input].

The universe’s integers are these. The Node clock is  $\Gamma = 6,000$ , a multiple of 3 so that the matter pair’s  $2/3$  is exact over it. The quantum’s norm is  $T = 2^{15}$ , the normalisation of the quadratic form. Two bounds are not physics: the integers’ width, 63 bits, from which the amplitude bound is derived, and the Link unit  $G = 16$ , the tension’s resolution.  $\Gamma$  and  $T$  are units, as  $c$  and  $\hbar$  are in nature. No line of Rule3 fixes  $T$ , since the rule is linear and the scale of its form is free. The energy line of Section 10.1 ties  $T$  to  $\Gamma$ ,  $E_s$  and the write weight  $k_w$  only where a plane family reads the holder of the sign, and no family of this universe does. Every holder of the file is a record stepped by Rule3 at its pair and written by the one write. A file’s starting rest is the fixed point of the division act, computed once before the first interval (ASSUMPTION) [1, the final unification][3, The world’s integers are the file’s; Rule3 is not changed].

## 6.3 What each family carries, and how gravity enters as a clock

What a family carries is set by what Rule3 reads and by what a record can write, not by a name (DERIVED, LATTICE; `families.py`). Rule3 reads two things of a Node and no third: the clock’s pace  $p_0$ , which is one number, and the Link’s paces, which are the three directions of the rule’s vector. So a holder read into the paces carries at most one time part and

**Table 4** The choices made for this world, Universe24, each with its status: declared (a number of the file, chosen), derived (from the line and the declarations), measured (a lattice reading of the archived runs, labelled so). The families' pairs and dimensions stand in Table 3, the calibrated weights in Table 5; the numbers already in the text are cited from here and not repeated.

| The choice                                       | Its value in this world                                                                                                                                                                                                               | Status                                                                        |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| The clock $\Gamma$ and the quantum of action $T$ | $\Gamma = 6,000$ , $T = 2^{15}$ , the width 63 ( <code>rule.json</code> , <code>light.json</code> )                                                                                                                                   | declared                                                                      |
| The families                                     | four: gravity [1, 1], the sign holder [1, 1], the binding holder [2400, 2401], matter [2, 3] of dimension 1; light the sign holder's record (Table 3)                                                                                 | declared; their bands derived                                                 |
| The weights                                      | gravity's $E_g$ and the sign holder's $k$ , read from $G$ and from $\alpha$ (Table 5)                                                                                                                                                 | calibration                                                                   |
| The vacuum content                               | 60 on gravity's row of every universe file, above the lattice's own floor (8 to 13 at $\Gamma = 6,000$ , Section 6.5) and above the measured swing (43 levels over the standing-body world's 400 intervals); no click bounds it above | declared above two floors; the floor derived, the swing measured              |
| The boxes                                        | the two slits $45 \times 48 \times 1$ with the $x$ faces receding; the body worlds the open 25-cube (the standing body) and the 31-cube (positronium); the neutron's 9-cube                                                           | declared                                                                      |
| The declared runs                                | 130 intervals for the two slits, 400 for the two bodies in a box, 1,000 for positronium, 32 for the neutron's conversion                                                                                                              | declared; the bodies' 400 by the fence of the vacuum content                  |
| The lays                                         | the lay tool's own output at the declared rest, from the file's declared count and Nodes: the packets at the paces of the rest, the bodies at the step's fixed point in their rests                                                   | derived by the tool; the fixed point measured not unique within its tolerance |
| The draw                                         | the world's one draw, seed 24 (and 25) for the two slits, shared by every NodeDetector with Nodes alone; the window [45, 130]                                                                                                         | declared                                                                      |
| The NodeDetectors                                | the screen's twelve regions of four rows each; the Bell and CHZ files' analysers at rational settings; the atom worlds' bodies with their own records                                                                                 | declared                                                                      |
| The ion                                          | the ion's world not built in this work                                                                                                                                                                                                | declared out                                                                  |

three axis parts, 1 + 3. A record of one real line, neutral matter or light, has a form  $D$  and a stress and no Wronskian. A record of a plane, charged matter, has a Wronskian  $W$  as well. A holder is always real, one line per source, since a plane's own  $W$  would be its own source. The shapes a row may declare follow from what Rule3 reads, and which family takes which is its row's declaration: *the scalar*, one real line, for matter, light and the binding holder; *the plane* for charged matter; *one plus three*, the time part with the three tensions, for gravity, which is sourced by the form and by the diagonal of the record's own stress (Section 4.3); and *two real lines never summed at a Node* for the pair family of light in Bell's file. A holder of the content takes no vector part and no off-diagonal stress, since Rule3 reads them nowhere; the holder of the sign under the rotation carries three odd axis lines sourced by the sign's current (S.33). For a real record no read of a current exists within the three tests, so there is no frame dragging of neutral bodies: gravity is a holder of the content under the act pace, with no odd lines (Section 10.3) [1, the final unification; the tension; every family has a dimension].

**How gravity enters: a slowed clock, and no curved space.** The model has no metric, no force and no law of motion. Gravity enters in three steps. (i) *The write* (the engine's act, the held write): at every Node, in each interval, every family of quanta writes its form over the action,  $D_i \text{ div } T$ , into gravity's holder over the divisor  $E_g$ , by the second line of Eq. (1). (ii) *The holder*: gravity's holder, [1, 1], steps by Rule3 at its pair [1, 1] and the divisor  $E_g$ . Around a static source it therefore rests at Poisson's solution,  $3G_0(r)s$ , with  $s$  the body's write per interval over the holder's divisor and  $G_0$  the lattice's Green's function (S.39), which is  $3s/(4\pi r)$  far away (DERIVED, LATTICE; S.39, S.42; `families.py`). A change travels at one Link per interval, retarded. (iii) *The read*: every family reads the holder's level  $\ell$  at its Node into its own coefficients, Eq. (3), the clock's pace once and each Link's pace twice. What general relativity calls curved space stands here as the Link's pace, light's index  $n = e^{2x}$  (DERIVED, LATTICE; S.2; `paces.py`); no space is bent in the model. The factor two on the Link is the one assumption of Section 3.4; the composition of the clock is derived from it (Section 3.4) [1, the paces; the clock once and the Link twice; the same clock and the same ruler for every act; the paces compose].

## 6.4 The reach of a holder, and the binding holder

A holder's rest solves  $\text{num } \Delta a - 6(\text{den} - \text{num}) a = -3 \text{ den } \sigma$ ,  $\sigma$  the source per interval. This is its line with the source once an interval;  $\Delta$  is the six-neighbour second difference. Outside a body the rest therefore falls by  $e^{-\kappa}$  per Link for  $0 < \text{num} < \text{den}$ , with  $\cosh \kappa = 3 \text{ den} / \text{num} - 2$ , which is the family's own band at zero rotation (at  $\text{num} = 0$  the rest is 0 outside the source, and for a negative numerator the tail alternates in sign, S.11). So the gap is one number that is both the mass of the holder's quanta and the inverse reach of the holder (DERIVED, LATTICE; `families.py`). [1, 1] has no gap, and its rest is Poisson's over the whole lattice. [1, 2] has  $\cosh \kappa = 4$  and falls by 0.127 per Link: a contact well with a reach of  $1/\kappa = 0.48$  Links. A pair with  $\text{num}$  just under  $\text{den}$  has  $\kappa^2 = 6 \text{ den} / \text{num} - 6$  to second order, and so a range  $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  Links, which is Yukawa's form [47] (S.11) [1, the well].

**Assumption 4** (the binding holder and gravity's divisor, decided with a reservation). The write of a record,  $D_i \text{ div } T$ , is laid into two holders. One is the binding holder: a family with a gapped pair  $[\text{num}_b, \text{den}_b]$ ,  $0 < \text{num}_b < \text{den}_b$ , at the divisor 1. It is stepped by Rule3 at its pair, as the massless holder is, and started at its rest. With the gap, that rest is a sum with a range,  $R = \sqrt{\text{num}_b/(6(\text{den}_b - \text{num}_b))}$  Links, and  $e^{-r/R}/r$  beyond it. It binds the bodies and is not the strong force (Section 10.3). Its own travelling events have the gap  $\cos \omega = \text{num}_b / \text{den}_b$ : a quantum with a mass whose Compton length is  $R$ , Yukawa's own line [47] and no input. The other is the massless holder [1, 1] at the divisor  $E_g$  declared by the file: gravity, long-ranged and weak by  $E_g$ . Every family reads the holders its declaration names, at the weights it names, and sources each at the weight it reads with: gravity's and the binding holder's levels are read by every family at the weight 1. The two holders are among the readers, each reading its own level among the content. The nuclear holders of Section 10.3 are read by the nucleon families alone (the law's line, Section 3.4; the selective read is a declaration, ASSUMPTION).

The first pair is [2400, 2401], with  $R = 20$  Links, and the holder's quantum has the rest frequency 0.029 per interval.

The range is a file's choice. In the archived files  $E_g$  is declared large, 1,000 in `rule.json` and  $10^6$  in `light.json` (Table 3), so that gravity does not enter a body's binding. Nature's  $E_g$  is read from  $G$ , as the sign holder's  $k$  is read from  $\alpha$ . The hierarchy of gravity and electricity is the product of the two declared weights and the quantum's energy against its unit of charge (Section 10.1). The decision carries a reservation: it may be wrong [1, the two rows; what is open].

## 6.5 The vacuum content

**Assumption 5** (the vacuum content is 60, this world's declaration above the law's two floors). The massless holder rests at a declared level everywhere: the vacuum content, the key `rest` of the universe file, which no line of the law derives. It is declared 60 at  $\Gamma = 6,000$  on gravity's row of every universe file, this world's declaration above two floors the law gives. The first, the lattice's own line: the massless band's corner mode sits on the guard's edge at the content 0, the plain rule's paces (the lattice's reference, not this world's rest),  $2 \cos \omega(\pi) = 2 - 4(p_a/\Gamma)^2 = -2$ , and a level below 0 grows; Rule3's margin at the corner mode of each held row is  $m(c) = 4 - 2g(p_0/\Gamma)^2 - 4(1 - g)(p_a/\Gamma)^2$  with  $g$  the row's gap,  $2g + 16c/\Gamma$  to the first order, and the start's kick  $K$  rings that mode at  $K/\sin \omega(\pi)$ , so the content stays inside the edge only above  $c_{\text{law}}(\Gamma)$ , the least  $c$  with  $c - K/\sin \omega(\pi) \geq 1$  over the world's rows: 8 to 13 at  $\Gamma = 6,000$  for  $K = 1$  to 2 levels (the kick measured at 0.5 on average and 2 at most; at 0.5 the rule gives 6),  $(K^2\Gamma/16)^{1/3}$  its closed form for the massless row at large  $\Gamma$ , below  $10^{-19}$  of  $\Gamma$  at  $2^{94}$  (the law's floor, derived, with the kick measured; LATTICE) [1, The vacuum content, item 22]. No click bounds it above: under the line every holder reads the content and a uniform rest is a change of units (below; S.12). The second floor, for a world with bodies in a closed box: its holders' measured swing over its declared run, 43 at the rest 60 over the standing-body world's 400 intervals, read at the box's symmetry points; 60 is this world's declaration above both, at the paces  $p_{\text{vac}}/\Gamma = (1 - 1/\Gamma)^{2c_{\text{vac}}} = 0.980$ , the same physics at an effective  $\Gamma$ . There are many worlds to build under the rule, and we build one.

A uniform rest  $c_{\text{vac}}$  is a change of units (Section 3.4), and only nature's measurements can tell it. Under the law's line every holder reads the content, as every family does. So the massless holder's own travelling events have light's paces identically in every level: gravity's events and light through the same wells arrive together, the difference of their two Shapiro delays is 0, and  $\gamma_{\text{GW}} = \gamma_{\text{EM}}$ . GW170817 bounds  $\gamma_{\text{GW}} - \gamma_{\text{EM}}$  between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [49] (DERIVED from the law's line; CLICKS; S.12; `families.py`). Inside matter the source is read through the pace: gravity gravitates (DERIVED, LATTICE; `weak_field.py`). Since the two speeds are one speed, the measurement bounds no rest, and the vacuum content stays a declaration, 60 in the file, above the law's floors (ASSUMPTION). The vacuum's write, which would give the faces a positive content, is a hypothesis of the law's documents (S.29) and is not claimed here. [1, every row reads the content; what is open][3, The vacuum content 60 is this world's declaration].

## 6.6 The file's numbers, and how the universe was chosen

Universe24 is one world built by the paper's method; the method and the claims that support it are the work, and no more; other worlds closer to nature can be built by the same method, and building one needs the calculations of Section 10.3, which this work does not do. The choices made for this world stand in Table 4, each with its status. The universe is chosen by one principle, stated before any row: one rule closes everything; the families are its inputs, each a band of the rule named by its pair; the units are a choice of scale; and the universe holds as few free numbers as the rule allows (ASSUMPTION) [3, The engine holds no number but Rule3's own]. The model's coefficients are of five kinds: three units (the Link, the interval, the quantum of action), which no theory computes; the declared dimensionless ratios (the elementary pairs over  $\Gamma$ , the holders' weights), from which  $\alpha$  and  $G$  are read; the derived ones (every composite family's pair from its constituents, every pure number of the predictions); the bounded ones (Cassini's bound on the pair, the Fermi Large Area Telescope's (Fermi-LAT) bound [87][89] on the lattice, which puts the electron's denominator at  $2^{94}$  and above); and the free rest. Whether the free rest is computable is a hypothesis under its own name and no claim of this paper (HYPOTHESIS): a computing line would need a selection principle for the integers that the law does not state; the one candidate tried, a record closing at its rotation's own period, selects the exact band alone and not the electron's pair (S.64). The road to them:

(i) a family is a band of the rule, and its row holds only its two keys; (ii) there are three exact massive rotations, the pairs [1, 2], [0, 1] and [-1, 2] (THEOREM, Section 6.1; LATTICE); (iii) the three of Rule3 is the three axes, so the massless holder of the content carries 1 + 3 parts and every other holder of the content is a scalar; (iv) light and matter have one denominator, their pairs written over  $\Gamma$ , a multiple of 3, and the binding holder's pair is the one pair not over it; (v) the pairs come from 1, 2 and 3 (ASSUMPTION): [1, 1] for gravity and the charge's holder, and [2, 3], the first pair past the exact bands, for matter, while nature's pair is far past it (Sections 8 and 10.1); (vi) every divisor is 1, under which the universe carried no free number but the mass; (vii) two numbers stand where nature's scale forced them, the binding holder's gapped pair and gravity's divisor  $E_g$  (ASSUMPTION, with the reservation of Section 6.4); (viii) the vacuum content is 60, this world's declaration above the law's floor and the measured swing, no click bounding it above (Section 6.5); (ix) the sign holder's write weight  $k$  is one key of the universe file, declared (1 in the rule's file), read from nature's  $\alpha$  as  $E_g$  is from  $G$  (Section 10.1).

What remains is this: two units,  $\Gamma$  and  $T$ ; two bounds, the integers' width and the Link unit; and five numbers of physics, which are matter's numerator  $m$ , the binding holder's pair, gravity's weight  $E_g$ , the sign holder's write weight  $k$  and the vacuum content. Under the hypothesis of Section 6.5 there is one more, the rate of the vacuum write. Every other divisor is 1. The two slits' file reads the universe file `light.json`, with the two holders of the content at the divisor  $10^6$ , so that light's own well stays under one level. That this universe is the universe is a decision and not a theorem (ASSUMPTION). The electron's mass over Planck's and the proton's mass over the electron's are open and not derived (Section 10.3) [1, the rule's own universe; the final unification].

## 7 Bodies

This section says what a body is and how it is computed (Section 7.1), when it is stable (Section 7.2), what of it is computed and what is not (Section 7.3), what its clicks give (Section 7.4) and how bodies form clusters (Section 7.5). In one sentence each: a body is a record standing in a holder's well, computed by the rule's own iteration and by no formula; what it gives in clicks is derived, no light from a steady rotation, the clock's rate slowed by the content, the resonance between two modes

and the share's drift under a moving well; what it does not give is named, a standing body, a lifetime, the reduced mass and the exchange (Section 10.3); and nothing of a body is claimed beyond these marks.

## 7.1 What a body is, and how it is computed

A body is its count and its record over its Nodes, together with the quanta that record carries; a record of count one may stand at one Node. In plain words, a body is a record held at its Nodes by the holders' wells. Its count is the number of its quanta, and it clicks, as Section 5.4 says, when a quantum from outside meets its own transition. What a NodeDetector over all its Nodes credits of it is the share of its record (Section 4.4), a reading and not a state; its count in the books moves only by clicks. Its form  $D_i \text{ div } T$  is written into the holders it sources, and the holders enter Rule3 through the paces alone. The record is bound by the wells its form sources, that is, by the binding holder's screened sum with the range  $R$ ; gravity's  $1/r$  is weak by  $E_g$  and enters no body's binding. No law is written for a body. A body is the support of a standing mode of the one rule, and its record moves by Rule3 alone. Four statements make a body (DERIVED from the line, with the decisions named; LATTICE). (a) The body is the fixed point of its binding holder, standing where the sources return themselves. (b) A holder's level is the static solution of the holder's line. The start solves that line from the bodies' forms to its own stop, after which Rule3 steps the holder. (c) A body's quanta are its record's share, carried wherever its record's current carries it. (d) A free packet or a cloud falls as  $a = \Delta |d\omega_0/d\ell| \nabla \ell$ , where  $\Delta = 1/m^*$  is the band's curvature at rest,  $d^2\omega/dk^2$  at  $k = 0$ ,  $\omega_0$  the family's gap and  $\ell$  the content level [1, what a body is; the four lines of the body].

**How a body is computed.** The holders' source is the record's form  $D_i \text{ div } T$ , the law's line. The body is computed by the rule's own iteration and by no other formula. Start from a spread of the declared count over the Nodes. Write the wells that spread lays into the holders, and let them rest. Find the record's top mode in those paces by Rule3's own read act, iterated. Scale its two levels so that the share carries the declared count. Write the wells that record lays. Repeat until the levels repeat, in integers, at the fixed point of the division act. The branch is chosen by the start's width alone: two fixed points at one count, the cloud and the compact body (S.13, the functional's two critical points); which start reaches which, a wide one the cloud and a narrow one the compact body, is the generator's reading, a lattice diagnostic [1, the generator; the four lines of the body; the generator is Rule3].

## 7.2 Stability

Let a body of  $C$  quanta have the record  $\phi$  over its region, with  $\sum \phi_i^2 = 1$  and the counts  $c_i = C\phi_i^2$ . Let every holder  $r$  it sources be a hollow ( $\sigma_r = +1$ ) or a hill ( $\sigma_r = -1$ ), with the static kernel  $\mathbf{G}_r$  of its line and the divisor  $E_r$ . The well at a Node, in levels, is then  $\ell_i = \sum_r \sigma_r (C/E_r) \sum_j G_{r,ij} \phi_j^2$  ( $\ell_i = \Gamma U_i$ ; Newton's potential is  $\Phi = -c^2 U$ , S.28). To first order in  $\ell_i/\Gamma$  the well raises the one-Node rotation  $2\cos\omega$  by  $4g\ell_i/\Gamma$ , where  $g = 1 - \cos\omega_0$  is the rest term's coefficient of Section 6.1. The body's fixed point is then a critical point on the sphere of

$$F[\phi] = \frac{\text{num}}{3 \text{ den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{C}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2, \quad (12)$$

the band's pressure less the wells' energy, each well counted once because the record is its own source (Hartree's form). The record is the top mode of Rule3's symmetric form at the paces, and  $dF/d\phi = 2\lambda\phi$  on the sphere is the fixed point's own equation, with  $2\cos\omega_b = 2\cos\omega_0 - \lambda$  (THEOREM, the functional's variational derivative, S.13's first step; LATTICE). Let  $\phi_\lambda(x) = \lambda^{-d/2} \phi(x/\lambda)$  be the body dilated about itself, with  $d$  the lattice's dimension. The pressure is then  $K/\lambda^2$  and each holder's well energy  $V_r/\lambda^{s_r}$ , with  $K$  and  $V_r$  the body's own pressure and well energies at  $\lambda = 1$ , where  $s_r$  is the holder's scaling power at the body's size:  $s_r = d$  beyond the holder's reach, and  $s_r = d - 2$  within it or for the massless holder [1, the functional and the dilation].

**Proposition 1** (the stable body, under the dilation and to first order in the well). *For power-law kernels,  $V_r \lambda^{-s_r}$  with  $s_r$  constant, at a fixed point of the binding holder the virial  $2K = \sum_r \sigma_r s_r V_r$  holds, and the second variation of  $F$  along the dilation is  $F''(1) = \sum_r \sigma_r s_r (2 - s_r) V_r$ . A body is stable along the dilation only where this sum is positive; the statement is along the dilation alone, and a full minimum of  $F$  would need the Hessian's mixed block, the dilation against the shape modes, or its Schur complement, which no derivation here gives.*

(Derived in S.48, by Derrick's scaling [50] and Pohozaev's identity [51] on the lattice.) Its mark is DERIVED (the continuous functional of Eq. (12), to first order in the well, the dilation taken as continuous; the integer step's version open; LATTICE), and not theorem; the Hessian beyond the dilation is open. For a screened kernel the exponent runs: a Yukawa kernel's log-derivative at one separation is  $1 + \kappa R$ , and a body's integrated well energy has the running exponent  $s(R) = 1 + \kappa R \langle u^2 \rangle / \langle u \rangle$  in the body's own weights,  $u$  the separation in units of the width, from 1 at  $\kappa R \ll 1$  to the contact kernel's 3 at  $\kappa R \gg 1$ , through 2 at  $\kappa R = 0.82$  for a Gaussian body;  $F''(1)$  then carries the term  $R s'(R) V$ , the dilation at  $\lambda = 1$  of a body of radius  $R$ , positive throughout, which the proposition's constant exponents omit; the transition is S.13's functional computed directly.

The factor  $s(2 - s)$  is  $+1$  at  $s = 1$ ,  $0$  at  $s = 2$  and  $-3$  at  $s = 3$ . So a hollow holds the body where its well falls more slowly than the pressure, and breaks it where the well falls faster. Three consequences follow. (a) One hollow beyond its reach binds every mass on a chain. In a box it has one critical point, a saddle: the body disperses beyond it and collapses within it. (b) A massless holder holds the body. Poisson's well in a box has  $s = 1$ , so  $F = a/R^2 - c/R$  has one critical point,  $R = 2a/c$ , a minimum for every mass; this is Pekar's body [52]. Beside it, a hollow beyond its reach adds a barrier:  $F = a/R^2 - b/R^3 - c/R$  on a Gaussian body, where the wide critical point is a minimum and the narrow one a saddle, and there is none where  $3bc > a^2$  ( $a, b, c$  the pressure's, the hollow's and the well's coefficients at unit size), which is the collapse (DERIVED, LATTICE; S.13; `bodies.py`). (c) A collapse ends in a frozen clock. Beyond first order the well saturates and the Link's pace weakens the pressure. Since no pace reaches 0 before the rounding, a collapse ends in a region of Nodes whose clocks are frozen to the unit, and never in a closed Link [1, the functional and the dilation; what is open].

### 7.3 The two branches, and the status of a standing body

The binding holder's fixed point has two branches, a wide cloud and a compact pixel, and the start's width alone chooses between them (Section 7.2; THEOREM for the equation, COMPUTED for its solution, S.13, the reader `pixel_mode.py`; LATTICE). No standing body is claimed from a run. The computed body is coarse in the lattice's pixel by construction (a few Links, about  $10^3$  levels per Node). So its drift and breathing are the lattice's budget and the lay's, and not the law's:  $(\text{Link}/R)^2$ , the pixel against the body's size, and  $\sqrt{n}/A$ , the integer step's walk over  $n$  intervals against the amplitude  $A$ . The first is  $2 \times 10^{-23}$  at nature's pair; the law's derivations are of the line without the division. The body's rows carry only what is computed: its definition (Section 7.1), the static fixed point and the modes by hand (S.46, S.60), the stability conditions as derived (Section 7.2) and its clicks (Section 7.4), and no sentence that a body stands under Rule3. Whether a laid body stands under Rule3 within that walk is the repository's gate and not this paper's claim [2, the gates].

**The atom: known from clicks, laid on the lattice.** The atom is known from nature's measurements: its spectra, its lifetimes and its scattering. Nothing of it is seen on the lattice. What is tested is whether the way from the clicks to the lattice and back carries the atom. This is a supporting finding, with its state named (S.46, S.52). In the model, an atom is a bound body of the holder of the sign; a hydrogen nucleus, one quantum of the proton's family, writes the time angle  $\theta_0 = Z\alpha c/r$ , with  $c = 1/\sqrt{3}$ , into the sign holder's field by its Wronskian. Electrons, which are quanta of the matter family, read that angle into their rotation. The standing records that Rule3 makes in that angle are the atom's levels, Schrödinger's in the slow limit. With the nucleus's angle alone, the levels have Bohr's form,  $E_n \propto 1/n^2$  with  $a_0 = \sqrt{3}/(m^*Z\alpha)$  (DERIVED in the slow limit, CLICKS; S.46; `atom.py`). A record that reads its own sign field (S.46 (c)) has the law's form at large  $Z$  only,  $1 - 1.25/Z$ , and not hydrogen. Under line (1) of S.60 a record reads every sign field but its own, and hydrogen's lines are those of S.60. The holders of the content keep the self-read (line (2) of S.60, ASSUMPTION). The atom of this paragraph is derived and not run, its derivations standing [1, the law's words].

**What the atom's clicks give, and what they do not.** What a click at the atom gives is derived: the born light is at the resonance  $\omega_e - \omega_g$  and not at the energy-conserving frequency, with the share's difference booked; absorption erases the photon's empty wave inside the click's light cone; and there is no recoil, consistently, since the write conserves no momentum and the nucleus pinned by line (2) of S.60 has infinite inertia, so nature's recoil shift is outside the claim. Bohr's levels, the threshold, the lines and the two-mode binding are unchanged by it, and the cause of a click is outside the paper's claims. The lines are beats of two levels, and the mass defect is the click layer's reading of a frequency. The orbital Zeeman splitting has Landé's  $g_L = 1$  (DERIVED; supporting findings with their numbers in S.46; CLICKS for the levels' ratios, LATTICE for the records; `atom.py`). The electron and the nucleons are quanta of their families. An electron is one quantum, of energy  $T \sin \omega_e$  with  $\omega_e$  here the electron family's rest rotation, ten orders below the compact threshold, and nothing smaller than a quantum clicks. In the model, mass is the family's gap,  $T \sin \omega_s$  ( $s$  the family), bound or free; what decays is a body, from which a click at its front parts a quantum; a quantum itself never breaks, since it has no part that the click credits (DERIVED; CLICKS; `clicks.py`). The criterion is one line, a rule of reading: what nature splits with energy is read as a body on the lattice, and what no energy splits is read as a quantum of a family. A nucleus of several nucleons, the atom and the molecule are bodies by the one mechanism the law has for a body, a record standing in the well of a holder: the sign holder for the atom and the nuclear holder for the nucleus (ASSUMPTION) [1, the bound body is one Node; what nature splits is a body; the nucleus; no mass without a bound body; the atom is a bound body of the holder of the sign][3, the masses are families]. A nuclear holder that the electron reads is set against hydrogen's  $1S$  and muonic hydrogen, a comparison that supports excluding it (DERIVED for the shift's form, the exclusion a comparison; one derivation; CLICKS; S.46; `atom.py`). So the nuclear holders of Section 10.3 are read by the nucleon families and not by the electron, a selective read that is declared. The table of S.60 sets the atom's clicks, the lines and the thresholds, against nature's under the two lines decided for the law, with how each was derived; this is a supporting finding and no more. What the atom of the model does not give is listed with its roots in Section 10.3 and S.52 [1, the atom is a bound body of the holder of the sign].

### 7.4 A body's clicks

What a NodeDetector reads of a body follows from the body's record at its Nodes, from the readings of Section 4.3 and from the paces, with no further line. Every formula below is a derivation, with its status and its fence; none is the result of a run of the implementation, and one check script of the paper is named where it is used [1, what a body does, in the words of the clicks].

**A steady rotation writes no light.** Take a body at rest with the plane record  $\text{re} = A \cos \omega_b t$  and  $\text{im} = eA \sin \omega_b t$ , where  $e$  is its ellipticity, the ratio of the two parts' amplitudes (S.9). Its Wronskian is  $W = -eA^2 \sin \omega_b$  at every interval. So the source of the sign holder is constant, and its holder's level stands at its rest: no travelling events, no click. Rule3 keeps a body's weighted total Wronskian  $\sum_i W_i/p_i^2$  at static paces, the plain total where the paces are uniform, and keeps an isolated Node's exactly. So a one-Node source cannot breathe, and a body's breathing moves  $W$  between its Nodes. Its light is written by the second moment of its charge; the monopole is constant by the continuity identity exactly where the paces are uniform over the body, and departs at first order in their variation across it,  $2 \Delta p/p$  of the total, below  $10^{-9}$  for an atom in a laboratory well (S.42 (c)). The light born therefore carries the frequency of the breathing and not the rotation, and the click rate of a far NodeDetector is the share of that current through its front (S.42). So a symmetric body at rest clicks nowhere. A body is seen by what it absorbs and by what it writes when it breathes or moves (DERIVED; the Wronskian and the source LATTICE, the far NodeDetector's rate CLICKS; `body_clicks.py`) [1, light is born by the write].

**The invariant, and the share's drift.** For one Node of a body,  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega(t) a_{\text{now}}$  with  $\omega$  changing slowly keeps  $A^2 \sin \omega_b$  invariant (S.14). This is the discrete adiabatic invariant of Rule3's line. The form  $D = A^2 \sin^2 \omega_b$  and the share are not invariant; the recurrence stepped over 20,000 intervals with  $\omega$  lowered from 0.841 to 0.600 keeps  $D/\sin \omega$  within  $4 \times 10^{-5}$  while  $D$  falls by a quarter (COMPUTED, `invariant_check.py`; LATTICE; S.14). The share in quanta at a Node is  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b/T$  (Section 4.4). Under a moving well it therefore moves as

$$\frac{e_i(t_2)}{e_i(t_1)} = \frac{\sin \omega_b(t_2)}{\sin \omega_b(t_1)} \cdot \frac{p_i^2(t_1)}{p_i^2(t_2)}, \quad (13)$$

with no current through any Link. The first factor is the rotation's own change under the clock's pace; the second is the Node's weight in the share (DERIVED for the invariant, to the first order in the slow variation, and for the drift; LATTICE; S.14; `body_clicks.py`). In the model the invariant stands where nature's number of quanta,  $E/\omega$ , stands, and the share stands where nature's energy stands. The same identity is Planck's relation on the lattice, share =  $I \sin \omega$  with  $I = A^2 \sin \omega / T$ . So  $E = \hbar \omega$  holds, with  $T$  for  $\hbar$  and  $\sin \omega$  in place of  $\omega$  (DERIVED; CLICKS; `body_clicks.py`) [1, the count is the record's share; the paces' own change moves the share with no current].

**The clock's click rate.** A bound body's rotation reads the content once. At rest  $\cos \omega_0(U) = \cos \omega_0 + 2gU$  to first order, so its rate falls by  $f(\omega_0) U$ , with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ : 1.063 at [2, 3] and 1 as the gap closes. In series,  $f = 1 + \omega_0^2/12 + \dots$ ; this is the finite-rotation correction in  $\omega$  of a rate that is exactly  $N$  in  $2 \sin(\omega/2)$  (S.31). A light clock between two faces falls by  $2U$ , the Link's pace read twice (Section 10.2; S.15) (DERIVED, CLICKS; `weak_field.py`).

**The resonance.** A bound record at the rotation  $\omega_b$  that reads a light level into its content is modulated by  $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$ , with  $x_0$  the body's content in units of  $\Gamma$  and  $b_L$  the light level's amplitude. The resonant part at  $\omega_L = 2\omega_b$  grows the amplitude at  $\mu = \epsilon/(4 \sin \omega_b)$  per interval, inside the full width  $\epsilon/\sin \omega_b$ . A body with two bound modes  $\omega_i < \omega_j$  transfers between them at the difference  $\omega_j - \omega_i$ , at Rabi's rate  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin \omega_i \sin \omega_j})$  (DERIVED, two derivations; S.16; CLICKS; `body_clicks.py`). Under light's declared read, the holder's rotation (Section 5.5), the Link phases couple the modes through the inter-mode current. The coupling is resonant at the difference line for drives below  $\omega_j + \omega_i$ , with Rabi's rate in the small-line form  $\Omega_R = \theta_L \omega_L |d|/2$ ,  $\theta_L$  the light's angle read on the Link (S.43's phase) and  $d$  the two modes' coupling element (S.43); it oscillates at every detuning with the envelope  $\Omega_R^2/(\Omega_R^2 + \Delta^2/4)$ . At the sum  $\omega_j + \omega_i$  the same coupling drives the two senses together. That is a pair line, outside the absorption and emission of this section (the atom's lines at optical frequencies lie far below it). The one-mode line at  $2\omega_b$  has no analogue there (DERIVED; S.43; CLICKS; `body_clicks.py`). The read's side is computed; the fraction absorbed, with the body's write back, is open (LATTICE for the record, CLICKS for the body's share) [1, the rows against nature, (y); the two-mode line].

**A body's count in clicks.** A body's count in clicks is its share summed over its Nodes, what a NodeDetector over all its Nodes credits per window (its clock's click rate, S.15); the credit books the conserved form's own current through the front, the Link's factor squared its one weight (the Node's  $1/p_i^2$  cancels against the read's  $p_i^2$ ), so what it counts is the conserved form's flux, exact at paces fixed in time; a gapped quantum's share in a well differs from  $W_{\text{rec}}$  by the rest term's shift  $4U(\text{den} - \text{num})/\text{den}$ , the law's physics read as the count (the mass defect, Section 10.3), and where the paces move in time the share changes by the work term, the body's and not the NodeDetector's, printed as the books' drift against the share (S.42, inputs; the law's line of the weighted reading; no step of Rule3 in it). Under a moving well it drifts by Eq. (13) (DERIVED, LATTICE for the drift; `body_clicks.py`). A declared region with Nodes alone credits the share: a bolometer's reading, Millikan's slope 0. What registers in nature is a body, and the model declares the click as one quantum's change in the NodeDetector's own record. This is the NodeDetector's own declaration, its record's parts, beside its Nodes, its own seed and the file's draw window (ASSUMPTION) [1, the whole of a click is the detector's own transition]. The NodeDetector's record answers at its rotation by the two-mode line above (S.43), one transition per click. The transition takes the share  $T(\sin \omega_j - \sin \omega_i)$ , which is one photon's share  $T \sin \omega_L$  to the order  $\omega_i \omega_j$ , below  $10^{-18}$  at nature's gap. So a body counts photons and reads  $\hbar \omega$  per click, Millikan's slope  $T$  (DERIVED; S.14; CLICKS; `body_clicks.py`). The share with the paces' weights stays the conserved form; the credit's reading at the vacuum's coefficients is the draw's measure; the two are one number at the vacuum's paces and differ by the Node's weight in a well, so a ratio of the checks is unchanged where its two records stand at one pace, a common unit cancelling (DERIVED, CLICKS; `clicks.py`). The credit is in the record's own unit  $W_{\text{rec}}$  (Section 5.3; CLICKS) [1, the count is the record's share].

## 7.5 Clusters: how they enter, and what each thing represents

**How a cluster enters.** The model writes no line for a cluster. Every body is bound and records may be free: a body is a record standing in a holder's well, so every body is bound; a record in no body is free, laid by the file or written by a click, and it gravitates by its form, falls as every family falls and clicks only where it meets a declared NodeDetector's transition (a definition of the law; the derived statement of the next paragraph). A cluster is bodies standing in one another's wells. Between two bodies there are the click, the light of the sign holder, the two real holders and the free records in flight [1, the bound body is one Node; every body is bound; records may be free; the algebra of clusters].

**Dark matter in kind.** What nature defines as dark matter, gravitating and unseen, is in Universe24 every record that meets no NodeDetector's transition: the free quanta of the gapped scalar matter family, with a rest rotation and the slow fall, and the bound bodies of a family no NodeDetector we declared meets. Each writes the content by its form (Section 3.5) and clicks nowhere, since a click is a meeting at a declared transition (Section 5); such a quantum has  $W = 0$ , reads no holder of the sign, writes and scatters no light, and is credited only by a NodeDetector whose record holds a transition for its family, and it cannot change family or decay on its own until a body holding a transition for its family absorbs it (Section 5.4). The massless scalar quanta a conversion lays free are dark radiation, free and uncredited, their kind derived; they are named as dark radiation and not as dark matter. The kind is derived under the universe's table as declared; its source, the cooling's shed quanta, is an explicit hypothesis; its amount and its gathering are not derived, and the flat rotation curves remain an explicit finding. Its three predictions in clicks: the universality of free fall, no light written or scattered, and a direct detection that names the transition it meets. The paper claims no identification with nature's dark matter: this is a supporting statement of what the model holds and how clicks would read it (DERIVED under the universe's table as declared, `dark_matter_check.py`; the source HYPOTHESIS; CLICKS) [1, dark matter in kind is a theorem of the lines].

**The clicks between bodies are clicks in the one sense.** A body is a region of Nodes with a front: the Ports where its tail meets its neighbour's. So a body declared a NodeDetector is one in the meeting's form; a body with no declaration does the Nodes' laws alone (Section 5.4). A click is not the NodeDetector's privilege: every body absorbs, emits and converts by the same meeting, a supernova's as a screen's; the NodeDetector is the one body whose clicks the run reports, and a body with no declaration clicks by the Nodes' laws alone, what draws there being open (Section 10.3). The click between two bodies is the pair  $(R, t)$ , one whole quantum credited to the region of the record that absorbs at the interval  $t$ , drawn by the share  $Q_R/W_{\text{rec}}$  of the inflow through its front on the forward orbit ( $Q_R/W_c$  for a laid record, Section 5.3). What a

NodeDetector credits of a body is its share summed over its Nodes. It moves only by the clicks at its front, up to the drift of Eq. (13) (DERIVED from the definitions; CLICKS, the drift lattice; `clicks.py`) [1, the click's algebraic definition; the click is the meeting of the future with the past].

**What each thing represents in a cluster.** The three levels at a Node are the nesting. The binding holder's level is the matter's tail, the sign holder's level is the charge's well, and the massless holder's level is gravity's well. These are three kinds of binding by rank: quanta into a body, bodies into a charged cluster, charged clusters into a gravitational cluster (DERIVED, LATTICE). The couplings are at the edge: every divisor is 1 but gravity's  $E_g$ , and the universe of clusters carries  $\Gamma$ ,  $T$ , the two holders' weights and the families' pairs, and nothing else (ASSUMPTION for the divisors) [1, the levels at a Node are the nesting; three kinds of binding, not three levels; the hierarchy is recursive; the click is the pointer; the observer is a cluster; the couplings are at the edge; what a body is; the rule's own universe].

**The clusters' cosmology.** The universe of the model is a finite ball of bodies flying through the static lattice. Every body is pulled by the inverse square of the others at Kepler's  $G_K$ , Newton's constant as the orbits read it (Section 10.1, after Eq. (20); Section 8, (j)). A shell of dust at the radius  $a$  feels the mass inside it, so  $\ddot{a} = -(4\pi G_K/3)\rho a$  and  $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - K/a^2$ . These are Friedmann's dust equations, exactly, for the ball's interior [53], with  $K$  the start's declaration, positive for a bound ball; the ball's bound part decelerates either way. A receding source's redshift is  $1 + z = 1 + \beta$  at first order: Doppler's and Hubble's known forms, derived here. The two derivations, the Doppler period of Section 8's row (m) times the moving clock of its row (e), agree exactly (DERIVED, two derivations; CLICKS; S.29; `cosmology.py`). The paper's cosmology is claimed at  $z \ll 1$  and not beyond. The second order, where the deceleration is read, is named among the findings of Section 10.3 [1, the clusters' cosmology].

## 8 The forms the rule arrives at, set beside nature's

Each form below follows from the rule's coefficients, read by the geometric optics of a packet's band, before any run. The form of nature beside it is never an input, except for the one assumption of Section 3.4. The coefficients contain no metric, no Lorentz transformation, no potential and no force. They contain only the paces of Section 3.4: the clock composes by the rule, and the Link is the clock twice by the one assumption.  $U = \ell/\Gamma$  denotes the model's potential at the point named, the clock's share per level, and  $U_b$  its value at the impact parameter  $b$  [1, the rows against nature].

**Two potentials.** The forms are stated in two potentials. The model's own potential is the clock's share,  $U = \ell/\Gamma$ , the content level  $\ell$  that the paces read; no NodeDetector reads it. Every test of gravity takes  $GM$  from Kepler's law of the orbits of matter and  $c$  from light, so it reads  $U_K = G_K M/(c^2 r)$ . In the model a free packet of matter of the pair [num, den] falls in a content gradient at  $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla U|$ , exactly from its band, Eq. (4) (S.18). So the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes.} \quad (14)$$

Each form below is lettered as the law's rows are. It is written first in  $U$ , where the composed paces' metric gives, in the post-Newtonian words,  $\gamma = 1$  and  $\beta = 1$  exactly, and an orbit at the quadratic band reads them as the post-Newtonian parameters; the metric is  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , and the redshift's factor  $f(\omega_0) = 1 + \alpha_U$  with  $\alpha_U = 0.063$  at [2, 3]. It is then written in  $U_K$  for  $0 < \text{num} \leq \text{den}$ , where the map  $U_K/U = 2\text{num}/(\text{den} + \text{num})$  inverts (0 at [0, 1], no parameter) and the same algebra gives  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$  and  $1 + \alpha_K = \tan \omega_0/\omega_0$ : 1.5, 1.25 (the small-gap closed form under the quadratic band; the lattice band's advance is in (d)) and 1.329 at [2, 3], and 1 as the gap closes; the computed pair's values are the lattice's illustration of the band at a coarse gap (LATTICE), and nature's rows stand beside the formulas at nature's pair alone. The two columns are one algebra, and nature's tests read the second (DERIVED for the formulas, CLICKS; S.18, S.19; `weak_field.py`) [1, the rows against nature].

- (a) *The redshift* (DERIVED, CLICKS; `weak_field.py`). The clock's pace is  $p_0/\Gamma = (1 - 1/\Gamma)^\ell = e^{-U}$  exactly, with  $U = -\ell \ln(1 - 1/\Gamma)$ , which is  $\ell/\Gamma$  to the first order in  $1/\Gamma$ . So the clocks of two Nodes tick in the ratio  $e^{-(U_1 - U_2)}$ ,  $1 - (U_1 - U_2)$  to first order. The rest rotation of a bound record, the clock this row names, shifts by  $f(\omega_0)U$  (Section 7.4):  $1.063U$  at [2, 3] and  $U$  as the gap closes. Against the Newtonian potential that the orbits measure, it is  $z = (1 + \alpha_K)U_K$ , 1.329 at [2, 3], where nature's clocks give  $|\alpha| < 2 \times 10^{-4}$  [38]. Beside it: Pound and Rebka's line [57][38].
- (b) *The bending of light* (DERIVED, CLICKS; `weak_field.py`). At a Node whose clock's share is  $U = \ell/\Gamma$  (the content level over  $\Gamma$ , Section 3.4), light's index is  $n = e^{2U}$ ,  $1 + 2U$  to first order, because the Link's pace reads the content twice. Far from a body, its level is the massless holder's rest,  $\ell(r) = 3s/(4\pi r)$ , where  $s$  is the body's write per interval over the holder's divisor  $E_g$ . A beam passing at the closest distance  $b$  turns by the transverse gradient of the index, integrated along the path (S.20):

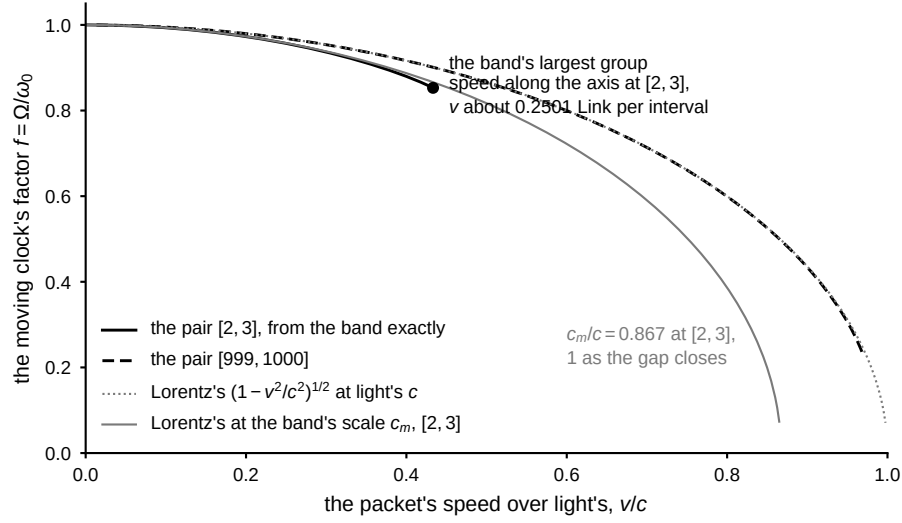
$$\alpha = \left| \int_{-\infty}^{\infty} \frac{\partial n}{\partial b} dl \right| = 4U_b, \quad U_b = \frac{3s}{4\pi\Gamma b}, \quad (15)$$

its magnitude, the deflection toward the source (S.49 (33)), twice the angle  $2U_b$  that a Newtonian pull of the same potential gives. The factor two is the ruler going with the clock,  $hN = 1$ , the one assumption of Section 3.4. Against the Newtonian potential that the orbits measure, the bending is  $(1 + \gamma_K) 2G_K M/(c^2 b)$  with  $\gamma_K = \text{den}/\text{num}$ , 1.5 at [2, 3], where Cassini's tracking gives  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$  [38][58]. Beside it: Einstein's  $4GM/(c^2 b)$  [38]. The bending is the  $\beta = 1$  case of the push on a moving record:

$$\mathbf{F} = \frac{E^2 + P^2}{E} \nabla U = E(1 + \beta^2) \nabla U, \quad \beta = \frac{P}{E} = \frac{v}{c_{\text{local}}} \quad (16)$$

with  $E = \omega$ ,  $P = cN^2 k$  the momentum at the composed paces, and  $\beta$  the record's speed over the local light speed  $cN^2$ ; Eq. (16) is the small-gap long-wave form of Eq. (4): at rest in the vacuum the exact pull is  $-d\omega/dU = 2 \tan(\omega_0/2)$ ,

- 0.8944 at [2, 3] against  $E = \omega_0 = 0.8411$ , the ratio the clock's  $f(\omega_0)$  of (a). At rest the push is  $E \nabla U$ ; for light it is  $2E \nabla U$ , and Eq. (15) is its  $\beta = 1$  case. Its bending is exact in the gap, while the orbit's second order at the computed pair carries the band's  $k^4$  term (S.19) (DERIVED from Eqs. (3) and (4) under the one assumption, fixed by one click; S.58; CLICKS; `weak_field.py`) [1, the push on a moving record]. The paper's momentum is the wave number  $\mathbf{k}$ , so the lattice form is  $(E^2 + c^2 \mathbf{k} \cdot \mathbf{k})/E$  with  $c^2 = 1/3$ . Written with the velocity  $\mathbf{v}$  in Link per interval,  $\mathbf{p} = E\mathbf{v}/c^2$ , it is  $E(1 + 3v^2)$  at  $N = 1$ , since  $\beta^2 = v^2/c^2 = 3v^2$ ; the post-Newtonian factor  $1 + \gamma\beta^2$  has  $\gamma = 1$  here, derived under the one assumption of Section 3.4. At the computed pair the coefficient of that  $\beta^2$ , the record's speed over light's, is  $\gamma_K - \alpha_K/2 = 1.335$  and the rest push is  $f(\omega_0)E$ ; read instead against the band's kinetic scale,  $\beta_v = v/c_m$ , the same push has the coefficient 1.0045, the two equal through  $f(\omega_0)$  and  $U/U_K = (1 + \cos \omega_0)/(2 \cos \omega_0)$ . The two readings of  $\beta$  in the box,  $P/E$  with the covariant variable  $\mathbf{\Pi} = E\mathbf{v}$  as S.58 writes the push ( $P = \mathbf{\Pi}/(cN^2)$  at the composed paces), and  $v/c_{\text{local}}$ , agree as the gap closes. As physics it is derived here, and it coincides with the known post-Newtonian deflection factor  $1 + \gamma\beta^2$  at  $\gamma = 1$ , derived under that assumption and not declared separately. Nature is met by its click formula, the ratio  $1 + \beta^2$  of the deflection angle to Newton's (CLICKS).
- (c) *The delay of light* (DERIVED, CLICKS; S.20; `weak_field.py`). Light's coordinate speed at the content  $U$  is  $e^{-2U}/\sqrt{3}$ , so its index is  $1 + 2U$  to first order. Take a signal grazing a body at the impact parameter  $b$  from  $x_1$  to  $x_2$ , with  $U = \mu/r$ ,  $\mu = GM/c^2$  in Links and  $b$ ,  $x_1$ ,  $x_2$  in Links. It is late by  $2\sqrt{3}\mu \ln(4x_1x_2/b^2)$  intervals: Shapiro's delay  $(1 + \gamma)(GM/c^3) \ln(4x_1x_2/b^2)$  at  $\gamma = 1$ . Beside it: Shapiro's delay [59].
- (d) *The perihelion* (DERIVED, CLICKS; `weak_field.py`). A slow body's orbit in the composed clock's metric,  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , is the post-Newtonian orbit with  $\gamma = 1$  and  $\beta = 1$  exactly in  $U$  by the metric, read at the quadratic band, the small-gap limit. So the periapsis advances by  $6\pi\mu/(a(1 - e^2))$  per orbit, with  $U = \mu/r$ ,  $\mu$  in Links,  $a$  the semi-major axis in Links and  $e$  the eccentricity. This is Einstein's value, which at nature's pair is Mercury's 43 arcseconds per century (a comparison). At the computed pair the band's  $k^4$  term, of the same order as the relativistic one, adds of the order of ten percent to the advance; the addition depends on the orientation and vanishes as  $\tan^2 \omega_0$  (DERIVED at the small gap; the quadratic band an ASSUMPTION at the computed pair; S.19, S.34; CLICKS). Against the Newtonian potential that the orbits measure, the advance is  $(2 - \beta_K + 2\gamma_K)/3$  of Einstein's, with  $\beta_K = (\text{den} + \text{num})/(2\text{num})$ , 1.25 at [2, 3], the small-gap closed form under the quadratic band (with the lattice band the advance gains of the order of ten percent, orientation dependent, and no single  $\beta_K$  stands for it). In nature the planets' and the Moon's orbits give  $|\beta - 1| < 10^{-4}$  [38]. Beside it: Einstein's  $6\pi GM/(c^2 a(1 - e^2))$  [38].
- (e) *The moving clock: the band's kinetic scale and the fourth order* (DERIVED, CLICKS; S.21; `moving_clock.py`). The moving clock's factor of a packet moving at the velocity  $v$  is  $f_v = \Omega(k)/\omega_0$ , where  $\Omega(k) = \omega(k) - k\omega'(k)$  is the rotation at the moving centre. From Rule3's exact band, for a packet moving along an axis,  $f_v = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0) v^4/(8c_m^4) + O(v^6)$ , with  $c_m^2 = \omega_0/(3 \tan \omega_0)$  the band's kinetic scale. The band's  $k^4$  term is anisotropic, so the fourth-order coefficient over Lorentz's along the unit vector  $\mathbf{n}$ , where the group velocity is parallel to  $\mathbf{k}$  (the axes and the two diagonals), is  $\omega_0 [(\sum_a n_a^4) \tan \omega_0 + \cot \omega_0]$ : 1.6926 along an axis at [2, 3], 1.2224 along a face diagonal, 1.0657 along the body diagonal, and 1 in every direction as the gap closes (Fig. 5, the axis). The kinetic scale  $c_m^2/c^2 = \omega_0/\tan \omega_0$  ( $1 - \omega_0^2/3$  at a small gap) is 0.752 at [2, 3], so  $c_m$  is 13 percent below light's. The band's largest group speed, along the cube's diagonal at  $\mathbf{k} = (\pi/2, \pi/2, \pi/2)$ , is  $c \cos \omega_0$ ,  $2/3$  of light's at [2, 3] (S.21); along an axis it is about 0.2501 Link per interval, close to  $\sqrt{3}/4$  of light's. At nature's matter pair  $c_m$  lies below light's by  $\omega_e^2/6$  and the diagonal's largest group speed by  $1 - \cos \omega_e = \omega_e^2/2$ :  $7 \times 10^{-29}$  and  $2.0 \times 10^{-28}$  at the MAGIC bound along an axis,  $2.2 \times 10^{-20}$  and  $6.7 \times 10^{-20}$  at LHAASO's (Section 10.1). The model's electron is subluminal. Nature bounds the difference of matter's and light's limiting speeds with its sign: the subluminal case at  $6 \times 10^{-20}$  by the Crab's synchrotron [62] (the cubic term bounded by [63]), the superluminal at  $5 \times 10^{-21}$  [64], and the earlier bounds at  $10^{-15}$  from photon decay to  $5 \times 10^{-23}$  for the proton from the absence of vacuum Cherenkov radiation [60]. The subluminal bound, where it holds, binds  $\omega_e$  near  $10^{-10}$ . Beside it: Lorentz's factor, reached as  $c_m \rightarrow c$  (S.21; Fig. 5).
- (h) *De Broglie's relation* (DERIVED; LATTICE for the band, CLICKS for the fringe spacing that reads it; S.22; `bands.py`). A free record of a matter family has the band  $\cos \omega = (\text{num}/(3 \text{den}))(2 + \cos k)$  and the group velocity  $v = (\text{num}/(3 \text{den})) \sin k / \sin \omega$ . To second order,  $k = \omega_0 v/c_m^2$ , with  $c_m$  the band's kinetic scale of (e). Beside it:  $\lambda = h/p$  with the rest rotation as the mass.
- (i) *The fall* (DERIVED, CLICKS; `weak_field.py`). In a pace gradient a free packet or a cloud accelerates toward the well at  $a = \Delta |d\omega_0/d\ell| (\ell_+ - \ell_-)/L$  Link per interval squared, where  $\Delta = 1/m^*$  is the band's curvature at rest,  $d^2\omega/dk^2$  at  $k = 0$ , in Links squared per interval and  $\ell_{\pm}$  is the content level at its two faces,  $L$  Links apart. So bodies with the same  $\Delta d\omega_0/d\ell$  fall alike. Beside it: Galileo's fall [38]. The binding energy writes a well as in nature, the active mass (lunar ranging's Nordtvedt test and Kreuzer's active-against-passive test beside it); the fall is the pace's and universal within a family whatever sources the well (Eötvös, Eöt-Wash and MICROSCOPE beside it); between families the law's statement is S.47's  $\eta = 32.3\omega_e^2$ . The seat of the electromagnetic binding energy is open (DERIVED; CLICKS; S.56 (12), S.38). Between families: a free packet of the pair [num, den] falls at  $c^2(1 - \tan^2(\omega_s/2)) \nabla U$  exactly, so two families of different gaps fall apart by  $(\omega_n^2 - \omega_p^2)/4$  at small gaps. With the proton and the neutron as two families, beryllium against titanium gives  $\eta = 32.3\omega_e^2$ , below  $1.3 \times 10^{-26}$  at the MAGIC bound ( $\omega_e < 2.0 \times 10^{-14}$  along an axis, from MAGIC's  $E_{\text{QG},2} > 6.3 \times 10^{10}$  GeV in the quadratic form, S.24) and  $4.4 \times 10^{-18}$  at LHAASO's. That, a free-packet estimate (S.47), is under nature's  $10^{-13}$  for that pair [38] and  $10^{-15}$  for titanium against platinum [37] (DERIVED, CLICKS; S.47; `weak_field.py`) [1, the rows against nature, (i)].
- (j) *Newton's pull and orbits* (DERIVED, CLICKS; `weak_field.py`). The massless holder rests at Poisson's solution with the sources, and at Laplace's outside a body: the content level is  $\ell(r) = 3s/(4\pi r)$  levels at  $r$  Links, where  $s$  is the body's write per interval over the holder's divisor. So the acceleration falls as  $1/r^2$ , and a circular orbit's  $v^2$  as  $1/r$ . Beside it: Newton's pull and Kepler's fall-off.
- (m) *A free record keeps its wave number* (THEOREM: Rule3 is invariant under the lattice's translations, the next sentence its



**Fig. 5** The moving clock's factor from the band (Section 8, (e); DERIVED, CLICKS; drawn from the formula, no run; `moving_clock.py`). A packet of a matter family moving at the velocity  $v = \omega'(k)$  along one axis ticks at the rotation at its moving centre,  $\Omega(k) = \omega(k) - k\omega'(k)$ , so its clock's factor is  $f_v = \Omega/\omega_0$ . It is drawn exactly from the band for the pair [2, 3] (solid), up to the band's largest group speed along the axis, about 0.2501 Link per interval (the dot), and for the pair [999, 1000] (dashed). Beside them are Lorentz's  $\sqrt{1 - v^2/c^2}$  at light's  $c = 1/\sqrt{3}$  (dotted) and at the band's own kinetic scale  $c_m$ ,  $c_m^2 = \omega_0/(3 \tan \omega_0)$ ,  $c_m/c = 0.867$  at [2, 3] (grey). The computed pair's clock follows Lorentz's at  $c_m$  to the second order, and the small-gap pair's clock follows Lorentz's at  $c$ . The factor is a property of the clicks; no boost is a symmetry of the lattice

proof; LATTICE). Rule3 is invariant under translation where the paces are uniform over the lattice. So a free record there keeps its wave number and rotation exactly in the rational line before the rounding (the integer step's remainders add their walk, Section 9.2), and a radiating body's period is read unchanged at every distance. Beside it: Hubble's law as Doppler clicks of matter flying apart through the static lattice, with no stretching of space (Section 7.5).

- (p) *The lens: a content bends light and matter alike, by one formula* (DERIVED, two derivations; LATTICE for the index, CLICKS for the focus and the clicks; S.28; `weak_field.py`). A stationary boundary conserves the rotation. So, for every family, a record of wave number  $k$  entering a content  $U$  takes the wave number  $k_{\text{in}}$  given by

$$1 - \cos k_{\text{in}} = \frac{1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}}{N^4}, \quad N = e^{-U} \quad (17)$$

light's index  $k_{\text{in}}/k \rightarrow e^{2U}$  at long wavelength (nature's  $n = 1 - 2\Phi/c^2$ , recovered; the one formula for light and matter with its chromatic term is the paper's own, listed among its additions in Section 11). For light it is exact in the law's own variable as  $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$  (matter keeps the rest term of Eq. (17)), with the lattice's chromatic term  $k^2(e^{6U} - e^{2U})/24$  at finite  $k$ . Slow matter's index is  $n_m^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$ , growing as  $1/k^2$  (nature's  $n^2 = 1 - 2\Phi/v^2$ , with the Kepler column's ratio at the computed pair, Eq. (14)). *What it does*: a slab of content bends every record toward its thick side, light a little and slow matter much. The fall of a body is the refraction of its own record. A slab of content thick in the middle is a lens, and it gathers the record's share at a focus. *One quantum through it* (Section 5.4): the whole record passes the lens, and its share gathers at the focal spot. The one quantum clicks once, at one Node of the spot. The NodeDetector draws that Node with the world's draw over the spot's shares and writes there: the levels that arrived are absorbed, and the count goes down by one. The next quantum draws again. The spot is the accumulation of the clicks, one per quantum, the image built click by click, as nature's single-photon images are (the formula derived from Eq. (4) at the composed paces; the one click per quantum the credit's declaration; the write at the one Node the NodeDetector's declaration) [1, the content's index for every family; the rows against nature, (p)].

**What goes in and what comes out.** The model puts in one rule, the paces' read under the one assumption of the forms (Section 3.4), the credit and the declarations listed in Section 10.3 and named where they stand, and the numbers of the files. In the model's own potential the bending (b) and the delay (c) are Einstein's exactly, and the perihelion (d) is Einstein's at the small gap;  $\gamma = \beta = 1$  in that column as the gap closes, light's ray metric. The post-Newtonian column at the computed pair departs with the gap. So at the rule's own matter pair the model does not describe the gravity of the solar system. The closing gap is nature's pair, where  $\gamma_K - 1$  is below  $2.0 \times 10^{-28}$  at the MAGIC bound and  $6.7 \times 10^{-20}$  at LHAASO's (Section 10.1); whether the model's bodies stand at that pair is not shown. Read as a metric, the composed clock gives  $N^2 = e^{-2U}$  and the ruler  $h^2 = e^{2U}$ , against the isotropic Schwarzschild chart's  $1 - 2U + 2U^2$  and  $1 + 2U + 1.5U^2$ : the spatial second order is 2 against 1.5, a prediction of the model that no measurement has read; one measurement in the deep level would read it (a comparison with the chart, no click claimed; CLICKS; S.34). Light's index  $e^{2U}$  places the photon sphere at  $r = 2m$  with  $U = m/r$ ,  $m = GM/c^2$  in Links (S.30), so the capture radius of a black hole's shadow is  $b = 2em = 5.44m$ ,  $e$  Euler's number, against Schwarzschild's  $3\sqrt{3}m = 5.20m$ , 4.6 percent larger, an explicit prediction (DERIVED under the one assumption, fixed by one click; CLICKS; S.30; `weak_field.py`). Beside it: the Event Horizon Telescope's shadows. Sgr A\*'s fractional deviation from Schwarzschild is  $-0.08 \pm 0.09$  [65], 1.4 standard deviations from the prediction under the Very

Large Telescope Interferometer mass prior, and M87\*'s is  $-0.01 \pm 0.17$ , about 0.3 standard deviations [67] (the shadow of [66]). Rule3's step times the record's conjugate gives an exact continuity identity at every Node and interval for any paces; summed over the body the conserved four-current is the weighted one, the Node density  $wW_i/p_i^2$  and the Link flux with  $q_{ij}^2/\Gamma^2$ , the plain pair at uniform paces (S.33); and the time turn with the three Link phases is one covector by minimal coupling. So the odd lines are sourced by the halved current over the same wall as the time level, with no number matched, and a co-moving reader feels Coulomb's force over Lorentz's  $\gamma_L$  (DERIVED; CLICKS; S.33; `sign.py`). A source and a reader of one family see one light speed, the band's own ( $c_s^2 = \text{num}_s/(3 \text{den}_s)$ ) in the variable  $2 \sin(\omega/2)$  and  $c_m^2$  in  $\omega$ , one as the gap closes; the family's Lorentz factor  $1/\gamma_s$  uses  $c_s$ ). In nature light's  $c$  is the same to  $10^{-28}$  at the MAGIC bound and  $10^{-20}$  at LHAASO's (DERIVED for the band's one speed, the bound a comparison; CLICKS; `constants.py`) [1, the odd line's source is the sign's own current; the rows against nature, (b2)].

## 9 The two slits and Bell: the one computed file and the derived check

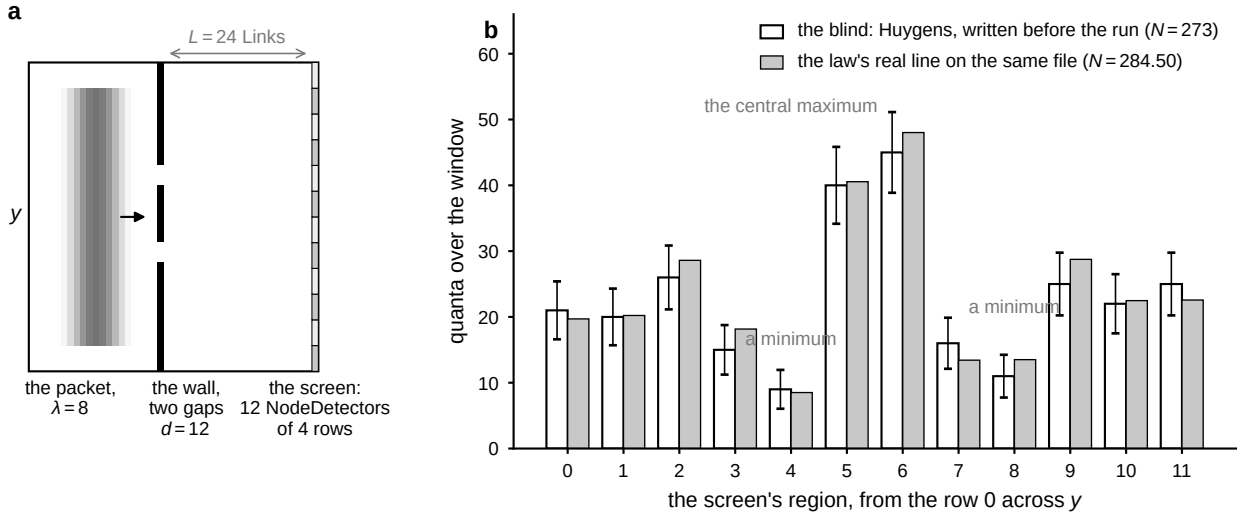
The paper contains no experiment against nature. The files below declare the Universe24 it computes; the two slits' row is read by the implementation's one run, and Bell's value is derived from the declared credit and from no run (Section 9.4). The two slits is the one computed file set against a blind written before its run. Bell's file is a check of the declared credit and not a test of it: its number is fixed by the credit and by the determinism of the parts. Nothing in a file's lay, geometry or NodeDetectors is built to bring the right result. The credit of Bell's reader is the declared and fitted exception, and Section 9.4 states it as such [3, Every run is blind first][1, one experiment and one gate].

### 9.1 The two slits: the file

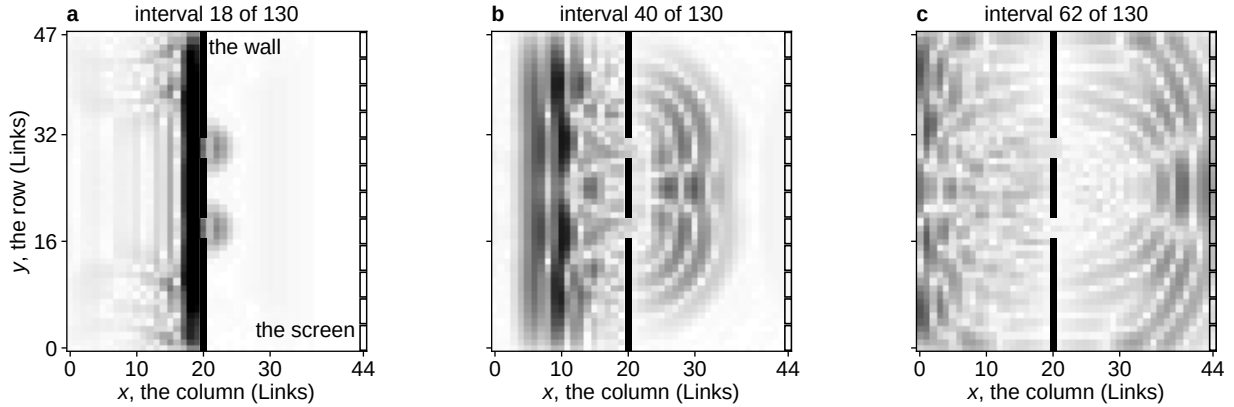
The source is a declared packet of light. Its wavelength is 8 Links along  $+x$ ,  $k = \pi/4$ . Its levels are a raised-cosine packet of about ten quanta per Node, with the peak level  $b = 1,241$ , over  $x = 4$  to 16, flat across  $y = 4$  to 44, 1,809 quanta, its before level the band's own exact level mode by mode at the declared vacuum content's paces (the law's lay tool). The wall is the inner face at  $x = 20$  with two gaps of three Nodes,  $y = 17$  to 19 and 29 to 31,  $d = 12$  Links apart. The lattice is flat,  $45 \times 48 \times 1$  with the third axis folded, its  $y$  faces open and its  $x$  faces receding. The screen is the last column,  $x = 44$ ,  $L = 24$  Links from the wall. It is twelve declared NodeDetectors of four rows each (Fig. 6a); the file declares a thirteenth, **gap**, a bare region of sixteen Nodes at  $x = 21$  to 24 and  $y = 16$  to 19 behind the first slit, backed by no face: a packet passes through it, its net inflow over the window is nothing (the real line's  $-0.58$ ), the credit floors it at 0 and never draws it, it reports no click, and its per-interval inflow is the reading the blind names **through**, beside the screen. The run is 130 intervals, read over the intervals 45 to 130. Fig. 7 shows the record at three intervals of that run, a lattice reading and no measurement: the run is Rule3 at every Node from the file's lay, with the wall's face and the screen's NodeDetectors the file declares, and nothing else; the one measurement is the screen's row (Fig. 6b). A record uniform along the folded axis has the three-dimensional band at  $k_z = 0$ , so the flat lattice carries the file's band exactly (DERIVED, LATTICE; `bands.py`). On this file the law's line gives 284.7 units of  $W_c$  through the screen in the physical labels (284.5 in the implementation's labels; 278.9 when the laid record's own integers, laid at the content 60's paces, are stepped by this line at the content 0's). The peak of the screen's inflow is at the interval 69.0, against 62.0 on the same file with the lattice refined fourfold (62.1 eightfold) (COMPUTED, CLICKS; `two_slits_real_line.py`) [1, the rows against nature, (g)][3, the two slits][1, one experiment and one gate].

### 9.2 The two slits: what the model says it must give

The one file is read twice before its run's clicks are set beside it: by the known forms, Huygens's sum and Young's spacing, applied to the geometry of the declared file and written before the run, and by the law's real line computed beside it. (a) *The fringe*. The spacing of the far field is  $\lambda L/d = 16$  Links. At  $L = 2d$  the fringe is the near field's, with the minima near the lines  $y = 15.3$  and  $32.7$ , in the regions 3 and 8 of the twelve four-row blocks counted from 0; the blind's regional sums dip at the regions 4 and 8 (the blind's minima), the pattern's centre, the line  $y = 24$ , being no mirror of the blocks' 23.5 (DERIVED, CLICKS; S.23; `two_slits.py`). Light's line at the vacuum's paces, stepped in real arithmetic on the declared file at its own lay, gives 284.7 units of  $W_c$  through the screen (Section 9.1), region by region as the script's table lists, and the integer step's deviation from the line is bounded exactly per step (the law's budget clause: below one level per Node per interval; over the run the carried remainders form a difference sequence, and summation by parts bounds the error per laid mode by  $(n-1)/\cos(\omega/2) + 1/\sin \omega + 1$ , about  $n$  at long wavelength and 135.6 at the carrier at  $n = 130$ , the band's two ends excluded by the division act's two sums and by the loader's refusal of the even periodic box) and walks, under the model of independent uniform remainders, as  $\sqrt{n/(12(1+\cos \omega_k))}$  per band mode, 2.4 levels at the carrier and 3.0 per Node over the lattice's zone at  $n = 130$ , against  $b = 1,241$ , 0.4 to 0.5 percent of the share and 1.1 to 1.4 units of  $N$  (COMPUTED, CLICKS; `two_slits_real_line.py`, the script of Section 3.7). The Huygens sum of the two gaps gives 273 with the same central fringe, a reference form (Fig. 6b). The run at that commit,  $N = 285$  in the region's unit  $W_c$  at both seeds (674 photons at  $\omega = 0.4366$ , a lattice reading, S.61), stands within that walk of the line's 284.7 and 12 units over the Huygens blind, the exact lay's own number and no scatter's, the run the check of the line and not the claim; the sum of the absolute differences of the seed-24 clicks from the blind's counts is 28 of 275 (275 the blind row's sum by its per-Node rounding, 273 its  $N$ ) and from the line's row apportioned at 285 by the largest remainders 32 of 285, one draw's scatter (the largest residual  $+1.3$  scatters at the central region 6, 56 against the line's 48.0), the law's real line's own shares from the blind's by 0.076, and the maxima fall at the blind's regions 2, 6 and 9, the clicks' minima at 1, 4, 7 and 10 against the line's 4, 7 and 10. This is a comparison and not a statistical test, since the blind is Huygens's sum and not a draw. (b) *The clicks*. The screen's shares, rounded to whole quanta, are the expectation. The row drawn by the shares at the declared seed is the clicks, one draw per unit credited, with the draw's scatter  $\sqrt{NP(1-P)}$  per region over the  $N = 285$  draws. Born's rule over the two seeds: the chi-square of the realised clicks against the window's shares on 11 degrees reads 3.7 at the seed 24 and 12.4 at the seed 25, the two summed, 16.1 on 22 degrees, inside the one percent band 8.6 to 42.8 (the upper tail 0.81); against the blind's counts scaled to 285, 4.6 and 15.0 (the same run; COMPUTED, `test_the_draw.py`; CLICKS). On the Huygens row at  $N = 273$  the central maximum stands above its two minima by 5.7 times its scatter (COMPUTED, CLICKS; `two_slits_real_line.py`). (c) *The lattice's own number* is the inflow peak's 69.0 against 62.0 (Section 9.1), a shift the



**Fig. 6** The two slits, the one computed file (Sections 9.1 and 9.2; `COMPUTED`, `CLICKS`; `two.slits.rows.py`). (a) Universe24 as its file declares it. The flat lattice has  $45 \times 48$  Nodes. The packet of light, at the wavelength 8 Links, is laid over the columns 4 to 16 (the grey raised cosine) and moves along  $+x$ . The wall at the column 20 has two gaps of three rows, 12 Links apart. The screen at the column 44, 24 Links from the wall, is twelve declared NodeDetectors of four rows each. (b) The screen's row per region over the window of the intervals 45 to 130. The Huygens blind, written before the run, is hollow ( $N = 273$ , its bars summing to 275 by the per-Node rounding, with the draw's scatter  $\sqrt{NP(1-P)}$  per region). Beside it, in grey, is the law's real line stepped on the same file, as printed in `two.slits.real.line.txt` by the paper's script `two.slits.real.line.py` at the lay as that commit holds it, the band's exact level ( $N = 284.50$  in the engine's labels, 284.7 in the physical ones; 278.89 when the laid record's own integers, laid at the content 60's paces, are stepped by this line at the content 0's). The run of the implementation gives  $N = 285$  at both seeds (Section 3.7). The numbers drawn are the blind file's and the script's; nothing is read from a run of the engine for the drawing



lattice reading (a diagnostic): the level of the light record, darker where larger; the measurement is the screen's row of Fig. 6

**Fig. 7** The two slits mid-run (`LATTICE`, a diagnostic and no measurement): the level of the light record over the declared  $45 \times 48$  lattice, darker where its magnitude is larger (the grey  $1 - \min(|\text{level}|/1,328, 1)^{0.6}$ , saturated above 1,328), at the intervals 18 (a), 40 (b) and 62 (c) of the 130, the packet's centre at the wall, the two wavelets past the gaps, the centre at the screen. The wall (column 20) with its two gaps and the screen's twelve regions (column 44) are drawn from the file's declarations; the grown layers beyond the receding faces are not. Every level is the engine's, stepped by Rule3 alone from the file's lay (the engine's own record of the archived file's run); the measurement is the screen's row of Fig. 6(b). The three frames are the engine's own levels of the archived file's run at the commit 1fe3790a over the 130 intervals, a lattice diagnostic drawn by `two.slits.frames.py`

law's real line reproduces (`COMPUTED`, `CLICKS`; `two.slits.real.line.py`); and the fringe row's own departure from the continuum: the lattice's transmission per laid quantum at  $\lambda = 8$  Links is 0.157 against its own continuum limit 0.136, the lattice 16 percent above the continuum (the sequence 0.157, 0.148, 0.142, 0.139 at the scales  $s = 1, 2, 4, 8$  converging as  $1/s$ , the one-Node wall and screen the first-order term; the plain lay's 0.149 against 0.127 says the same, 17 percent); the raw  $N$  carries the lay's count and moves with the scale (284.7, 308.3, 308.1, 304.9 at  $s = 1, 2, 4, 8$ , the limit 303.8, the script's section 6); per laid quantum the transmission is the lattice's number, 0.157, 0.148, 0.142, 0.139 converging as  $1/s$  to 0.136 (the exact lay, the fit from  $s = 4$  and 8 checked against 2 and 4) (`COMPUTED`, `LATTICE`; `two.slits.real.line.py`, its section 6). The Huygens blind (15 and 11 at those regions) is itself a per-Node lattice sum, so the run's agreement with it checks the lattice's arithmetic at  $\lambda = 8$  Links and not Young's continuum pattern. [1, the rows against nature, (g); the click is the meeting of the future with the past].

### 9.3 Bell: the file

The entanglement the model computes is one event with two trees. A pair is two records from one root. Each is read in two directions: forward, the future, and backward, the past. A click is where the two readings of one record meet. The pair's correlation is the partner's record read through the same root, re-weighted by what was realised. The pair is laid as the pair family: two real lines of light, never summed at a Node, with equal parts. Each side's NodeDetector is a region at least half a wavelength across. It reports the signed level sums of the two parts over its region and knows nothing of angles; the ports enter at the credit. The NodeDetector applies the declared setting, a pair of integers  $(p, q)$ , with the coefficients of the analyser's two channels, written + and - (the analyser's outputs, not a Node's Ports),  $e(+) = (p, q)$  and  $e(-) = (-q, p)$ . These are exactly orthogonal, so each side's marginal is one half for any setting. The settings of the Clauser-Horne-Shimony-Holt test are  $(1, 0)$ ,  $(1, 1)$ ,  $(12, 5)$  and  $(5, 12)$ , at 0, 45, 22.62 and 67.38 degrees. The credit is the one form of the click: the joint share pairs the parts by their label across the two sides through the root,

$$J(p_A, p_B) = \sum_{\text{window}} \left| \sum_k c_k e_k(p_A) \alpha_k(p_A) e_k(p_B) \beta_k(p_B) \right|^2, \quad c = (1, 1), \quad (18)$$

with  $\alpha_k$  and  $\beta_k$  the amplitude of the part  $k$  at the port on each side,  $e_k(p)$  the port's integer coefficient on the part  $k$ , and  $c_k$  the parts' laid weights. The window's sum stands outside the square, on both members of the level pair. The joint share is never the square of a local sum; such a sum one NodeDetector could form alone, and it gives  $S$  at most 2. One draw is made per pair. It is the NodeDetectors' draw with the declared seed, and the lattice draws nothing: the first side is drawn from its marginal and the partner from the conditional. Each side's write at the Node of its exchange spreads by Rule3 and changes no number of the row, one click per record (Section 5.4) (ASSUMPTION, CLICKS) [1, the pair family and Bell's gate].

### 9.4 Bell: what the model says it must give

(1) Rule3 gives each part's amplitudes at the ports. The two parts of one beam, laid equal, are two records of one family stepped by the same line along the same path, and they are identical bit for bit (THEOREM: one integer map on equal integers, the determinism of equal parts; LATTICE). (2) The meeting, with the linearity of Rule3, gives the joint amplitude. The realised click at A's port weights the parts by A's analyser. It is read back to the root, where the two parts were one event, and forward to B, where B's analyser weights them. The root sums the two branches:  $\sum_k c_k e_k(a) e_k(b) = \cos(a - b)$  for the normalised vectors  $e(a)$  and  $e(b)$ ,  $k = 1, 2$  the part's label, with  $c$  the parts' laid weights,  $(1, 1)$  for the symmetric lay (DERIVED, CLICKS; `bell.py`). (3) That the share is the density of the meeting, its square, is Born's rule for a pair [44]. Nothing in Rule3 forces it. Everything after it, the fractions,  $S$  and GHZ's  $M$ , is derived in the algebra (DECLARATION, FITTED; CLICKS). From the three, with the pair's common phase  $\theta$  uniform over the pairs, A's + port at  $a$  and B's at  $b$ , with  $E$  the correlation of the two sides' signs and  $S$  the Clauser-Horne-Shimony-Holt sum,

$$\begin{aligned} \text{the meeting:} & P(++) = \frac{1}{2} \cos^2(a - b), \quad E(a, b) = \cos 2(a - b), \quad S = 2\sqrt{2}; \\ \text{a local credit by the shares:} & P(++) = \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \quad E = \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\ \text{a local credit by the sign:} & E = 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \quad S = 2; \\ \text{the contrast credit, one side weighted:} & E = \cos 2(a - b) \text{ at one side's efficiency } 2/\pi, \quad S = 2\sqrt{2}, \end{aligned} \quad (19)$$

The second, third and fourth lines follow by averaging over the pair's phase  $\theta$  (DERIVED, CLICKS; S.10; `bell.py`). The fourth counts one side weighted by its contrast and always counts the other side's sign, by  $\langle \cos x \operatorname{sgn} \cos(x - 2\delta) \rangle = (2/\pi) \cos 2\delta$  with  $\delta = a - b$ ; the symmetric readings give no cosine (S.10). So  $E(a, b) = \cos 2(a - b)$  under the meeting. For the declared pairs it is exact, by Lagrange's identity of the NodeDetector on equal parts (S.10). At the four settings, then,  $E = 119/169$ ,  $-119/169$ ,  $120/169$  and  $120/169$ , and  $S = 478/169 = 2.8284$  exactly. The efficiency is 1 by construction, and the marginal is one half on each side whatever the partner's setting (DERIVED, CLICKS; S.10; `bell.py`). For unequal parts, with  $r$  the ratio of their products at the ports, the same credit gives  $E = \cos 2a \cos 2b + \rho \sin 2a \sin 2b$  with  $\rho = 2r/(1 + r^2)$ , and  $S = \sqrt{2}(1 + \rho)$  at the ideal angles (DERIVED; CLICKS; `bell.py`). The maximum of  $S$  over the four settings is  $2\sqrt{1 + \rho^2}$ : Tsirelson's bound [83] at  $\rho = 1$ , and never the algebraic 4. The declared pairs' 478/169 sits below  $2\sqrt{2}$  because the angles are rational (DERIVED, CLICKS; `bell.py`).

Without the declaration the lattice gives the second and third lines of Eq. (19), two results on local credits on the same file (DERIVED under the uniform phase, CLICKS; `bell.py`). A credit of each side by the shares of its ports gives  $S = \sqrt{2}$ . A credit by the sign of its larger port gives the sawtooth, whose sum is exactly 2, Bell's bound. Bell's value therefore sits in the credit rule, a declaration, and it is no independent evidence that the rule accounts for entanglement. The same credit on four lines gives the three-quantum GHZ correlation [82],  $E_3 = \cos 2(a + b + c)$ , with every pairwise  $E$  equal to 0 and Mermin's  $M = -4$ , where every local credit gives at most 2 (DERIVED under the declared credit, CLICKS; S.40; `bell.py`) [1, the GHZ gate]. No entanglement other than the pair's and the swap's is named as met (S.51). Between separately prepared bodies joined by a coupling or by a transferred quantum, the law's credits give at most 2. Nature's gate-joined ions [32] and transmons with a transferred photon [34] exceed it: the largest discrepancy the paper names between the law as written and measurement (DERIVED for the bound, the comparison a named finding; CLICKS; S.56 (14); `bell.py`). No signal passes between the sides, since each side's marginal is one half. No settings are chosen by the lattice's own past, so a superdeterministic world [70] is not this one [1, the rows against nature, (h); the click is the meeting of the future with the past; the pair family and Bell's gate].

## 10 Discussion: the constants, the calibration and the findings

### 10.1 The constants

In the Universe24 it computes, the quantum of action is one click, whose action is the quantum's norm  $T$  and whose energy is  $T \sin \omega$  at the frequency  $\omega$ ,  $T$  at  $\omega = \pi/2$ , the share's ceiling, which the band contains along the diagonal (light's band's

top,  $\omega = \pi$ , reads 0; a gapped family's top,  $\pi - \omega_0$ , reads  $T \sin \omega_0$ ). The engine's units are the Node's: the interval, the Link, and the two integers  $\Gamma$  (the clock's count per interval) and  $T$  (the invariant's unit,  $2A^2 \sin \omega = T$  per quantum); every energy the clicks read is a multiple of  $T \sin \omega$ , every charge's Wronskian  $T/2$  per quantum, every speed a fraction of one Link per interval; nature's constants enter only through the calibration series of Section 10.2, where a measured click pattern fixes the pairs and  $T$  and bounds a family's denominator from below, and  $\Gamma$  with it where the file declares  $\text{den} = \Gamma$ ,  $\hbar$  standing against  $T$  in the energies  $T \sin \omega$  and  $c$  against the Link over the interval, and nothing else of nature is in the line.  $\alpha$  and  $G$  enter as the two declared weights  $k$  and  $E_g$ , read from nature; the lattice fixes the prefactors  $3\sqrt{3}/8\pi$  and  $3/4\pi$ . A body whose write is  $s$  quanta per interval over the divisor  $E_g$  (the  $s$  of Section 8, (b)) makes the rest of the massless holder  $3G_0(r)s$  at  $r$  Links,  $G_0$  the lattice's Green's function; far away this is  $3s/(4\pi r)$ . The level slows the clock by Newton's potential  $\Phi = -c^2 U = -c^2 \ell / \Gamma$ , with Newton's sign, so (DERIVED, LATTICE; S.25; `constants.py`)

$$\boxed{\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi \Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi \Gamma E_g} \text{ per quantum of well, in Links and intervals}}, \quad (20)$$

a reading of  $\Gamma$ , the file's  $E_g$  and the lattice. Kepler's  $G$ , read from the orbits of matter as clicks, is  $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (14), which is 0.8 of it at  $[2, 3]$  (DERIVED, CLICKS; `constants.py`). The vacuum content is 60 in the file, a declaration above the law's floors that no click reads (Section 6.5). A cosmological constant, if the law has one, would be the rate of the vacuum write, a hypothesis not claimed here (S.29), and not this rest  $[1, \text{the constants}]$ .

**The fine-structure constant, a coefficient of the file; the units.** Under the holder's rotation (Section 5.5) the quantum of a family is one unit of the invariant,  $2A^2 \sin \omega_s = T$ . Its energy is  $T \sin \omega_s$ , and its Wronskian is  $A^2 \sin \omega_s = T/2$  for every family; this is the unit of charge. Charge universality follows from these two definitions, the quantum one unit of the invariant and the sign holder's source the Wronskian:  $|q_p| = |q_e|$  by construction, no test, which nature's measurements give to  $10^{-21}$  [69] (DERIVED from the definitions; CLICKS; S.26; `constants.py`). The holder of the sign carries one declared integer weight  $k$  in its write, as gravity's holder carries  $E_g$ : the written level is  $k$  times the plain one, and the turn is the level over  $\Gamma$ . The law's coupling between two quanta is then  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) k / (\Gamma E_s)$ , independent of the family's gap (DERIVED under the rotation,  $k$  ASSUMPTION; CLICKS; S.26; `constants.py`). The force on a body of  $C$  quanta and sense  $\pm$  is  $\pm C \nabla \theta$ : Coulomb's  $q_1 q_2$  with  $q = \pm Ce$ , where  $e$  is one quantum's Wronskian  $T/2$  and which sense moves apart is the declared bit (Section 5.5, S.26 (c)). The reader sources the holder it reads, so the pair is reciprocal (DERIVED; S.26 (c); CLICKS; `constants.py`). Nature's  $\alpha = 1/137.036$  fixes  $k / (\Gamma E_s) = 1/28.33$  and nothing else. So  $\alpha$  is a coefficient of the file, read from one click as  $G$  is; it is neither derived nor a finding. A charged body and its light read one energy by a line of the law,  $E_s T \cos \omega_s = k_w \Gamma$ : the energy its clicks read and the holder's form move together under the turn and the write, both up for a body in its own level and both down for a pair of opposite senses, and agree to first order in the turn if and only if  $k_r E_s T \cos \omega_s = k_w \Gamma$  ( $k_w$  the write's weight and  $k_r$  the read's,  $k = k_w k_r$ ,  $k_r = 1$  on the integer lattice): one energy read twice, by the clicks and by the form, exact in no form, at the vacuum's paces of S.26 (DERIVED for the one energy read twice; the line ASSUMPTION; S.26 (d'); LATTICE; `constants.py`). The loader enforces it for a plane family reading a holder under the rotation,  $E_s T \text{ num} = k_w \Gamma \text{ den}$  exactly. The line holds per family, through  $\cos \omega_s$ ; one write weight cannot hold it for two charged families of different gaps, and the loader as built admits one moving charged family per holder of the sign; in nature  $\cos \omega$  differs from 1 by under  $10^{-18}$  for every family, so one weight serves all to that order, and no exact two-family identity is claimed. Under it  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) T \cos \omega_s / \Gamma^2$ , and at nature's gap  $T = \Gamma^2 / 28.33$  (a calibration from  $\alpha$ );  $E_s$  is then a unit of the level, and the file has one integer fewer. The line binds the plane families that read the holder of the sign, and not light or a real-line family. The rule's own universe has no plane family, so its  $T = 2^{15}$  stands. Under the law's line a record of a family that reads a holder of the sign has the count 1, so that many quanta of charge are that many records of count 1; the charged families' derivations stand under that line, per record and additive over records (DERIVED, one derivation; CLICKS; S.26; `constants.py`). [1, no record reads its own write of the sign]. The two holders are one kind of family with one kernel: the sign holder reads a quantum's Wronskian  $T/2$ , and the content holder reads its form  $\sin \omega_s$ . So gravity's coupling between two quanta of the family  $s$  is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2 \omega_s / (\Gamma E_g)$ , and the ratio of the two forces at that family is  $F_e / F_g = k E_g / (2E_s \sin^2 \omega_s)$ . Between two families the weights cancel:  $(F_e / F_g)_e / (F_e / F_g)_p = (m_p / m_e)^2$  with no number, and nature's  $4.1656 \times 10^{42} / 1.2356 \times 10^{36} = 3.3713 \times 10^6$  against  $1836.15^2 = 3.3714 \times 10^6$ , to the four digits of the inputs. This is an identity in the small-gap limit (the gap factors  $\cos \omega_s$  and  $2 \cos \omega_s / (1 + \cos \omega_s)$ , 1 to  $10^{-18}$  at nature's gaps, S.44), forced by charge universality and  $F_g \propto m^2$ , by construction and no test (DERIVED from the definitions; CLICKS; S.44; `constants.py`). On the lattice there is no  $10^{42}$ . The number appears when a NodeDetector compares the two forces on one quantum, and most of it is the smallness of the electron's energy against its unit of charge (S.44). The units are the electron's: one interval is  $\omega_e \hbar / (m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_e$ , and one Link is  $\sqrt{3} \omega_e \bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_e$  ( $\bar{\lambda}_e$  the electron's reduced Compton wavelength). What a click reads of a quantum of frequency  $\omega$  is its share,  $T \sin \omega$  (S.14), at most  $T$ , against the electron's rest energy  $T \sin \omega_e$ . LHAASO's photons of 1.4 PeV [71], read by a calorimeter as a sum of such shares, give  $\omega_e < 3.7 \times 10^{-10}$ ,  $\text{den} > 1.5 \times 10^{19}$  and  $\gamma_K - 1 = \text{den}/\text{num} - 1 = \omega_e^2 / 2$  below  $6.7 \times 10^{-20}$  for every orientation of the lattice (the band's top,  $\omega = \pi$  along the diagonal, is a lattice statement and no click's energy) [1, what a click reads of a quantum]. The Link's bound of Section 10.2, from MAGIC's subluminal dispersion limit [72], the one the subluminal band obeys, the Link over the unit, gives  $\omega_e < 2.0 \times 10^{-14}$  along an axis and  $3.1 \times 10^{-14}$  on the average over directions,  $\gamma_K - 1 < 2.0 \times 10^{-28}$  and  $4.9 \times 10^{-28}$ ; the dispersion vanishes on the eight body diagonals and weakens continuously toward them, so each source's bound is its direction's against the lattice, unknown; the axis's bound leads and LHAASO's is slack by four orders, the Link bound weakening tenfold only within 4 degrees of a body diagonal, one percent of the sky, so the comparison holds under the mean's premise (S.24) (DERIVED, two derivations; CLICKS; S.24, S.27; `constants.py`).  $\Gamma$  reads nothing of  $\omega_e$  or of  $\alpha$ , and nothing of the clocks bounds it (DERIVED, LATTICE; `constants.py`): the clocks read the mean of the write's dithered sub-unit level, the exact rest, and not  $\Gamma$  (an argument on the mean; the dynamics under a dithered level are open; S.26). No click of the clocks bounds  $\Gamma$ . The gap bound binds the matter family's own denominator,  $\text{den} > 1.5 \times 10^{19}$  for the least pair; it binds  $\Gamma$  only in a file that writes that pair over  $\Gamma$ , as the archived file does. The energy

**Table 5** The calibration series: each coefficient of the universe’s file, the one measurement of nature that reads it, and the layer it lives in (DERIVED for every form of reading, the integer the file’s; LATTICE; `constants.py`). The lattice reads none of them.

| The coefficient                         | The click that reads it, or why none does                                                                                                                                                                                                                          | The layer                                  |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| $T$ , the unit of action                | Planck’s constant: the counter’s unit of credit and Millikan’s slope (Section 7.4)                                                                                                                                                                                 | the click                                  |
| The Link and the interval               | $\omega_e$ ’s bounds from a photon of 1.4 PeV (LHAASO’s 2021 record; a 2.5 PeV photon was reported in 2024 [90]), and light’s dispersion, MAGIC’s bound [72] (S.24, S.27)                                                                                          | the lattice, through the electron’s family |
| $\Gamma$ , the lattice’s clock          | bounded by no click of the clocks (Section 10.1); the energy line ties it to $T$ , $E_s$ and $k_w$ (S.26)                                                                                                                                                          | clicks                                     |
| The gaps, one per family                | the masses: $\omega_e$ by the electron’s mass, the rest terms’ coefficients $g = 1 - \cos \omega_0$ in the ratio $g_p/g_e = (m_p/m_e)^2$ under one denominator (HYPOTHESIS, S.53), read as click masses $T \sin \omega_s$                                          | the lattice, through bound bodies          |
| $E_g$ , gravity’s weight                | Newton’s $G$ , the fall and the orbits, Eq. (20)                                                                                                                                                                                                                   | the lattice, through the fall              |
| $k$ , the sign holder’s weight          | Coulomb’s force between two quanta, $k/(\Gamma E_s) = 1/28.33$ (S.26)                                                                                                                                                                                              | the lattice, through two free quanta       |
| $c_{vac}$ , the vacuum content          | declared 60 at $\Gamma = 6,000$ , bounded below by the lattice’s own floor (8 to 13 levels) and by the measured swing (43 levels); no click reads it or bounds it above, every family reading the content; GW170817 bounds $\gamma_{GW} - \gamma_{EM}$ and no rest | the lattice                                |
| The nuclear holders’ ranges and weights | the nuclear binding and the nuclei’s sizes, if the holders are admitted (Section 10.3; HYPOTHESIS)                                                                                                                                                                 | the lattice, through the nucleus           |

line ties it to  $T$ ,  $E_s$  and  $k_w$  (CLICKS) [1, a body and its light keep one energy; the law’s alpha is a coefficient of the file; the units are the electron’s and the share’s ceiling is the highest photon energy; what is open; the strong force].

## 10.2 Nature’s numbers enter only as clicks

**What the lattice fixes with no number.** Before any file, the lattice fixes the following, each DERIVED with its status in S.54 (LATTICE for the forms, CLICKS where a click reads them; `constants.py`). The geometry, with its group of 48, hence one preferred frame and  $\alpha_1 = -8$ . The dynamics, reversible to the bit with no loops and, with no pair created between clicks (S.37), no running of  $\alpha$  and no Lamb shift. The conservations per family, with the paces’ weights at static paces. The band, with light at  $1/\sqrt{3}$  Link per interval under the causal bound 1. The paces, the exponential metric with the finest level  $1/\Gamma$ . The holders’ kernel  $3/(4\pi r)$ . The click layer’s bilinear credit. And, beside the file’s one scale  $\Gamma$ , pure numbers of order one, every large number being a power of  $\Gamma$  or a declaration, each with its status in S.54. In one sentence: the lattice is a machine of forms; it fixes every form and every ratio of order one, conserves, reverses and makes no noise but the remainders; it fixes no strength, no scale and no large number; those are the file’s [1, the properties of the lattice].

**The calibration series.** Every measured number of an experiment is first written as a click pattern on a declared NodeDetector against a declared clock. The families’ pairs and the units  $\Gamma$  and  $T$  would be fixed by that match, and by nothing read off the lattice; the archived file’s integers are the rule’s own (Section 6.6). The two ratios the model fixes without it are light’s speed of  $1/\sqrt{3}$  Link per interval, from Rule3’s vacuum band (Eq. (4)), and the causal bound of 1, the step’s reach (DERIVED for the speed, LATTICE; the bound is postulate 4; `bands.py`) [1, the reading; nature’s numbers enter the board only as clicks]. No coefficient of the file is computed from the lattice. Each is read from one measurement of nature on a bound body; the lattice fixes the form of the reading, and the file holds the integer, Table 5 (S.54). No number comes from the lattice itself (HYPOTHESIS for any such number, S.54) [1, the calibration series]. One check stands beside the table, the two clocks. A cavity of bodies and an atom’s beat shift alike by  $N$  at long wavelength, as nature’s clocks do, atom against atom and cavity against maser [38]; only a cavity of declared, bodiless faces would shift by  $N^2$  (DERIVED; CLICKS; S.31; `paces.py`).

**One prediction of the model, untested, one check and one open question.** One number follows from the law’s lines where general relativity and quantum theory do not meet. It is derived from the law’s lines and entered under its own name as untested, no measurement having read it (DERIVED; S.41; CLICKS; `weak_field.py`): *the gravity of light* (CLICKS, untested). In the model a light packet’s integrated write is a static body’s of the same count,  $1 \times E/c^2$  under the static kernel (the packet’s own field open, S.41), where general relativity’s pencil pulls at  $2 \times E/c^2$  [74]. The check: a bound body laid as one record on two paths at two heights has its beat scale as  $N = 1 - U$ , so the visibility of the fringes falls as  $|\cos(\Omega \Delta U \tau_{\text{hold}}/2)|$ , with  $\Omega = \omega_j - \omega_i$  the body’s internal beat and  $\tau_{\text{hold}}$  the time the two paths are held apart. This is 0.989 for a strontium clock at one metre for one second, the magnitude of the semiclassical expectation [73], and it is unmeasured (DERIVED, CLICKS; S.41; `weak_field.py`). The open question: a collapse ends in frozen clocks that read nothing, are read, and are dark by the share’s declaration (Section 7.2). Whether such a region radiates, Hawking’s temperature [78] and its entropy are open; no line puts the frozen Node’s six-Port flux at 0, so  $T_H = 0$  is not claimed (S.41, Section 10.3).

## 10.3 Findings against nature: where the model misses, and the open questions

**Lattice readings, not findings.** The rule’s own universe at [2, 3] is no finding against nature, because nature measures at its own pair (LATTICE; Section 8, (e)). The masses are families. Nature’s mass ratios are not whole numbers ( $m_p/m_e = 1836.153$ ), and a cluster’s defect is bounded by the law at  $C\omega_0 x_{\text{self}}$  with  $x_{\text{self}}$ , the self-read fraction of S.38, of the order of  $10^{-22}$  at nature’s Links (S.38), so a cluster’s mass is whole in the electron’s quantum to  $10^{-20}$  and no particle of nature can be read in the model as an aggregate of the electron’s quanta, the muon’s 206.768 among them. The proton is one quantum of its own family, and hydrogen’s clicks read it as one unit of  $W$  over 0.84 fm (a comparison for the ratios, ASSUMPTION for the families; CLICKS; S.53). What the click’s structure derives of the masses is in S.53: the rest energy  $m^* c_m^2$  exactly, the ratio of two declared rotations, the ceiling at LHAASO’s bound and the kinetic excess of a conversion, and no cause for 1,836 (DERIVED, two derivations; CLICKS; S.53; `constants.py`). Charge universality follows from the writes exactly, and the equivalence between families as the gaps close, the free packet’s fall coefficient  $2\nu/(3(1+\nu))$ ,  $\nu = \text{num/den}$ , differing

between families at finite gaps, 4/15 at [2, 3] against 2/7 at [3, 4] (DERIVED, CLICKS; S.53; `constants.py`). If the families share one denominator, the masses are the square roots of the gaps (HYPOTHESIS). [1, the proton's two readings]

**The declarations.** The model declares the following and does not derive it. This is the law's own list, and each item carries what is derived once it is declared (ASSUMPTION for every declaration). (1) A family: the name of its row, its pair, its reads and its dimension, and the integers  $\Gamma$ ,  $T$  and the width (derived: its band, its tensions and the forms of its clicks). (2) A NodeDetector with Nodes alone: the region of two Nodes or more and half a wavelength, the file's draw window and the seed (the random generator belongs to the credit, the floor is the credit's own and the whole count is the record's; derived under the declaration: the share, Born's form for one quantum, one click per record, and the uncertainty relation as these three). (3) A NodeDetector with a record of its own: the record laid over its Nodes, one or more, its parts the record's parts, the modes, held by its declaration and never bound by a well, its write at the Node the draw picks (derived: the exchange, the absorption, the null window's write, Zeno's blind and the telegraph's blind). (4) The emission's rate: the beat's current over  $W_c$  is derived, the rate's constant is declared, and there is no floor. (5) The conversion's table and rate: a hypothesis under its own name, declared to happen and derived in what it does. (6) The lay. (7) The receding face, a declaration of the file. (8) The credit's one draw through the root, the one declared non-local element. (9) The erasing front: the law's line by two derivations and the model's decision. The marks of this paper agree with the list one to one. Nothing in the list is marked derived, and what each item gives is marked derived under it [1, the declarations by name; what produces a click is not known; the receding face][3, nature is not said to have clicks]. *The four forces, as declared.* On the lattice there are only records stepped by Rule3 and holders written by the write. On the lattice the weak force is nothing but the conversion click, a change of family at a click, and the strong force inside a baryon is nothing but one record of three lines read whole; both appear only in clicks and after them. The nuclear binding is a bound body in the binding holder's well. Electromagnetism and gravity are the holders' continuous reads and writes, which a body's standing reads between clicks. The paper's account of the forces stands under its own name and claims nothing beyond its marks. The weak force is the conversion click (HYPOTHESIS under its own name): a click that absorbs a quantum of one family and emits it as quanta of others, the rotations adding. It is the act of one Node, with no family of its own and no range but the Node. In the model the neutron's end is such a click. Its table and rate are declared and read from nature, as the lifetimes are (Section 5.4; item 5). The strong force inside a baryon is one record, and neither a force nor a body. The baryon is the quantum of a family of dimension 3, and the quarks are its three lines; no quark family is declared in the model and none is needed by it, no count per line and no click of one line. Confinement is the record's indivisibility under the click (DERIVED from its declared row; S.37; CLICKS). The three lines are three real lines stepped by the one rule. They are rotated among themselves only by the colour holder's phases on each Link (S.37), as the plane's two lines are by the sign holder's, and never by the lattice's rotations. The testable consequence is that the three fractions of a hard click sum to one and no line is reported alone. The baryon number, the flavours, the quark masses and the fractional charges are not derived and not claimed; the mesons are named open. The strong force between nucleons is, under the hypothesis below, a bound body in the gapped binding holder's well, whose row [2400, 2401] is declared; the range  $R = 20$  Links and Yukawa's form follow from that declared row, so the range is a declaration by that row; the gap, and the full nuclear holders are named open. The nucleus is then a bound body in that holder, as the atom is in the sign holder's. In the model the proton is three planes and the neutron three real lines, two families, with the conversion the change between them (Section 5.4; S.56 (3)). Electromagnetism is the sign holder, Poisson's rest and the rotation, with  $k$  declared. Gravity is the massless holder, the content and the paces, with  $E_g$  declared. Their forms are derived from their declared rows, continuous reads and writes of the Nodes' laws (Section 3). Only the weak force is a click, which is why it alone changes a family [1, the forces as the law holds them; the strong force; the nuclear holder].

**Findings against nature: the model's misses.** The massless holder's travelling events carry the breathing polarisation and the lattice-aligned + and no  $\times$ . That is derived under the static read (DERIVED; CLICKS; S.35, S.36) and is a finding [1, what is open]: nature's gravitational waves are read as the two tensor modes + and  $\times$  at every orientation, GW170814's three-detector test favouring pure tensor polarization over pure vector or pure scalar [48] (EXPERIMENT; CLICKS). Gravity's readers are real records with no read of a current, so in the model a gyroscope precesses by 0 from the motion of a rotating body. Gravity Probe B measured  $37.2 \pm 7.2$  milliarcseconds per year, against general relativity's 39.2 [79] (DERIVED, CLICKS; `frame.py`) [1, the current's read]. A metric with no  $g_{0i}$  in its preferred frame has  $4\gamma + 4 + \alpha_1 = 0$ , so  $\alpha_1 = -8$  at  $\gamma = 1$ . Nature has  $|\alpha_1| < 4 \times 10^{-5}$  from the binary pulsar PSR J1738+0333 [80]. This is a finding (DERIVED, CLICKS; S.35; `frame.py`) [1, what is open]. With  $g_{0j} = 0$  in the lattice frame,  $g_{00} = -e^{-2U}$  and  $g_{ij}$  isotropic, which is the standard post-Newtonian gauge, the two  $g_{0j}$  coefficients vanish separately. They give  $\alpha_1 = -8$  and  $\alpha_2 = -1$  together, and their sum is the gauge-invariant statement above. Nature has  $|\alpha_2| < 4 \times 10^{-7}$  from the Sun's spin [61] and  $< 1.6 \times 10^{-9}$  from solitary pulsars [81]. Like  $\alpha_1$ , this is conditional on the metric read for a static source (named, not claimed; DERIVED under the static read, CLICKS; S.35; `frame.py`). At the second order in  $z$  the model's finite ball of bodies decelerates, bound or free,  $q_0 = \Omega_m/2 > 0$  (DERIVED, CLICKS; S.29; `cosmology.py`), where the supernovae read  $q_0 \approx -0.5$ , which is the flat  $\Lambda$ CDM model's (cold dark matter with a cosmological constant)  $\Omega_m/2 - \Omega_\Lambda$  [54][55][56]. Dark energy, in the model's words: Rule3 has no repulsive sector, every held level a hollow, so an accelerating Hubble diagram is either the vacuum family's declared uniform write, one number read from the supernovae, the cosmological constant of Newtonian cosmology exactly, S.29 (the hypothesis the law's documents carry under its own name [1, the vacuum writes], not this paper's claim), or a finding against nature by name; a hierarchical or off-centre start lowers the deceleration toward 0 and never below it, a declaration of the start and no line (an order-of-magnitude estimate, one derivation). The paper claims nothing beyond  $z \ll 1$  there and no identification with nature's dark energy: a supporting statement of what the model holds (the deceleration derived in S.29; the write's one number declared; nature's acceleration without it an explicit finding; CLICKS).

**The lattice's frame, and its mend as a hypothesis.** At long wavelength and small gap every record's band is a Lorentz form,  $\omega^2 = \omega_0^2 N^2 + c_m^2 N^4 k^2$ , the Lorentz form in the Node's own units ( $\omega/N$  against  $Nk$ ) (DERIVED, LATTICE; S.21, S.34). So the moving clock's  $\gamma$  follows, and no Michelson-Morley shift appears at nature's pair beyond  $c_m$ 's own difference from  $c$ ,  $2.2 \times 10^{-20}$  at LHAASO's bound (DERIVED, CLICKS; Section 8 (e); `moving_clock.py`) (at the computed pair  $c$  and  $c_m$  differ, and a matter-armed interferometer would shift at order  $v^2$ ). But the content's Poisson rest in the lattice's frame gives a moving mass no  $g_{0a}$  and no velocity-dependent  $g_{ab}$  (DERIVED; CLICKS; `frame.py`) [1, the lattice's frame is read by gravity alone, and its carrier is a pair of Links; what is open]. The mend is an explicitly named hypothesis with no line

written (HYPOTHESIS): a carrier on a pair of Links, sourced by the mass current. Against it stands this: among the five forms tried, no real, local, explicit, bijective lattice line on Rule3's clock carries  $g_{0a}$  (DERIVED for the five forms, no theorem over every form; S.36; CLICKS; `frame.py`). [3, Rule3 is not changed]

**The strong force, by derivation.** The strong force is two things (the declarations above). One is confinement, with no coupling. The other is the nuclear binding under the declared holders, read by the nucleon families alone; under it the size of the nucleus, its binding and its light are derived in form, and its shells are not. Confinement is not a force (the declarations above). A hard click reads three fractions that sum to one, the lines' shares; how they move with the momentum transfer is uncomputed, the read's fractions depending on the lines' lay and the hard click's read operator not derived, where nature's measurements show logarithmic violations (uncomputed, an open item; CLICKS; S.37 (c), S.56 (13)). No derived body of the law reaches the nuclear binding (CLICKS). A body's binding per quantum is bounded by  $0.76 CW/(\Gamma E)$  at one Node, and is about  $0.29 CW/(\Gamma E r)$  for a uniform ball of  $r$  Links (0.36 at its centre), with  $C$  the body's count,  $W$  the holder's declared write weight and  $E$  the holder's divisor (DERIVED, a bound; LATTICE; S.38; `bodies.py`). Under the families' reading the nucleon is one quantum whose binding is the nuclear holder's well,  $W \geq 1.17 \Gamma \kappa / \omega_p^3$  at each record's own inertia about the pinned centre,  $\mu = m^*$  by postulate 2, 591 at [2, 3] and 502 at the exact inertia; under the pair hypothesis alone the relative part carries  $m^*/2$  and the threshold doubles,  $2.35 \Gamma \kappa / \omega_p^3$ , 1,183 (S.45). The nucleus is therefore a hypothesis under its own name: a bound body of nucleon quanta in the well of a nuclear holder, Yukawa's kernel (S.39) with its range and weight declared, read by the nucleon families alone (Section 7), together with a core holder of shorter range and negative weight; four declared integers per nucleus (HYPOTHESIS). Under it the threshold, the binding at a body's size, fusion and fission with the photodisintegration shape of Bethe and Peierls [46], and the size law are computed in form (one derivation; S.45). The findings are these, each DERIVED under the hypothesis and set beside nature (CLICKS; S.45; `nuclear.py`). One well gives  $R \propto 1/A$ , where nature's nuclei grow as  $A^{1/3}$  and saturate; the core holder is the saturating declaration. The shells and the magic numbers (Pauli's exclusion) and the neutron's lifetime are explicit findings: the model derives none of them (CLICKS). [1, the strong force; the nuclear holder; the nucleus; what is open].

**The weak force.** The model has no electroweak theory; its weak interaction is the conversion click (the declarations above). Every rotation holder, massive or not, couples at  $\alpha$  at the Link; its gap is its range and never its strength (DERIVED; LATTICE; S.39; `families.py`). No Fermi constant  $G_F$  follows. A three-line record read as  $(\nu, e)$  under one  $O(3)$  has no fourth rotation, so no  $Z$  boson, where nature has the neutral current [84] and the  $Z$  [85]. Parity is conserved in the model, where nature's weak interaction violates it [86]. And the law has no spontaneous click, so the conversion has no derived rate and the neutron's  $878.4 \pm 0.5$  s [68] has no source. These four are explicit findings, the model's misses (CLICKS). What decays in the model is a body and never a quantum (Section 7), so the neutron's end is the conversion [3, the weak force is the conversion click]. The conversion is one list of the one click act: a record is converted whole at its Node at the weights [window, rate – window], and the records out are laid by the count at the massless pair; the neutron's file is the repository's and no claim of this paper. The table and the rate are nature's numbers, and nothing of the neutron's mass or its excess is claimed (ASSUMPTION for the table and the rate; CLICKS; S.56 (3)) [1, the click writes on the lattice; what the law declares and does not derive][2, the input].

**The gaps in two columns.** The first column is the file's, and the method invites another file:  $\alpha$  through  $k$ ,  $G$  through  $E_g$ , the Link and the masses, the hierarchy of the two weights, the spin holder and the click's unit. The second column is the rule's. Each of its gaps has its cause in the findings above, and none is the fault of a number [1, why the gaps]: no loops ( $\alpha$ 's running, read as  $\alpha(M_Z)$  in [85]; the Lamb shift;  $g - 2$ ), no spontaneous click (the lifetimes), no vector for the exclusion, no line for  $g_{0a}$ , and a step that commutes with the cube's reflections (the  $Z$  and parity).

**The record's parts are its particles.** An explicitly named hypothesis beside the families' reading (S.56 (7)) and the account of the forces, not the law; a gate file decides it (HYPOTHESIS; CLICKS). Nature's quanta are the parts of the matter records and the records of the three holders. The baryons are the parts of the three-line record (S.37). The charged leptons are families of the plane record, and the neutrino is a real-line family, neutral by the law's own line. They are converted into each other only at the conversion click, by a declared table at a declared rate, read from nature as the lifetimes are. The neutrino's record meets no transition of the bodies that stand for nature's counters except through the conversion, and a declared NodeDetector of its family would click it. Postulate 2 holds. Its test is the neutron's end, the conversion of one part at a declared rate, the neutron's file of the repository. It gives neither the rates nor the mixing nor parity's violation, and this section's findings stand [1, the hypotheses under their own names].

**Open.** The hypothesis that only a click moves a quantum between families (Section 5.4; HYPOTHESIS), with its gates below. The standing body: no standing body is claimed from a run (Section 7.3). The closed box's filling: a body laid in a box of faces that read the rest beyond breathes from the first interval, and the holders' swing grows linearly in the run, the far Nodes about 45 levels by 400 intervals (43 at the box's symmetry points, the swing of Tables 4 and 5) and 90 to 105 by 1,200 on the open 25-cube, the record's share in the books rising 0.42 quanta per interval, the same at every vacuum content; the box returns everything the holders radiate, and what remains by exclusion, measured and not derived, is the box's own loop, the record and the content modulating each other at the box's modes near  $2\omega_b$ , seeded by the start's kick and focused at the box's symmetry points, four external sources measured and excluded, each with its reason (the tension, the lay's rounding, the lay's second mode, the rounding inside the form); the vacuum content fences it over the declared run (the first hill near the interval 470 at 60) and does not cure it; the one cure named is the law's one-way face, a face beyond which a held row reads its own level continued without reflection so that the radiation leaves (not the open axis of Section 3.1, whose beyond reads 0), derived in its integer form and not built (a lattice reading, the one open physics finding of the archive's runs) [2, known issues]. The black hole's quanta (S.41). The lattice's scale: the massless band's quartic term  $-(3 \sum_a n_a^4 - 1) k^4 / 108$  ( $-1/54$  on an axis, 0 on the body diagonal) gives the quadratic Lorentz-violation scale  $E_{QG,2} = \hbar c / (\ell_{\text{Link}} \sqrt{\varepsilon_4})$ ,  $\ell_{\text{Link}}$  the Link's length and  $\varepsilon_4 = (3 \sum_a n_a^4 - 1) / 36$  the band's quartic coefficient (S.65); the Fermi-LAT bound [87][89] puts the Link at  $\ell_{\text{Link}} < 6.4 \times 10^{-27}$  m, and the electron's pair then needs  $\text{den} \geq 2^{94}$ , beyond the file's width  $2^{63}$ : a finding on the engine's width as a file integer, the law fixing no number (DERIVED, LATTICE; S.65; `counterexamples.py`). The shelved ion's telegraph [75]: bimodal counts with exponential dark periods. The law without the NodeDetectors' writes does not give it (no recorded reading exists; the ion's world is not built in this work), and a

NodeDetector with the ion's own record over its Nodes, at a declared rate, gives the rate equations' statistics in derivation (S.56 (15)). The shelving is the absorption click of the weak drive's quantum, at the Zeno-limited rate  $\Omega_w^2/R$ ,  $\Omega_w$  the weak drive's Rabi rate and  $R$  the strong transition's scattering rate [76]. The return is the emission click, at a declared rate. The bright periods have the mean  $R/\Omega_w^2$ , and the dark periods the declared  $\tau_D$ . The shape of the statistics is derived from the known form of the rate equations [76] (DERIVED under a declaration; the rate itself a HYPOTHESIS under its own name; S.56 (15); CLICKS; `clicks.py`). The three gates of the click's write are the ion's body with its modes as parts and two declared rates, the anticoincidence (S.57) and the Zeno effect (S.59): their blind rows are derived here, their files are the repository's, and a reading that misses its blind is a named finding there and no claim of this paper. A gate-built register stands outside by postulate 2 (Section 3.6).

**Where a click happens: declared, and open in its cause.** The model does not derive from the Nodes' laws where a click happens. It declares where, a NodeDetector, with or without a record of its own, and derives what the click does once declared. In the model a bound body with no declared NodeDetector never jumps, it only follows Rule3; that nature's bodies jump is the method's widest open gap. A spread body's click, the exchange at its entry Node and its settling a reading, stands with it. The click's picture is closed in its form and open in its cause. What is open is this: the draw, which meeting is realised and at which Node; where a click happens, declared; the conversion's table and rate; the resonance against the share at the computed pair; and the carried remainder of the credit's window with its own blind on  $N$ . The form is closed and claimed; the cause is open and not claimed. The law's one principle for every write from outside Rule3 into a massless record, that it wakes no zero mode of the massless holder (the record's two sums over the lattice as the act found them, by the division act), is a declaration of the law (ASSUMPTION). A single quantum's track of thousands of ionisations, as a cloud chamber shows it, is not derived and not claimed; a click absorbs one whole quantum of the record. The click's write conserves no momentum, since the quanta are laid with a phase and no wave number. So the recoil is the NodeDetector's, where nature's counter recoils, as in the Mössbauer line and the Compton shift: a named finding beside what produces a click (CLICKS; S.61). The clicked record's empty wave beyond the front is a price of locality (Section 5.7).

**Also open, and the atom.** Also open: an exactly conserved integer charge; gravity's divisor  $E_g$  (a calibration, Table 5); the matter pair's gap; what fixes  $\Gamma$ ; and a current's read for real records [1, what is open]. The relations among the file's integers are the program of future work, each an explicitly named hypothesis tested by clicks: one denominator for all the families, and whether  $k$ ,  $E_g$ ,  $T$  and  $\Gamma$  are tied, with the energy line of Section 10.1 already a declared line of the law (ASSUMPTION, Section 10.1) [1, no free number but the mass]. The atom: the law does not support the reduced mass, since it holds no amplitude over two positions, no wave of two records on pairs of Nodes (every record's levels stand at its own Nodes, jointly with the other records there and coupled to them by the read, and the pair's joint amplitude enters at the click, in the credit's pairing, S.51; postulates 1 and 2's locality), and the nucleon's centre is one classical variable under line (2) of S.60. So the isotope shifts and the muonic atoms are outside what it derives. The many-body correlation is not reproduced by a product of records (helium 1.4 percent, positronium 44 percent of nature's, COMPUTED, LATTICE, lattice readings by `pixel_mode.py` and `body_rest.py`; S.60), and Pauli's exclusion has no line (S.56, S.60) [1, the atom is a bound body of the holder of the sign].

**What building a world closer to nature needs, and this work does not do.** Universe24 is one world built by the method, and the method with its supporting claims is the work; a world closer to nature is built by the same method and needs these calculations, named here and not done. This world differs from nature's along two axes, each with its rule. The box's axis: a world's box is finite and its faces return the holders' radiation; a body in a closed box of 25 Nodes a side has its radiation back every 42 intervals, 28 times in 1,200, and the box fills from the body's own loop (above); a box larger than the run's light-crossing ( $10^{12}$  Nodes, a side of  $10^4$ , 17,000 intervals at  $1/\sqrt{3}$  Link per interval) behaves as open for that run, and the law's rule for it on any box is the one-way face of the paragraph above.  $\Gamma$ 's axis: the integers' fineness is  $\Gamma$ 's and not the box's; the vacuum content's floor  $c_{\text{law}}(\Gamma)$ ,  $(K^2\Gamma/16)^{1/3}$  at large  $\Gamma$ , is 8 to 13 levels at  $\Gamma = 6,000$  by the law's rule whatever the box and  $10^{-19}$  of  $\Gamma$  at nature's  $2^{94}$ , so the discreteness' marks shrink as  $\Gamma^{-2/3}$ , and a world closer to nature is a larger  $\Gamma$  with the one-way face, not a larger count of Nodes, which costs as the box's volume times the run ( $3.8 \times 10^{-6}$  seconds per Node and interval on the machine used). The calculations: (1) The closed box's filling (above): the one-way face as the law's line, a held row beyond the face continued without reflection, its one-way fill derived in integers and its bounded growth not measured. (2) The vacuum content as a derived number: the lattice's floor  $c_{\text{law}}(\Gamma)$  is derived in closed form with the start's kick  $K$  measured on this world alone, and the integer is not run at a second  $\Gamma$ . (3) The lays: the body's fixed point under the step's own iteration (Section 7.1) is not unique within the lay's tolerance, so a lay is made from the file's declared count and Nodes and never from a laid body. (4) The tensions: inert in these files (gravity's axis lines write nothing over the runs), so the Link's stress enters no reading here; a world whose axis lines write states their weights. (5) A standing run beyond 400 intervals: the integers' budget, the least  $T$  the lay's tolerance and the run admit, needs  $T = 2^{16}$  for 1,000 intervals, and the fence of 60 covers about 470 on the 25-cube. (6) The ion's world, not built in this work. (7) A thermal world:  $N$  bodies and one light record in a closed box, the lattice's temperature from the count of its states (equipartition over the level pairs), the blind Boltzmann's ratio in the click fractions and Planck's form with  $T \sin \omega$ , derived if the emission's hazard and the absorption satisfy detailed balance, else measured; the lattice has no temperature of its own, Rule3 being a bijection, and the clicks have one. Each is a calculation of the method, none a claim of this paper.

## 10.4 What can follow from here, and how to compute it

Every sentence of this subsection is inspiration and no claim: a definition in the law's terms, as the law holds it; what could follow; and how one would compute it on the lattice, with its blind. Nothing in it is a result, and the paper's claim, what the model computes of nature, rests on none of it. *The black hole.* The definition. The law's static metric is the exponential one,  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$  with  $U = m/r$ . It has no horizon and no singularity, and the clock  $p_0$  is never 0 at any content, so a collapse ends in a frozen clock and not at a horizon. In the model, a black hole at nature's scale is a deep clock, defined by its light ring at  $r = 2m$  and its shadow's capture radius  $2em$ , the conditional prediction of Section 8 under the one assumption, the interior open; this adds nothing to it. What could follow. A Node whose content passes  $(\Gamma/2) \ln(2\Gamma)$  has its Link's pace rounded to 0: it reads nothing and is read by its neighbours. It is a one-way valve that holds its level and gravitates, has no share and clicks nowhere. This is the law's one-Link black hole,  $(\Gamma/2) \ln(2\Gamma)$  levels of content at a

Node, about  $6.2\Gamma E$  quanta under the one-Node kernel, a mass that is a number of  $\Gamma$ , which no click bounds (Section 10.1); dark matter as frozen matter is a named hypothesis; what is derived is only that a record meeting no declared transition stays dark (Section 7.5). Whether a frozen Node radiates is open: no line puts the flux through its surface at 0, and none derives Hawking's temperature. Under the composed paces the compact body passes the content  $\Gamma \text{ div } 2$ , the quadratic form's horizon, still standing, and no Link's pace closes to 0, so no body of the generator reaches the frozen content; that content is reached by declaration and not by a collapse. How to compute it. One declared Node at the frozen content inside an active region, with the loader's refusal lifted for that one file and said so. The lattice reading of the outside level  $m/r$  and of the shares at the neighbours, with the blind of no click at the frozen Node and the neighbours' clicks unchanged. The form's flux through the frozen Node's six Ports over a long run, against the blind 0 and against Hawking's temperature in the law's units, a reading and not a gate, since nature has no click of that radiation. And the compact body's fixed point at rising count under the composed paces, the clock's depth read against  $\Gamma \text{ div } 2$ , with the blind standing at every count the width admits (INSPIRATION; LATTICE) [1, the paces compose; what is open; the rows against nature].

## 11 Conclusion

**The contribution.** One line of integer arithmetic, Rule3, with the declarations named beside it and no number of nature put in but the one click that fixed the one assumption and the correlations the credit was chosen for, gives in lattice form the forms nature's measurements test: the reversibility, the conserved form with its current, the share identity and the Wronskian as the charge (Section 5.5); the bands, the Klein-Gordon equation and Schrödinger's slow limit (Section 3.3); the bound body's stability along the dilation, to first order, for the continuous functional (Section 7.2); Lorentz's factor at  $c_m$  to the second order; Einstein's weak field in the clock's potential under the one named assumption, with the bending, Shapiro's delay and the perihelion; de Broglie's and Planck's relations and  $E = mc^2$  in the band's form (Sections 8 and 7.4); Coulomb over  $\gamma$  and Yukawa's range; Rabi's rate and charge universality; Newton's pull and  $G$  as a reading; Hubble's law and Friedmann's dust forms at  $z \ll 1$  (Section 7.5); the two slits' row, computed at one file; and, under the declared credit and not from the line, Bell's, Tsirelson's and GHZ's values (Section 9). Born's rule is not recovered but put in, as the credit's declaration, with Bell's  $2\sqrt{2}$  its evaluation. Beside the recovered forms stand the lattice's own, Table 6: the band's inertia and kinetic scale, one speed as the gap closes; the post-Newtonian column with the gap, 1 as it closes; the exponential metric's spatial second order; the shadow's  $2em$ ; the moving clock's fourth order, anisotropic by  $\sum_a n_a^4$ ; the lens's chromatic term; the fine-structure constant's form; the frozen content. Each follows from the line and the assumptions named beside it, with its mark, its fence and its number, and is right or wrong at the measurement its column names; the calibrated inputs (Table 5) and the misses (Section 10.3) stand beside the hits. The contribution is the explicit chain from the rule through the declarations and the derivations to the clicks, every input and every miss named, so that each step can be checked and each reading repeated.

**The mechanism.** Universe24 is built from three things. The lattice is a cubic array of Nodes, each holding two integer levels per family and a remainder, stepped by Rule3, one line of integer arithmetic on a Node's present, its past and its six neighbours through one integer division: reversible to the bit, conserving a quadratic form exactly in the rationals at paces fixed in time, covariant under the cube's 48 symmetries, drawing nothing. The NodeDetector is a declared region of Nodes, two or more where it reads with Nodes alone and one or more where it holds a record of its own: it reads the inflow through its Ports over a window, draws once by the shares with the world's one draw (a body's NodeDetector with its own seed), and writes one quantum at one Node of its region; its draw is the only randomness in the model and the one non-local element. The clicks are the NodeDetector's reports, and Universe24 is those clicks, the model's name for what an experiment's counter registers, defined by the model and not by nature; a click is the meeting of the future, the record run forward from its root, with the past, the transition that waits for it read backward from the click. The lattice computes, the NodeDetector is the gate, the clicks are the world of the model, and the formulas the clicks give are set beside nature's.

**What is new.** The formulas the paper adds, the ones the title names, stand in Table 6, each with its number, its mark and the click that would read it; the findings against nature ( $\alpha_1$ ,  $\alpha_2$ , the running of  $\alpha$ , the nuclear binding) stand in Section 10.3 and not here. None of the rows is confirmed by nature: each follows from the line and the assumptions named beside it, and that is all its mark says; where a measurement exists it bounds the form (Cassini's  $\gamma_K$ , the shadow's diameter) and where none exists the form is a prediction, so each row is right or wrong at the measurement its column names and nowhere else. The table is a sample and not the line's extent: every row came from one file or one reading, and the line admits many more files and readings than the paper tried; what they would give is the program of Section 10.3 and the directions of Section 10.4, named as inspiration and not claimed.

**The claim, and its limits.** The paper has one claim: that Universe24 helps compute things in nature, that is, what it computes of nature set beside measurement. Its content is what the model computes of nature against measurement, each quantity with its status mark, its fence and its number, the hits and the misses alike; the method, a mechanism of lattice, NodeDetector and clicks, is how it computes and not the claim; and none of it is a proof about nature. The law holds no amplitude over two positions, no wave of two records on pairs of Nodes: every record's levels stand at its own Nodes, jointly with the other records there and coupled to them by the read, and the pair's joint amplitude enters at the click, in the credit's pairing (S.51). The model is a wave mechanics of records on the Nodes, their reads of each other and their meetings, and the reduced mass, exchange and the correlation of two bodies (helium's threshold and positronium, S.60; the violations between separately prepared bodies, S.56 (14)) stand outside it, named findings. What is put in is named (the one assumption of the forms, the credit chosen with Bell's correlations in view, the declarations listed in Section 10.3, lines (1) and (2) of S.60, the selective read, the energy line, the file's numbers read from nature); where the model misses nature's measurement the miss is named with its number, its  $\alpha_1 = -8$ , its hydrogen 1s under the self-read, its telegraph of the shelved ion without the NodeDetector's write (Section 10.3). The form of a click is claimed and its cause is not: why a click happens is the draw, a declaration named in Section 10.3, on which no claim rests. Why nature is so, we do not claim to know.

**Outlook.** Nature's formulas are many, and the paper computes no complete set of them. It samples a few places: the atoms (Sections 7.3 and 10.3; S.46, S.60), gravity and Newton's potential (Section 8), and the two slits and Bell (Section 9). Some derivations meet nature's measured values and some miss, and the misses are items of Section 10.3 with their causes

**Table 6** The formulas the paper adds: the lattice’s own forms, with no continuum counterpart or with one they depart from, each with its number at the computed pair [2, 3] or at nature’s, its mark and fence, and the click that would read it. The status column uses the key’s marks; “untested” names a prediction of the model that no measurement has read. The forms recovered in lattice form, with the known form beside them, are listed in the section’s first paragraph.

| The formula                                                                                                                                                                                                                            | Its number                                                                                                                                 | Status, fence                                                                                  | The click that reads it                                                                          | Where                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------|
| The band’s inertia and kinetic scale,<br>$m^* = 3 \tan \omega_0$ , $c_m^2 = \omega_0 / (3 \tan \omega_0)$ ,<br>$E = m^* c_m^2 = \omega_0$ : one band read<br>three ways                                                                | $c_m/c = 0.867$ at [2, 3],<br>one speed as the gap<br>closes                                                                               | derived; lattice for<br>the band, clicks for<br>the clock                                      | a matter-armed<br>interferometer shifts at<br>order $v^2$                                        | 8 (e); S.18,<br>S.21, S.22 |
| The post-Newtonian parameters with<br>the gap, for $0 < \text{num} \leq \text{den}$ ,<br>$\gamma_K = \text{den}/\text{num}$ ,<br>$\beta_K = (\text{den} + \text{num})/(2\text{num})$ ,<br>$1 + \alpha_K = \tan \omega_0 / \omega_0$    | 1.5, 1.25, 1.329 at [2, 3];<br>1 with the gap                                                                                              | derived under the<br>one assumption;<br>clicks                                                 | Cassini’s<br>$\gamma_K - 1 = (2.1 \pm 2.3) \times 10^{-5}$<br>bounds $\text{den}/\text{num} - 1$ | 8 (b), (d);<br>S.19        |
| Yilmaz’s exponential metric,<br>$N^2 = e^{-2U}$ , $h^2 = e^{2U}$ , $\gamma = \beta = 1$<br>exactly in $U$ , reached from the<br>composed paces; its spatial second<br>order 2 against the Schwarzschild<br>chart’s 1.5 is the addition | exact in $U$ ; the redshift’s<br>factor $f(\omega_0) = 1 + \alpha_U$ ,<br>$\alpha_U = 0.063$                                               | derived under the<br>one assumption;<br>lattice, the second<br>order clicks                    | one click in the deep level                                                                      | 3.4, 8;<br>S.34            |
| The shadow’s capture radius<br>$b = 2e m$ , against Schwarzschild’s<br>$3\sqrt{3} m$                                                                                                                                                   | 5.44 $m$ against 5.20 $m$ ,<br>4.6 percent                                                                                                 | derived under the<br>one assumption,<br>bounded by<br>measurement and not<br>confirmed; clicks | the shadow’s diameter (Sgr<br>A* 1.4, M87* 0.3 standard<br>deviations)                           | 8; S.30                    |
| The gravity of light: a light packet’s<br>integrated write at $1 \times E/c^2$ (static<br>kernel; the packet’s field open),<br>against general relativity’s $2 \times E/c^2$                                                           | the factor 1                                                                                                                               | derived, untested;<br>clicks                                                                   | no measurement has read<br>it                                                                    | 10.2; S.41                 |
| The lens, Eq. (17): one formula for<br>light and matter, for light exact as<br>$\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$ , with the<br>chromatic term $k^2(e^{6U} - e^{2U})/24$                                                          | light’s index $e^{2U}$ and<br>slow matter’s<br>$n_m \propto 1/k^*$ at long<br>wavelength                                                   | derived, two<br>derivations; lattice<br>for the index, clicks<br>for the focus                 | a chromatic bending of<br>light at finite $k$                                                    | 8 (p); S.28                |
| The fine-structure constant’s form,<br>$\alpha = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$ , $k$ the<br>declared write weight                                                                                                                   | $k/(\Gamma E_s) = 1/28.33$ at<br>nature’s $\alpha$                                                                                         | derived form, the<br>weight declared;<br>clicks                                                | the one click that fixes $k$                                                                     | 10.1; S.26                 |
| The moving clock’s fourth order,<br>$\omega_0 [(\sum_a n_a^4) \tan \omega_0 + \cot \omega_0]$ ,<br>anisotropic by $\sum_a n_a^4$                                                                                                       | 1.6926 along an axis,<br>1.2224 along a face<br>diagonal, 1.0657 along<br>the body diagonal at<br>[2, 3]                                   | derived; clicks                                                                                | a clock flown along the<br>lattice’s axes against its<br>diagonals                               | 8 (e); S.21                |
| Bell’s and GHZ’s values at rational<br>settings, $S = 478/169$ and $M = -4$                                                                                                                                                            | 478/169 = 2.82840<br>against $2\sqrt{2} = 2.82843$                                                                                         | an evaluation of the<br>declared credit,<br>fitted, no prediction;<br>clicks                   | the credit’s own check                                                                           | 9.4; S.10,<br>S.40         |
| The frozen content, $U_f = \frac{1}{2} \ln(2\Gamma)$ ,<br>$(\Gamma/2) \ln 2\Gamma$ levels                                                                                                                                              | 28,178 levels at<br>$\Gamma = 6,000$ by the<br>formula, 28,206 by the<br>law’s integers ( $p_0$<br>rounded to the unit<br>before squaring) | derived under the<br>one assumption;<br>lattice                                                | no click bounds it                                                                               | 3.4, 10.4;<br>S.41         |
| The clusters’ deceleration,<br>$q_0 = \Omega_m/2 > 0$ at $z \ll 1$ , bound or<br>free                                                                                                                                                  | the sign                                                                                                                                   | derived at $z \ll 1$ ;<br>clicks                                                               | the supernovae’s Hubble<br>diagram stands against it,<br>a miss named in<br>Section 10.3         | 7.5; S.29                  |
| A body’s binding per quantum,<br>bounded by $0.76 CW/(\Gamma E)$ at one<br>Node and about $0.29 CW/(\Gamma E r)$ for a<br>ball of $r$ Links                                                                                            | the lattice’s bound; that<br>it lies below nature’s<br>nuclear binding is a miss<br>named in Section 10.3                                  | derived, a bound;<br>lattice                                                                   | none                                                                                             | 10.3; S.38                 |
| The click’s booking identity,<br>$\Delta(wQ) = w(x-v)(x-b)$ , with the<br>record’s own quantum $W_{\text{rec}}$                                                                                                                        | exact at the Node at any<br>paces                                                                                                          | derived; clicks                                                                                | the share’s change under<br>any write                                                            | 5.4; S.50                  |

where known. The open list is the program, each item a derivation to complete or a gate file to run. The paper gives the direction and not the whole, so that others can take the method, with this set of rules or another, and continue.

# Statements and Declarations

**Funding.** None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data and code availability.** The law's document, the implementation and the scripts of the figures are developed in the public Git repository <https://github.com/Closer24/Universe24>. They are archived at Zenodo under the concept DOI 10.5281/zenodo.22738746 [43], the archive's version 1.1.0, under the MIT licence. **Author contributions.** The author conceived the model, carried out the derivations and the computations with the tools stated below, and wrote the paper. **The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) extensively as a tool, under the author's direction, for the software, the computations, the derivations and their checks, and the drafting of the text. The model is not an author. The author reviewed every statement and takes full responsibility for the content.

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